LeanForControl

Anand Gokhale Francesco Bullo

University of California, Santa Barbara

October 28, 2026

Chapter 1

Analysis toolbox

Facts about real functions, limits and integrals with no differential-equation content of their own. Each section says which later results consume it.

1.1 Continuity toolbox

Continuity gives the last root before a sign change and the first exit of a curve from an open region. First sphere crossing is the open-ball specialization of the general exit lemma.

Lemma 1 (Greatest zero of a sign-changing continuous function)**.** *Let g be continuous on $[t_0, t_1]$ with $g(t_0) \le 0 < g(t_1)$. Then g has a* last *root: there is $a \in [t_0, t_1)$ with $g(a) = 0$ and $g(t) > 0$ for every $t \in (a, t_1]$.*

Proof. □

Lemma 2 (Boundary crossing by a continuous curve)**.** *A continuous curve on $[t_0, t_1]$, with $t_0 \le t_1$, starting in an open set D and ending outside D meets ∂D on that interval.*

Proof. The image of the interval is connected. If it avoided the boundary, meeting D would force the whole image to lie in D. □

Lemma 3 (First exit from an open set)**.** *If a continuous curve on $[t_0, T]$ starts in an open set D and is outside D at $t_1 \in [t_0, T]$, some $\tau \in [t_0, t_1]$ satisfies $\varphi(\tau) \in \partial D$ and $\varphi(t) \in D$ for $t \in [t_0, \tau)$.*

Proof. The nonempty compact set of exit times has a minimum. A boundary crossing by that time must occur at the minimum itself. □

Theorem 4 (First crossing of a sphere)**.** *Let φ be continuous on $[t_0, T]$ with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < \rho$, and suppose some $t_1 \in [t_0, T]$ satisfies $\rho \le \|\varphi(t_1) - x_{\mathrm{eq}}\|$. Then there is a* first *hitting time $\tau \in [t_0, T]$ of the sphere of radius ρ:*

$$\|\varphi(\tau) - x_{\mathrm{eq}}\| = \rho, \qquad \|\varphi(s) - x_{\mathrm{eq}}\| \le \rho \quad \forall\, s \in [t_0, \tau].$$

Proof. Apply lemma 3 to the open ball. Its boundary lies in the sphere; before the first exit the curve remains in the ball. □

A field cut off outside a set stays Lipschitz.

Lemma 5 (Cutting off a field that is Lipschitz on a set)**.** *Let g be L-Lipschitz and bounded by B on S, and let χ be M-Lipschitz with $|\chi| \le 1$, vanishing off S. Then χg is $(L + MB)$-Lipschitz on the whole space.*

Proof. □

1.2 Dini derivatives

The upper right Dini derivative replaces the ordinary derivative where a function is only known to be continuous, which is the situation of the comparison lemma. As in the literature it takes values in $[-\infty, +\infty]$: the forward difference quotients may be unbounded, and the extended-real lim sup always exists, so no boundedness hypotheses are needed. Bounds against a real number are characterized by the difference quotients themselves.

Definition 6 (Upper right Dini derivative)**.** The *upper right Dini derivative* of $f : \mathbb{R} \to \mathbb{R}$ at t is

$$D^+ f(t) \;=\; \limsup_{h \to 0^+} \frac{f(t+h) - f(t)}{h} \in [-\infty, +\infty].$$

When f is differentiable at t this equals $f'(t)$.

1.2.1 Relationship to the classical derivative

Lemma 7 (Forward difference quotients of a right-differentiable function)**.** *If $f : \mathbb{R} \to E$ has right derivative f' at x, then $h^{-1}(f(x+h) - f(x)) \to f'$ as $h \to 0^+$.*

Proof. □

Lemma 8 (Backward difference quotients of a left-differentiable function)**.** *If $f : \mathbb{R} \to E$ has left derivative f' at x, then $h^{-1}(f(x+h) - f(x)) \to f'$ as $h \to 0^-$.*

Proof. □

Lemma 9 (Dini derivative of a differentiable function)**.** *If f has right derivative L at t, then $D^+ f(t) = L$: the Dini derivative agrees with the classical one wherever the latter exists.*

Proof. □

1.2.2 Upper bounds

Lemma 10 (Upper bound on the Dini derivative)**.** *$D^+ f(t) \le L$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ is eventually at most $L + \varepsilon$ as $h \to 0^+$.*

Proof. □

Lemma 11 (Dini derivative of a norm)**.** *If x has right derivative v at t and $f(s) - f(t) \le \|x(s) - x(t)\|$ for all s, then $D^+ f(t) \le \|v\|$; in particular for $f = \pm\|x\|$.*

Proof. For $h > 0$ the forward quotient of f is at most $\|h^{-1}(x(t+h)-x(t))\|$, which tends to $\|v\|$. □

Lemma 12 (Monotonicity of the Dini derivative)**.** *If $f(a) = g(a)$ and $f \le g$ on a right neighbourhood of a, then $D^+f(a) \le D^+g(a)$.*

Proof. The forward quotients of f at a are eventually below those of g, and lim sup is monotone. □

Lemma 13 (Monotonicity of the right derivative)**.** *If $f(a) = g(a)$, $f \le g$ on a right neighbourhood of a, and f, g have right derivatives f', g' at a, then $f' \le g'$.*

Proof. Both right derivatives are Dini derivatives (lemma 9); apply lemma 12. □

1.2.3 Grönwall's inequality in Dini form

Theorem 14 (Grönwall's inequality, Dini form)**.** *Let v be continuous on $[a,b]$ with $D^+v(t) \le K\,v(t)$ for $t \in [a,b)$. Then $v(t) \le v(a)\,e^{K(t-a)}$ on $[a,b]$.*

Proof. □

Lemma 15 (Dini derivative from a dissipation inequality)**.** *If g is continuous on $[r,\infty)$ and $f(z) + \int_r^z g \le f(r)$ for z in a right neighbourhood of r, then $D^+f(r) \le -g(r)$.*

Proof. $\frac{f(r+h)-f(r)}{h} \le -\frac{1}{h}\int_r^{r+h} g \to -g(r)$. □

1.3 Interval integrals

Estimates and integrability side conditions for the interval integral, used wherever a differential equation is handled in integral form. The last three concern an integral whose endpoints move: normalizing by the window length makes it a mean value, after which continuity, antitonicity and decay each need only a hypothesis on the endpoints themselves. They support the class $\mathcal{KL}$ construction of the chapter on comparison functions.

Lemma 16 (Difference of integrals over adjacent intervals)**.** *For μ continuous on $[a,t]$ and $s \in [a,t]$,*

$$\int_a^t \mu - \int_a^s \mu = \int_s^t \mu.$$

Proof. □

Lemma 17 (Integral bound from a pointwise multiplicative bound)**.** *Let u take values in a normed space and v be real, both integrable on $[a,b]$ with $a \le b$. If $\|u(s)\| \le L\,v(s)$ for all $s \in [a,b]$, then*

$$\Big\|\int_a^b u\Big\| \le L \int_a^b v.$$

Proof. □

Lemma 18 (Integral bound from a constant pointwise bound)**.** *If $\|u(s)\| \le C$ for all $s \in [a,b]$ with $a \le b$, then*

$$\Big\|\int_a^b u\Big\| \le C\,(b-a).$$

Proof. □

Lemma 19 (Integral of a constant)**.** $\int_a^b C\,\mathrm{d}s = (b-a)\,C$ *for a real constant* C*.*

Proof. □

Lemma 20 (Interval integrability of a continuous composition)**.** *If* f *is jointly continuous and* z *is continuous on the segment between* t_0 *and* t_1*, then* $s \mapsto f(s, z(s))$ *is interval-integrable between them. This is the integrability side condition every integral-form solution argument needs.*

Proof. □

Lemma 21 (Differentiating an integral in its upper limit)**.** *For* μ *continuous on* $[a, b]$ *and* $t \in (a, b)$*,*

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big|_{x=t} \int_a^x \mu(\tau)\,\mathrm{d}\tau \;=\; \mu(t).$$

Proof. □

Lemma 22 (Differentiating an integral with variable limits)**.** *For* w *continuous on* $\mathbb{R}$ *and* g*,* h *differentiable at* t*,*

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Big|_{\tau=t} \int_{g(\tau)}^{h(\tau)} w(s)\,\mathrm{d}s = h'(t)\,w(h(t)) - g'(t)\,w(g(t)).$$

Proof. $\int_g^h w = \int_0^h w - \int_0^g w$; differentiate each term by the fundamental theorem of calculus and the chain rule. □

Lemma 23 (Translating an integral over a half-line)**.** *For any* w *and* $a, c \in \mathbb{R}$*,* $\int_a^\infty w(\sigma + c)\,\mathrm{d}\sigma = \int_{a+c}^\infty w(s)\,\mathrm{d}s$*.*

Proof. □

Lemma 24 (Continuity of an integral in its upper limit)**.** *If* f *is integrable on* $[a, t]$ *with* $a \le t$*, then* $s \mapsto \int_a^s f(\tau)\,\mathrm{d}\tau$ *is continuous on* $[a, t]$*.*

Proof. □

Lemma 25 (Continuity of an integral in both endpoints)**.** *Let* f *be locally integrable on* $(0, \infty)$*, and let* a, b *be continuous on a set* S *with* $a(\eta), b(\eta) > 0$ *in* S*. Then*

$$\eta \;\longmapsto\; \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is continuous on S*.*

Proof. □

Lemma 26 (The sliding average of an antitone function is antitone)**.** *Let f be antitone and locally integrable on $(0,\infty)$, and let a, b be monotone on a set S with $0 < a(\eta) < b(\eta)$. Then the mean value*

$$\eta \;\longmapsto\; \frac{1}{b(\eta)-a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is antitone on S.

Proof. □

Lemma 27 (Sliding average of an eventually vanishing function)**.** *Let f be nonneg, antitone and locally integrable on $(0,\infty)$ with $f(s) \to 0$ as $s \to +\infty$, and let $a(\eta) < b(\eta)$ with $a(\eta) \to +\infty$ as $\eta \to +\infty$. Then*

$$\frac{1}{b(\eta)-a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x \;\longrightarrow\; 0 \qquad (\eta \to +\infty).$$

The mean is squeezed between 0 and $f(a(\eta))$; no hypothesis relating a to b is needed.

Proof. □

1.4 Monotonicity above a level

Lemma 28 (Nonincreasing above a level)**.** *Let v be continuous on $[a,b]$ with right derivative $v'(t)$ at every $t \in [a,b)$, and suppose $v'(t) \le 0$ whenever $v(t) > \eta$. Then $v(t) \le \max\{v(a), \eta\}$ for all $t \in [a,b]$.*

Proof. Let $m = \max\{v(a), \eta\}$ and $\varepsilon > 0$. The barrier $B(t) = m + \varepsilon\,(1 + t - a)$ starts above v, and wherever v touches it, $v > \eta$, so $v' \le 0 < \varepsilon = B'$. The barrier comparison theorem gives $v \le B$ on $[a,b]$; let $\varepsilon \to 0$. □

1.5 Norm inequalities

1.6 Positive definite functions

Positive definiteness in the sense of functions, not matrices: a function that vanishes at a point and is positive elsewhere. It is the shape every Lyapunov function and every comparison bound takes.

Definition 29 (Positive definite function)**.** A function W is *positive definite* on S about x_0 when it is continuous on S, $W(x_0) = 0$, and $W(x) > 0$ for every $x \in S \;\; \{x_0\}$.

Lemma 30 (Compositions of positive definite functions)**.** *If g is positive definite about 0 on $[0,\infty)$ and W is positive definite about x_0 on S (definition 29), then $g \circ W$ is positive definite about x_0 on S.*

Proof. □

Lemma 31 (The distance is positive definite)**.** *$x \mapsto \|x - x_0\|$ is positive definite about x_0 on every set (definition 29).*

Proof. □

Lemma 32 (The squared distance is positive definite)**.** *For $c > 0$, $x \mapsto c\|x - x_0\|^2$ is positive definite about x_0 on every set (definition 29).*

Proof. □

Chapter 2

Ordinary differential equations

2.1 Gronwall–Bellman inequality

A function obeying an integral inequality is bounded by the solution of the corresponding integral equation. The two specializations are what the rest of the development actually calls: constant Λ gives exponential growth, and constant Λ and μ gives the $Ce^{\mu(t-a)}$ bound behind every continuous-dependence estimate.

Theorem 33 (Gronwall–Bellman inequality)**.** *Khalil, Lemma A.1. Let $\Lambda, \mu : [a,b] \to \mathbb{R}$ be continuous with $\mu \geq 0$, and let $y : [a,b] \to \mathbb{R}$ be continuous satisfying*

$$y(t) \;\leq\; \Lambda(t) + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then

$$y(t) \;\leq\; \Lambda(t) + \int_a^t \Lambda(s)\, \mu(s)\, e^{\int_s^t \mu(\tau)\,\mathrm{d}\tau}\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Proof. An integrating-factor argument. Write $z(t) = \int_a^t \mu y$, $M(t) = \int_a^t \mu$, and $v = z + \Lambda - y$, so that $v \geq 0$ is exactly the hypothesis. The transform $w = e^{-M} z$ has, by the product rule and lemma 21,

$$\dot{w} = e^{-M}\mu\,(y - z) = e^{-M}\mu\,(\Lambda - v) \;\leq\; e^{-M}\mu\Lambda,$$

the inequality because e^{-M}, μ and v are all nonnegative — this is the only place $\mu \geq 0$ is used, and it is essential: a sign change would reverse it. Integrating from a, where $w(a) = 0$, gives $w(t) \leq \int_a^t e^{-M(s)}\mu(s)\Lambda(s)\, ds$. Multiplying by $e^{M(t)}$ and folding $e^{M(t)}e^{-M(s)} = e^{\int_s^t \mu}$ turns this into the stated bound on $z(t)$, and $y \leq \Lambda + z$ finishes. □

Lemma 34 (Gronwall–Bellman with a constant rate)**.** *Khalil, Lemma A.1 (the case $\Lambda(t) \equiv \Lambda$ constant). Let μ be continuous and nonnegative on $[a,b]$, and let y be continuous with*

$$y(t) \;\leq\; C + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then $y(t) \leq C\, e^{\int_a^t \mu(\tau)\,\mathrm{d}\tau}$ on $[a,b]$.

Proof. theorem 33 at $\Lambda \equiv C$ gives $y(t) \le C + \int_a^t C\mu(s) e^{\int_s^t \mu}\, ds$, and the integral evaluates in closed form: $s \mapsto -C\, e^{M(t)-M(s)}$ is an antiderivative of the integrand, so the integral is $Ce^{M(t)} - C$. The two C's cancel. □

Lemma 35 (Gronwall–Bellman with constant data)**.** *Khalil, Lemma A.1 (the case Λ and $\mu \ge 0$ both constant). Let $\mu \ge 0$ be constant and let y be continuous with*

$$y(t) \;\le\; C + \int_a^t \mu\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a, b].$$

Then $y(t) \le C\, e^{\mu(t-a)}$ on $[a, b]$. This is the form the continuous-dependence estimates use, and the reason their bounds carry a factor $e^{L(t-t_0)}$.

Proof. lemma 34 with $\int_a^t \mu = \mu(t-a)$. □

2.2 Existence, uniqueness and continuous dependence

Definition 36 (Integral solution)**.** A function x, defined on the segment between t_0 and t_1, is an *integral solution* of $\dot{x} = F(t, x)$ with initial value x_0 when

$$x(t) \;=\; x_0 + \int_{t_0}^t F(s,\, x(s))\, \mathrm{d}s$$

for every t between t_0 and t_1, the integrand being integrable.

Lemma 37 (Re-anchoring an integral solution)**.** *Let x be an integral solution of $\dot{x} = F(t, x)$ with initial value x_0 on the segment between t_0 and t_1. Then for any s, r on that segment, x is an integral solution on the segment between s and r, with initial value $x(s)$:*

$$x(t) \;=\; x(s) + \int_s^t F(w, x(w))\, \mathrm{d}w \qquad \text{for } t \text{ between } s \text{ and } r.$$

Leaving both endpoints free means the domain need not shrink — taking $s = t_1$ and $r = t_0$ re-anchors at one end of the original segment while retaining all of it.

Proof. Subtract the defining equations at t and at s and split the integral at s, which is legitimate because $w \mapsto F(w, x(w))$ is integrable on each piece, by the definition of an integral solution. □

Lemma 38 (Time reflection of an integral solution)**.** *Let x be an integral solution of $\dot{x} = F(t, x)$ with initial value x_0 on the segment between t_0 and t_1. Then $\sigma \mapsto x(t_0 + t_1 - \sigma)$ is an integral solution, with the same initial value x_0, of the reflected field*

$$(r, y) \;\longmapsto\; -F(t_0 + t_1 - r,\; y)$$

on the segment between t_1 and t_0.

The map $\sigma \mapsto t_0 + t_1 - \sigma$ is an involution exchanging the two endpoints, so reflection turns a statement anchored at t_0 into one anchored at t_1 and nothing more. Its purpose is to let an argument that is inherently forward-marching — a Grönwall bootstrap, say — be applied unchanged in backward time.

Proof. Substituting $r \mapsto t_0 + t_1 - r$ in the defining integral reverses the orientation of the interval, contributing one sign, and the negation of the field contributes another; the two cancel. □

2.2.1 The integral and differential formulations

An ODE can be written as a differential equation, $\dot{x} = F(t, x)$, or as the equivalent integral equation. The continuous-dependence and Gronwall arguments want the integral form, which presupposes no differentiability and carries its initial condition; the Lyapunov arguments want the differential form, so that $V \circ \varphi$ can be differentiated. The results below are the two directions of the fundamental theorem of calculus relating them.

They are provision rather than current practice: nothing in the stability chapters has yet been routed through this equivalence, so it is the intended path from a trajectory to the results of this chapter rather than an exercised one.

Lemma 39 (Integral curves are integral solutions)**.** *Let x be an integral curve of F on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral solution on that segment, anchored at its own value $x(t_0)$:*

$$x(t) \;=\; x(t_0) + \int_{t_0}^{t} F(s, x(s))\,\mathrm{d}s.$$

Continuity of x itself is not assumed — it follows from differentiability.

Proof. The fundamental theorem of calculus. The derivative hypothesis holds on an open neighbourhood of each interior point, so it yields the right-derivative form the theorem needs, and continuity of F along x gives the integrability side condition. □

Lemma 40 (Integral solutions are integral curves)**.** *Let x be an integral solution of $\dot{x} = F(t, x)$ on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral curve of F on that segment: it is differentiable, with $\dot{x}(t) = F(t, x(t))$.*

Proof. Differentiating the primitive: the integral term has derivative $F(t, x(t))$ by the fundamental theorem of calculus, and the constant x_0 contributes nothing. Transport that along the defining equation, which says x agrees with the primitive. □

Lemma 41 (Integral and differential forms agree)**.** *Given continuity of $s \mapsto F(s, x(s))$ on the segment between t_0 and t_1, the two formulations agree: x is an integral solution anchored at $x(t_0)$ if and only if it is an integral curve of F there. Continuity of F along x is exactly the price of the equivalence — without it the integral form is the weaker notion.*

Proof. The two implications are lemma 40 and lemma 39. □

Lemma 42 (Integral and differential forms agree, forward time)**.** *lemma 41 in forward time: for $t_0 \le t_1$, the equivalence holds over $[t_0, t_1]$. This is the interval the finite-forward stability predicates are phrased over, so this is the form the stability track would use to reach the results of this chapter.*

Proof. □

Lemma 43 (Integral form for an autonomous field)**.** *Let g be a continuous* autonomous *field and let x be an integral curve of g on $[t_0, t_1]$, $t_0 \le t_1$. Then x is an integral solution there, anchored at $x(t_0)$.*

This is the autonomous, forward-time specialization of lemma 42, and so the shape a trajectory of an autonomous system has: continuity of the field is the only hypothesis, the continuity of x along which F must be continuous being supplied by the curve itself.

Proof. □

2.2.2 Time invariance and local existence

A time-invariant field has no preferred time origin: a solution can be shifted to start at time 0.

Lemma 44 (Time invariance)**.** *If x is an integral curve of an autonomous field g on a set s, then $t \mapsto x(t + \Delta t)$ is an integral curve of the same g on $s - \Delta t$, for every Δt.*

An autonomous field has no preferred time origin, so translating a solution translates its interval of definition and changes nothing else. For a time-varying field the translated solution would solve the translated *equation instead.*

Proof. □

Lemma 45 (Shifting a solution segment to the origin)**.** *lemma 44 in the form the finite-segment predicates need: a solution segment of an autonomous field on $[t_0, t_1]$ becomes one on $[0, t_1 - t_0]$ under $s \mapsto x(s + t_0)$.*

This is why the autonomous stability predicates may fix the initial time at 0 without loss: any segment can be shifted there.

Proof. □

Lemma 46 (Constant integral curves at equilibria)**.** *If $v(t, p) = 0$ for every $t \in s$, then the constant curve $t \mapsto p$ is an integral curve of $\dot{x} = v(t, x)$ on s.*

Proof. □

Lemma 47 (Local existence for a C^1 field)**.** *Let g be an autonomous field that is C^1 at x_0. Then there is a $T > 0$ and a curve φ with $\varphi(0) = x_0$ that is an integral curve of g on $[0, T]$: a nontrivial solution segment exists from every point at which the field is C^1.*

The anchor 0 is a construction choice rather than a restriction — by lemma 44 a segment from any other anchor follows.

Reference: the Picard–Lindelöf local existence theorem.

Proof. Picard–Lindelöf local existence: being C^1 at x_0 gives a closed ball on which g is Lipschitz, and hence a solution on some $(-\varepsilon, \varepsilon)$; take $T := \varepsilon/2$. □

Theorem 48 (Picard–Lindelöf on a compact interval)**.** *Let $f : \mathbb{R} \to E \to E$ be jointly continuous and K-Lipschitz in the state variable, uniformly in time, on a Banach space E. Then for every $t_0 \le t_1$ and every $x_0 \in E$ there is an α with $\alpha(t_0) = x_0$ solving $\dot{\alpha} = f(t, \alpha)$ on all of $[t_0, t_1]$.*

Global — rather than merely local — Lipschitz continuity is what makes the solution reach t_1: it forces linear growth in the state, ruling out finite-time blowup.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2 (Global Existence and Uniqueness), proved in his Appendix C.1. Difference: Khalil assumes f piecewise continuous in t; here it is jointly continuous.*

Proof. Mathlib's Picard–Lindelöf theorem supplies a solution on a time interval of length Δt only when $\|f\| \le L$ on a ball of radius a about x_0 with $L\,\Delta t \le a$. For a K-Lipschitz field the sharpest available bound on that ball is $L = M + Ka$, where M bounds $\|f(t, x_0)\|$, so the condition becomes $M\Delta t \le a(1 - K\Delta t)$ — solvable in a exactly when $K\Delta t < 1$. One application therefore never covers a long interval.

Take instead a step h with $Kh \le 1/2$; then $a = 2Mh$ and $L = 2M$ satisfy the hypotheses, since $\|f(t, x)\| \le M + Ka = M(1 + 2Kh) \le 2M$ and $Lh = 2Mh = a$. This gives a solution on $[s, s + h]$

from any initial state. Chain n such steps, splicing consecutive segments at their shared endpoint: away from a junction the glued curve agrees with one piece on a neighbourhood, and at a junction the two one-sided derivatives combine. Choosing n with $nh \geq t_1 - t_0$ covers $[t_0, t_1]$. □

Lemma 49 (Right derivatives of an integral curve)**.** *If α is an integral curve of f on $[t_0, t_1]$, then for every $s \in [t_0, t_1)$ it has right derivative $f(s, \alpha(s))$ at s, i.e. α has derivative $f(s, \alpha(s))$ within $[s, \infty)$ at s.*

Proof. □

Lemma 50 (Gronwall separation of two solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$. Then*

$$d(\alpha(t), \beta(t)) \leq d(\alpha(t_0), \beta(t_0))\, e^{K(t-t_0)} \qquad \forall\, t \in [t_0, t_1].$$

Uniqueness (lemma 51) is the case where the initial distance is zero; continuous dependence on the initial state is the statement read as a bound on the whole interval.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.4, in the unperturbed case: two solutions of the* same *field, so the perturbation bound is zero.*

Proof. □

Lemma 51 (Uniqueness of solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$ with $\alpha(t_0) = \beta(t_0)$. Then $\alpha = \beta$ on $[t_0, t_1]$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), the uniqueness half of Theorem 3.2.*

Proof. The initial distance in lemma 50 is zero, so the bound forces the distance to vanish throughout. □

Lemma 52 (Integral curves of a bounded field move at bounded speed)**.** *Let γ be an integral curve of v on a convex set s, with $\|v(\tau, \gamma(\tau))\| \leq B$ for $\tau \in s$. Then $\|\gamma(\tau_2) - \gamma(\tau_1)\| \leq B\,|\tau_2 - \tau_1|$ for $\tau_1, \tau_2 \in s$.*

Proof. □

Theorem 53 (Picard–Lindelöf on the forward ray)**.** *Let f be jointly continuous and K-Lipschitz in the state variable, uniformly in time. Then for every t_0 and x_0 there is an α with $\alpha(t_0) = x_0$ solving $\dot{\alpha} = f(t, \alpha)$ on all of $[t_0, \infty)$, and any other such solution agrees with it on $[t_0, \infty)$.*

Uniqueness is agreement on $[t_0, \infty)$, not equality of functions: the defining condition says nothing about $t < t_0$, so solutions may differ there.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, whose t_1 is arbitrary, extended to the forward ray.*

Proof. For each n take a solution α_n on $[t_0, t_0 + n]$ (theorem 48). By uniqueness (lemma 51) these agree wherever two of them are both defined, so $\Phi(t) := \alpha_{\lceil t - t_0 \rceil}(t)$ is well defined and agrees with α_n on all of $[t_0, t_0 + n]$. Near any $t \geq t_0$ choose $n > t - t_0$; then $[t_0, t_0 + n]$ is a neighbourhood of t within $[t_0, \infty)$ on which $\Phi = \alpha_n$, so Φ inherits the derivative there. Uniqueness on the ray follows by applying the compact-interval uniqueness at each n. □

Theorem 54 (Picard–Lindelöf, scalar integral form)**.** *Let $g : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ be jointly continuous and L-Lipschitz in its state argument, uniformly in time, and let $t_0 \le t_1$. Then for every x_0 there is a z, continuous on $[t_0, t_1]$, that is an integral solution of $\dot{z} = g(t, z)$ with $z(t_0) = x_0$ and has right derivative $g(s, z(s))$ at each $s \in [t_0, t_1)$.*

This is the scalar specialization of theorem 48; it discharges the perturbed-solution hypothesis that theorem 58 takes for generality.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, scalar case.*

Proof. Apply theorem 48 with $E = \mathbb{R}$. Continuity on $[t_0, t_1]$ is automatic for an integral curve, which converts the differential form into the integral one (lemma 42); the right derivatives are lemma 49. □

2.2.3 Continuous dependence

Two solutions of the same equation, started at different points or subject to different perturbations, separate no faster than exponentially — at a rate fixed by the Lipschitz constant of the field. Everything downstream that needs uniqueness gets it from here, by taking the two solutions to agree initially and the perturbation to vanish, which collapses the bound to zero.

Theorem 55 (Continuous dependence on initial states and parameters)**.** *Khalil, Theorem 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on the segment between t_0 and t_1, let y be an integral solution of $\dot{y} = f(t, y)$ with $y(t_0) = y_0$, and let z be an integral solution of the perturbed equation $\dot{z} = f(t, z) + g(t, z)$ with $z(t_0) = z_0$, where $\|g(t, x)\| \le \mu$. Then for every t on that segment,*

$$\|y(t) - z(t)\| \;\le\; \|y_0 - z_0\|\, e^{L|t-t_0|} \;+\; \frac{\mu}{L}(e^{L|t-t_0|} - 1).$$

Two solutions may separate, but no faster than exponentially, at a rate set by the Lipschitz constant alone; the two error sources — a different starting point and a perturbed field — contribute additively. The bound is stated on the unordered segment, so it runs in either time direction.

The perturbation g is assumed only bounded and integrable along z, never continuous, so z need not be differentiable anywhere. That is why the theorem is phrased over the integral formulation rather than the differential one.

Proof. The forward case is a Grönwall bootstrap. Subtracting the two integral equations and applying the triangle inequality gives

$$\|y(\tau) - z(\tau)\| \;\le\; \|y_0 - z_0\| + \mu(\tau - t_0) + L \int_{t_0}^{\tau} \|y(s) - z(s)\|\, \mathrm{d}s,$$

the Lipschitz hypothesis bounding the f-difference and the uniform bound on g the rest. Adding μ/L to both sides absorbs the linear term $\mu(\tau - t_0)$ into the integral, putting the inequality in the shape lemma 35 expects, and Grönwall delivers the exponential.

That bootstrap is inherently forward-marching, so rather than repeat it for the backward case, reflect: lemma 38 sends $\sigma \mapsto t_0 + t_1 - \sigma$ and negates the field, turning the backward instance into a forward one on $[t_1, t_0]$ to which the same argument applies unchanged. □

The same mechanism bounds a single solution from below: an equilibrium cannot be approached faster than exponentially.

Theorem 56 (Continuous dependence on parameters)**.** *Khalil, Theorem 3.5. If y solves $\dot{y} = f(t, y)$ and z solves $\dot{z} = f(t, z) + g(t, z)$ with $\|g(t, x)\| \le \alpha$ and $\|z_0 - y_0\| \le \alpha$, and $\alpha(1 + 1/L)e^{L(t_1 - t_0)} \le \varepsilon$, then $\|y(t) - z(t)\| \le \varepsilon$ for all $t \in [t_0, t_1]$.*

Proof. Specialise theorem 55 with $\mu = \alpha$ and $\|y_0 - z_0\| \le \alpha$: for $t \in [t_0, t_1]$ it gives $\|y(t) - z(t)\| \le \alpha e^{L(t-t_0)} + \frac{\alpha}{L}(e^{L(t-t_0)} - 1) = \alpha(1 + 1/L)e^{L(t-t_0)} - \alpha/L$. Drop the $-\alpha/L$ and bound $e^{L(t-t_0)}$ by $e^{L(t_1-t_0)}$ to get at most ε. □

2.3 Comparison lemma

The comparison lemma bounds a function satisfying a differential *inequality* by the solution of the corresponding differential *equation*. The inequality is stated with the Dini derivative, so v need only be continuous — which is what makes the lemma applicable to $\|x(t)\|$, not differentiable where a trajectory crosses the origin.

Lemma 57 (Integral form from a pointwise derivative)**.** *Let u be continuous on $[t_0, t_1]$ with $u(t_0) = u_0$ and $\dot{u}(t) = f(t, u(t))$ at every interior point, and let f be jointly continuous. Then u is an integral solution:*

$$u(s) = u_0 + \int_{t_0}^{s} f(\tau, u(\tau))\, \mathrm{d}\tau.$$

.

Proof. The fundamental theorem of calculus on $[t_0, s]$; the integrand is continuous there, being a composition of continuous maps. □

Theorem 58 (Comparison lemma)**.** *Khalil, Lemma 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on $[t_0, t_1]$, and suppose*

1. *u is continuous on $[t_0, t_1]$ and solves $\dot{u} = f(t, u)$ with $u(t_0) = u_0$;*
2. *v is continuous on $[t_0, t_1]$ with $v(t_0) \le u_0$, satisfies the Dini subsolution inequality $D^+v(t) \le f(t, v(t))$;*
3. *for every $\lambda > 0$ the perturbed equation $\dot{z} = f(t, z) + \lambda$, $z(t_0) = u_0$, has a solution on $[t_0, t_1]$.*

Then $v(t) \le u(t)$ for all $t \in [t_0, t_1]$.

The perturbed solutions in (3) are taken as a hypothesis for generality, so that callers with solutions already in hand need not re-derive them. Under the hypotheses already listed they always exist: $f + \lambda$ is Lipschitz exactly when f is, so theorem 54 discharges (3).

Proof. Fix $\lambda > 0$ and let z_λ be as in hypothesis (3).

Claim: $v \le z_\lambda$ on $[t_0, t_1]$. Otherwise $v(t_{\mathrm{bad}}) > z_\lambda(t_{\mathrm{bad}})$ for some t_{bad}, and since $v(t_0) \le z_\lambda(t_0)$ the continuous difference $v - z_\lambda$ has a last zero $a < t_{\mathrm{bad}}$, with $v > z_\lambda$ on $(a, t_{\mathrm{bad}}]$. Being below v immediately to the right of a forces $\dot{z}_\lambda(a) \le D^+v(a)$, so

$$f(a, z_\lambda(a)) + \lambda \;\le\; D^+v(a) \;\le\; f(a, v(a)) = f(a, z_\lambda(a)),$$

contradicting $\lambda > 0$.

Now let $\varepsilon > 0$ and choose λ small enough that theorem 56 — which applies because f is Lipschitz in its state argument, and which treats λ as a perturbation of size λ — gives $\|u - z_\lambda\| \le \varepsilon/2$ on $[t_0, t_1]$. Then $v \le z_\lambda \le u + \varepsilon$. As ε was arbitrary, $v \le u$. □

2.4 Lie derivatives

The rate of change of a function $V(t,x)$ along the integral curves of a time-dependent vector field, with no differential-equation content beyond the chain rule: what every Lyapunov, Chetaev and invariance argument differentiates.

Definition 59 (Lie derivative along a vector field)**.** For $V : \mathbb{R}\times E \to \mathbb{R}$ and a time-dependent vector field $v : \mathbb{R}\times E \to E$, the *Lie derivative of V along v* is

$$L_v V(t,x) = \frac{\partial V}{\partial t}(t,x) + \frac{\partial V}{\partial x}(t,x)\, v(t,x),$$

the derivative of V at (t,x) in the direction $(1, v(t,x))$: the Lie derivative of V, as a function on $\mathbb{R}\times E$, along the autonomous field $(t,x)\mapsto(1,v(t,x))$.

Lemma 60 (Lie derivative of a time-invariant function)**.** *If V does not depend on t and is differentiable at x, then $L_v V(t,x) = DV(x)\, v(t,x)$.*

Proof. □

Lemma 61 (Lie derivative of $V(x)+ct$)**.** *If V does not depend on t and is differentiable at x, then the Lie derivative of $(t,x)\mapsto V(x)+ct$ along v is $DV(x)\, v(t,x)+c$.*

Proof. □

Lemma 62 (Chain rule along an integral curve)**.** *Let γ be an integral curve of v on I, and V differentiable at $(t,\gamma(t))$ for $t\in I$. Then for $t\in I$, within I,*

$$\frac{d}{dt}\, V(t,\gamma(t)) = L_v V(t,\gamma(t))$$

(definition 59).

Proof. □

Lemma 63 (The Lie derivative along a solution)**.** *Let γ be an integral curve of v on $[t,\infty)$, V differentiable at $(t,\gamma(t))$, and let $\tau\mapsto V(\tau,\gamma(\tau))$ have right derivative w at t. Then $L_v V(t,\gamma(t)) = w$.*

Proof. By lemma 62 the right derivative is also $L_v V(t,\gamma(t))$, and right derivatives are unique. □

Lemma 64 (The Lie derivative of an integral along a solution)**.** *Let γ be an integral curve of v on $[t,\infty)$, V differentiable at $(t,\gamma(t))$, ψ continuous, and $V(\tau,\gamma(\tau)) = C + \int_t^\tau \psi$ for $\tau\ge t$. Then $L_v V(t,\gamma(t)) = \psi(t)$.*

Proof. By the fundamental theorem of calculus the right-hand side has derivative $\psi(t)$ at t; apply lemma 63. □

Lemma 65 (Lie derivative along a perturbed field)**.** *If V is differentiable at (t,x), then for any two fields v and w,*

$$L_w V(t,x) = L_v V(t,x) + \frac{\partial V}{\partial x}(t,x)\,(w(t,x)-v(t,x))$$

(definition 59).

Proof. Linearity of the derivative of V at (t, x) in the direction $(1, w) = (1, v) + (0, w - v)$; its value on $(0, e)$ is the state derivative applied to e. □

Lemma 66 (Nonpositive Lie derivative along an integral curve)**.** *Let γ be an integral curve of v on an interval s, V differentiable at $(t, \gamma(t))$ for $t \in s$, and $L_v V(t, \gamma(t)) \le 0$ at every interior point t of s. Then $t \mapsto V(t, \gamma(t))$ is nonincreasing on s.*

Proof. The mean value theorem, with the derivative given by lemma 62. □

Lemma 67 (Integrated Lyapunov inequality)**.** *Let γ be an integral curve of v on $[t_0, t_1]$, V differentiable at $(t, \gamma(t))$ for $t \in [t_0, t_1]$, W continuous on $[t_0, t_1]$, and $L_v V(t, \gamma(t)) \le -W(t)$ for $t \in (t_0, t_1)$. Then*

$$V(t_1, \gamma(t_1)) + \int_{t_0}^{t_1} W(t)\, dt \le V(t_0, \gamma(t_0)).$$

Proof. Integrate the derivative bound of lemma 62. □

Lemma 68 (Exponential bound from a linear Lie-derivative bound)**.** *Let γ be an integral curve of v on $[t_0, t_1]$, V differentiable at $(t, \gamma(t))$ for $t \in [t_0, t_1]$, and $c \in \mathbb{R}$ with $L_v V(t, \gamma(t)) \le -c\, V(t, \gamma(t))$ for $t \in (t_0, t_1)$. Then $V(t, \gamma(t)) \le e^{-c(t-t_0)}\, V(t_0, \gamma(t_0))$ on $[t_0, t_1]$.*

Proof. The derivative of $e^{ct} V(t, \gamma(t))$ is $e^{ct}(cV + L_v V) \le 0$ (lemma 62), so it does not increase. □

Lemma 69 (Continuity of the Lie derivative)**.** *If V is C^1 and v is continuous, then $(t, x) \mapsto L_v V(t, x)$ is continuous.*

Proof. □

2.5 ω-limit sets

The points a curve or sequence returns arbitrarily close to, arbitrarily late: its cluster points at infinity, which is the ω-limit of topological dynamics for a single orbit. These are the topological facts behind LaSalle's invariance principle, in continuous and in discrete time alike: a continuous function converging along the orbit is constant on its ω-limit set, an orbit trapped in a closed set has its ω-limit set there, and an orbit trapped in a compact set converges to its ω-limit set.

Definition 70 (ω-limit set)**.** The *ω-limit set* of $\varphi : \tau \to X$, for τ a preordered index set such as $\mathbb{R}$ or $\mathbb{N}$, is the set of cluster points of φ as $t \to \infty$:

$$\omega(\varphi) = \bigcap_T \overline{\{\varphi(t) : t \ge T\}},$$

the points y such that $\varphi(t)$ is frequently in every neighbourhood of y as $t \to \infty$.

Lemma 71 (Membership in the ω-limit set)**.** *$y \in \omega(\varphi)$ (definition 70) if and only if for every neighbourhood U of y there are arbitrarily large t with $\varphi(t) \in U$.*

Proof. □

Lemma 72 (The ω-limit set as Mathlib's ω-limit)**.** *definition 70 is the ω-limit, in the sense of topological dynamics, of the one-member family $\{\varphi\}$.*

Proof. □

Lemma 73 (A limit of $g\circ\varphi$ is the value of g on $\omega(\varphi)$)**.** *Let $g : X \to Y$ be continuous, Y Hausdorff, and $g(\varphi(t)) \to L$ as $t \to \infty$. Then $g(y) = L$ for every $y \in \omega(\varphi)$ (definition 70).*

Proof. $g(y)$ is a cluster point of $g\circ\varphi$, which converges to L; in a Hausdorff space the two coincide. □

Lemma 74 (The ω-limit set of a curve trapped in a closed set)**.** *If Ω is closed and $\varphi(t) \in \Omega$ for all large t, then $\omega(\varphi) \subseteq \Omega$.*

Proof. A point outside Ω has the open neighbourhood $X \setminus \Omega$, which φ eventually avoids. □

Lemma 75 (Convergence to a set containing the ω-limit set)**.** *If $\varphi(t)$ lies in a compact set for all large t and $\omega(\varphi) \subseteq M$, then $\varphi(t) \to M$: for every open $W \supseteq M$, $\varphi(t) \in W$ for all large t.*

Proof. Mathlib's `eventually_mapsTo_of_isCompact_absorbing_of_isOpen_of_omegaLimit_subset` (lemma 72): a family eventually absorbed by a compact set is eventually inside every open set containing its ω-limit set. □

Chapter 3

Systems

The most general systems the library considers are $\dot{x} = F(t, x, u)$ in continuous time and $x^+ = F(k, x, u)$ in discrete time, each with an output $y = h(t, x, u)$. As for the linear systems of the next chapter, there is one object per time domain: the data has the same shape, but a continuous trajectory is an integral curve and a discrete one is a recursion.

Special cases are constructors on the general object, each with a predicate recognising it: *time invariance* drops t, and *autonomy* drops u. Autonomy is used here in the control-theory sense, *without input*; much of the dynamical-systems literature uses the word for time invariance instead.

3.1 Continuous time

Definition 76 (Continuous-time system)**.** A *continuous-time system* with state space E, input space U and output space Y is the data of two maps

$$F : \mathbb{R} \times E \times U \to E, \qquad h : \mathbb{R} \times E \times U \to Y,$$

to be read as $\dot{x} = F(t, x, u)$, $y = h(t, x, u)$. The structure is only the data.

Definition 77 (Time-invariant continuous-time system)**.** The continuous-time system $\dot{x} = f(x, u)$, $y = g(x, u)$, whose equations do not depend on time.

Definition 78 (Time invariance, continuous-time system)**.** A continuous-time system is *time-invariant* when $F(t, \cdot, \cdot)$ and $h(t, \cdot, \cdot)$ are the same for every t.

Lemma 79 (Time-invariant continuous-time systems are time-invariant)**.** *The system of definition 77 is time-invariant (definition 78).*

Proof. □

Lemma 80 (Time-invariant continuous-time systems come from time-free data)**.** *A time-invariant continuous-time system (definition 78) is* timeInvariant *f g for some f and g. The converse of lemma 79.*

Proof. Read f and g off at $t = 0$; time invariance says every other time agrees with it. □

Definition 81 (Autonomous continuous-time system)**.** The continuous-time system $\dot{x} = f(t, x)$, $y = g(t, x)$, which has no input: its input space is the one-point type.

Definition 82 (Autonomy, continuous-time system)**.** A continuous-time system is *autonomous* when $F(t, x, u)$ and $h(t, x, u)$ do not depend on u.

Lemma 83 (Autonomous continuous-time systems are autonomous)**.** *The system of definition 81 is autonomous (definition 82).*

Proof. □

Lemma 84 (Autonomous continuous-time systems come from input-free data)**.** *If a continuous-time system is autonomous (definition 82) and its input space is nonempty, there are f and g with $F(t, x, u) = f(t, x)$ and $h(t, x, u) = g(t, x)$ for every input u.*

Proof. Fix any input u_0 and read f and g off at it. □

Definition 85 (The system with its input held constant)**.** For u_0 in the input space, the autonomous system (definition 81) $\dot{x} = F(t, x, u_0)$, $y = h(t, x, u_0)$ obtained by holding the input at u_0.

Definition 86 (The unforced system)**.** The system with its input held at 0 (definition 85): $\dot{x} = F(t, x, 0)$, $y = h(t, x, 0)$. Stability of a system with input means stability of this system.

Lemma 87 (Trajectories of the unforced system)**.** *A map is a trajectory of the unforced system (definition 86) on I exactly when it is a trajectory of s under the zero input on I.*

Proof. □

Definition 88 (Cascade of continuous-time systems)**.** For a system s_1 whose input space is the state space of a system s_2 with input w, the *cascade* is the system on pairs (x_1, x_2) with input w

$$\dot{x}_1 = F_1(t, x_1, x_2), \qquad \dot{x}_2 = F_2(t, x_2, w),$$

with output $(h_1(t, x_1, x_2), h_2(t, x_2, w))$: the state of s_2 drives s_1.

Lemma 89 (Trajectories of a cascade)**.** *A map (x_1, x_2) is a trajectory of the cascade (definition 88) under the input w on I exactly when x_1 is a trajectory of s_1 under the input x_2 on I and x_2 is a trajectory of s_2 under w on I.*

Proof. □

Lemma 90 (Right trajectories of a cascade)**.** *A map (x_1, x_2) is a right trajectory of the cascade (definition 112, definition 88) under w on I exactly when x_1 is a right trajectory of s_1 under the input x_2 and x_2 is a right trajectory of s_2 under w, on I.*

Proof. □

Well-posedness is kept apart from stability. The stability notions quantify over every solution, so they need no uniqueness; forward completeness enters only where a limit as $t \to \infty$ is taken. Theorems that need either property assume it.

Definition 91 (Forward completeness from a set)**.** A continuous-time system is *forward complete* from a set S at time t_0 under the input u when every trajectory on a segment $[t_0, t_1]$ (definition 108) with $x(t_0) \in S$ agrees on that segment with a trajectory on $[t_0, \infty)$.

Definition 92 (Forward completeness)**.** A continuous-time system is *forward complete* under the input u when it is forward complete (definition 91) from every state at every initial time.

Definition 93 (Uniqueness of solutions)**.** A continuous-time system has *unique solutions* under the input u when any two trajectories on a segment $[t_0, t_1]$ (definition 108) that agree at t_0 agree on all of $[t_0, t_1]$.

Theorem 94 (Lipschitz systems have unique solutions)**.** *If $x \mapsto F(t, x, u(t))$ is K-Lipschitz for every t, the system has unique solutions under u (definition 93).*

Proof. Grönwall's inequality, in the form of Mathlib's `ODE_solution_unique_of_mem_Icc_right`: a derivative within $[t_0, t_1]$ at a point of $[t_0, t_1)$ is a right derivative there. □

Theorem 95 (Globally Lipschitz systems are forward complete)**.** *If $(t, x) \mapsto F(t, x, u(t))$ is continuous and K-Lipschitz in x uniformly in t, the system is forward complete under u (definition 92).*

Proof. From the endpoint $x(t_0)$ of a segment, Picard–Lindelöf gives a solution on $[t_0, \infty)$ (theorem 53), which agrees with the segment by uniqueness (theorem 94). □

Theorem 96 (Equilibria are not reached in finite time)**.** *If the state equation is Lipschitz in the state, uniformly in time, and a trajectory on $[t_0, t_1]$ satisfies $F(t, x(t_1), u(t)) = 0$ for all t, then $x(t) = x(t_1)$ for every $t \in [t_0, t_1]$.*

Proof. x and the constant curve at $x(t_1)$ (lemma 46) are two solutions agreeing at t_1; backward uniqueness (Grönwall) makes them equal. □

Theorem 97 (Exponential bounds on a trajectory)**.** *If x is a trajectory on $[t_0, t_1]$ (definition 108) with $\|F(t, x(t), u(t))\| \le L\|x(t) - x_c\|$ for $t \in [t_0, t_1)$, then for every $t \in [t_0, t_1]$*

$$\|x(t_0) - x_c\|\, e^{-L(t-t_0)} \le \|x(t) - x_c\| \le \|x(t_0) - x_c\|\, e^{L(t-t_0)}.$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Exercise 3.17(b).*

Proof. $D^+\|x - x_c\| \le \|\dot{x}\| \le L\|x - x_c\|$ and $D^+(-\|x - x_c\|) \le \|\dot{x}\| \le (-L)(-\|x - x_c\|)$ (lemma 11); apply Grönwall (theorem 14) to each. □

3.1.1 The state transition map

The state transition map $\varphi(t; t_0, x_0)$ is the solution from x_0 at time t_0, evaluated at time t, on its maximal interval of existence. The state transition matrix of a linear system is its linear case.

Definition 98 (Existence interval)**.** The *existence interval* of the trajectory of s under u from x_0 at time t_0 is the set of $t \ge t_0$ for which a trajectory on $[t_0, t]$ (definition 108) with $x(t_0) = x_0$ exists: the maximal interval of existence.

Definition 99 (State transition map)**.** The *state transition map* $\varphi(t; t_0, x_0)$ of s under u is the value at time t of a trajectory from x_0 at time t_0, for t in the existence interval (definition 98); with unique solutions (definition 93) it does not depend on the trajectory chosen. Outside the existence interval it is x_0, by convention.

Lemma 100 (Existence intervals of a forward complete system)**.** *If s is forward complete under u (definition 92), the existence interval (definition 98) from every (t_0, x_0) is $[t_0, \infty)$.*

Proof. The trajectory on $[t_0, t_0]$ is trivial; forward completeness extends it to $[t_0, \infty)$. □

Lemma 101 (Trajectories are given by the state transition map)**.** *If s has unique solutions under u (definition 93), every trajectory x on $[t_0, t_1]$ satisfies $x(t) = \varphi(t; t_0, x(t_0))$ there (definition 99).*

Proof. x itself witnesses that t is in the existence interval, and the trajectory chosen by φ agrees with x on $[t_0, t]$ by uniqueness. □

Lemma 102 (The state transition map is a trajectory)**.** *If s has unique solutions under u (definition 93), $t \mapsto \varphi(t; t_0, x_0)$ (definition 99) is a trajectory on the existence interval (definition 98).*

Proof. Near each time t of the existence interval, φ agrees with a trajectory on a segment $[t_0, t']$ containing a neighbourhood of t in the interval (lemma 101), so it has that trajectory's derivative. □

Lemma 103 (The semigroup property)**.** *If s has unique solutions under u (definition 93), then for $t_0 \le \tau \le t$ with t in the existence interval from (t_0, x_0),*

$$\varphi(t; \tau, \varphi(\tau; t_0, x_0)) = \varphi(t; t_0, x_0).$$

Proof. $\varphi(\cdot; t_0, x_0)$ is a trajectory on $[\tau, t]$ (lemma 102), so it is given by φ from $(\tau, \varphi(\tau; t_0, x_0))$ (lemma 101). □

Theorem 104 (Lipschitz dependence on the initial state)**.** *If the system is forward complete under u (definition 92) and $x \mapsto F(t, x, u(t))$ is L-Lipschitz for every t, then for $t \ge t_0$*

$$\|\varphi(t; t_0, x_0) - \varphi(t; t_0, y_0)\| \le \|x_0 - y_0\| \, e^{L(t - t_0)}$$

(definition 99).

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.4, without perturbation.*

Proof. Both maps are trajectories on $[t_0, \infty)$ (theorem 94); apply Grönwall's inequality to their difference on $[t_0, t]$. □

Theorem 105 (Lipschitz dependence on the initial time and state)**.** *If the system is forward complete under u (definition 92), and $x \mapsto F(t, x, u(t))$ is L-Lipschitz and bounded by B for every t, then for $\sigma \ge 0$*

$$\|\varphi(t_0 + \sigma; t_0, x_0) - \varphi(t_0' + \sigma; t_0', x_0')\| \le (2B + 1) \, e^{L\sigma} \, \|(t_0, x_0) - (t_0', x_0')\|$$

(definition 99), the norm on $\mathbb{R} \times E$ being the maximum.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Section 3.2.*

Proof. Say $t_0 \le t_0'$. The first solution moves at most $B|t_0 - t_0'|$ between the two initial times and between the two final times; from t_0' on, the two solutions separate at most like $e^{L\sigma}$ (theorem 104). □

Definition 106 (State equation of a continuous-time system)**.** Under an input signal $u : \mathbb{R} \to U$, the right-hand side of the state equation is the time-dependent vector field $(t, x) \mapsto F(t, x, u(t))$.

Definition 107 (Output equation of a continuous-time system)**.** Under an input signal u, the output equation is $(t, x) \mapsto h(t, x, u(t))$. An instantaneous relation: no derivative appears.

Definition 108 (Trajectory of a continuous-time system)**.** A map $x : \mathbb{R} \to E$ is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when

$$\dot{x}(t) = F(t, x(t), u(t)), \qquad \forall\, t \in I,$$

the derivative being taken within I. Stated on a set: a map can satisfy the equation on an interval without being defined for all time.

Definition 109 (Trajectory for all time of a continuous-time system)**.** A trajectory (definition 108) defined on all of $\mathbb{R}$.

Lemma 110 (Trajectories of a continuous-time system on the whole line)**.** *definition 108 at $I = \mathbb{R}$ is definition 109.*

Proof. □

Lemma 111 (Restriction of a trajectory of a continuous-time system)**.** *A trajectory for all time is a trajectory on every set.*

Proof. □

Inputs that switch — steps, sample-and-hold, bang-bang — produce solutions with corners at the switching times. With the input taken right-continuous, such a solution still has a right derivative everywhere, given by the new value of the input; these right trajectories are the solution concept for input-to-state stability, and every trajectory is one.

Definition 112 (Right trajectory of a continuous-time system)**.** A map $x : \mathbb{R} \to E$ is a *right trajectory* of s under the input u on $I \subseteq \mathbb{R}$ when it is continuous on I and has right derivative

$$\dot{x}(t^+) = F(t, x(t), u(t)), \qquad \forall\, t \in I,$$

taken within $I \cap [t, \infty)$. It is the solution concept for piecewise continuous, right-continuous inputs: at a switching time the solution has a corner, but its right derivative is given by the new value of the input. Every trajectory (definition 108) is one.

Lemma 113 (Trajectories are right trajectories)**.** *A trajectory on I (definition 108) is a right trajectory on I (definition 112).*

Proof. □

A set is forward invariant when trajectories that start in it stay in it.

Definition 114 (Forward invariant set)**.** A set $S \subseteq E$ is *forward invariant* (or positively invariant) for s under the input u when every trajectory on a segment $[t_0, t_1]$ (definition 108) that starts in S stays in S:

$$x(t_0) \in S \;\Rightarrow\; x(t) \in S \quad \forall\, t \in [t_0, t_1].$$

The derivative of a function along the system is the rate at which it changes along trajectories. It takes the input value as an argument: stability uses the input-free system, input-to-state notions a general input.

Definition 115 (Derivative along the system)**.** For $V : \mathbb{R} \times E \to \mathbb{R}$ and an input value $u \in U$, the *derivative of V along the system $\dot{x} = F(t, x, u)$* is

$$\dot{V}(t, x, u) = \frac{\partial V}{\partial t}(t, x) + \frac{\partial V}{\partial x}(t, x)\, F(t, x, u),$$

the Lie derivative of V along the field $F(\cdot, \cdot, u)$ (definition 59).

The derivative form above excludes the solutions produced by discontinuous inputs, which have corners where the input jumps. The integral form admits them, and is the one input-driven notions such as input-to-state stability are stated with; for continuous right-hand sides the two agree.

Definition 116 (Carathéodory trajectory of a continuous-time system)**.** A map $x : \mathbb{R} \to E$, with E a Banach space, is a *Carathéodory trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when, for all $a \leq b$ in I, the map $\tau \mapsto F(\tau, x(\tau), u(\tau))$ is integrable on $[a, b]$ and

$$x(b) - x(a) = \int_a^b F(\tau, x(\tau), u(\tau))\, \mathrm{d}\tau.$$

Unlike definition 108 it does not require x to be differentiable, so it admits the solutions produced by discontinuous or merely measurable inputs, which input-to-state stability quantifies over.

Definition 117 (Carathéodory trajectory for all time)**.** A Carathéodory trajectory (definition 116) on all of $\mathbb{R}$.

Lemma 118 (Restriction of a Carathéodory trajectory)**.** *A Carathéodory trajectory on I is one on every subset of I; in particular one for all time is one on every set.*

Proof. □

Theorem 119 (Carathéodory and classical trajectories agree on a segment)**.** *Let $t_0 \leq t_1$ and let $\tau \mapsto F(\tau, x(\tau), u(\tau))$ be continuous on $[t_0, t_1]$. Then x is a Carathéodory trajectory on $[t_0, t_1]$ (definition 116) if and only if it is a trajectory there (definition 108).*

Proof. Both are equivalent to the integral equation anchored at t_0 (definition 36), by lemma 41. The anchored equation gives the equation between any $a \leq b$ by subtracting its values at b and a. □

Theorem 120 (Carathéodory and classical trajectories agree on an interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval (order-connected) and let $\tau \mapsto F(\tau, x(\tau), u(\tau))$ be continuous on I. Then x is a Carathéodory trajectory on I (definition 116) if and only if it is a trajectory there (definition 108).*

Order-connectedness cannot be dropped: on $\{0\} \cup [1, 2]$ the integral form relates $x(1)$ to $x(0)$, the derivative form does not.

Proof. Both notions reduce to closed segments inside I, where they agree by theorem 119. For the integral form, the equation between $a \leq b$ only involves $[a, b] \subseteq I$. For the derivative form, at $t \in I$ choose $a \leq t \leq b$ in I, taking $a < t$ if I has a point left of t and $a = t$ otherwise, and likewise for b. Then $[a, b] \subseteq I$ agrees with I near t, so a derivative within $[a, b]$ at t is one within I. □

3.2 Discrete time

Definition 121 (Discrete-time system)**.** A *discrete-time system* with state space E, input space U and output space Y is the data of two maps

$$F : \mathbb{N} \times E \times U \to E, \qquad h : \mathbb{N} \times E \times U \to Y,$$

to be read as $x^+ = F(k, x, u)$, $y = h(k, x, u)$. The same data as definition 76 with time in $\mathbb{N}$; what differs is the reading, a shift rather than a derivative.

Definition 122 (Time-invariant discrete-time system)**.** The discrete-time system $x^+ = f(x, u)$, $y = g(x, u)$, whose equations do not depend on time.

Definition 123 (Time invariance, discrete-time system)**.** A discrete-time system is *time-invariant* when $F(k, \cdot, \cdot)$ and $h(k, \cdot, \cdot)$ are the same for every k.

Lemma 124 (Time-invariant discrete-time systems are time-invariant)**.** *The system of definition 122 is time-invariant (definition 123).*

Proof. □

Lemma 125 (Time-invariant discrete-time systems come from time-free data)**.** *A time-invariant discrete-time system (definition 123) is* timeInvariant *f g for some f and g. The converse of lemma 124.*

Proof. Read f and g off at $k = 0$; time invariance says every other time agrees with it. □

Definition 126 (Autonomous discrete-time system)**.** The discrete-time system $x^+ = f(k, x)$, $y = g(k, x)$, which has no input: its input space is the one-point type.

Definition 127 (Autonomy, discrete-time system)**.** A discrete-time system is *autonomous* when $F(k, x, u)$ and $h(k, x, u)$ do not depend on u.

Lemma 128 (Autonomous discrete-time systems are autonomous)**.** *The system of definition 126 is autonomous (definition 127).*

Proof. □

Lemma 129 (Autonomous discrete-time systems come from input-free data)**.** *If a discrete-time system is autonomous (definition 127) and its input space is nonempty, there are f and g with $F(k, x, u) = f(k, x)$ and $h(k, x, u) = g(k, x)$ for every input u.*

Proof. Fix any input u_0 and read f and g off at it. □

Definition 130 (State equation of a discrete-time system)**.** Under an input sequence $u : \mathbb{N} \to U$, the right-hand side of the state equation is the time-dependent next-state map $(k, x) \mapsto F(k, x, u(k))$. The same expression as definition 106; what differs is what a trajectory does with it, a shift rather than a derivative.

Definition 131 (Output equation of a discrete-time system)**.** Under an input sequence u, the output equation is $(k, x) \mapsto h(k, x, u(k))$.

Definition 132 (Trajectory of a discrete-time system)**.** A sequence $x : \mathbb{N} \to E$ is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{N}$ when

$$x(k+1) = F(k, x(k), u(k)), \qquad \forall\, k \in I.$$

Membership constrains the step *out of* k, so $I = [k_0, \infty)$ is the forward recursion from k_0, and the horizon $[k_0, k_1]$ corresponds to $I = [k_0, k_1)$.

Definition 133 (Trajectory for all time of a discrete-time system)**.** A trajectory (definition 132) satisfying the recursion at every $k \in \mathbb{N}$.

Lemma 134 (Trajectories of a discrete-time system on the whole domain)**.** *definition 132 at $I = \mathbb{N}$ is definition 133.*

Proof. □

Lemma 135 (Restriction of a trajectory of a discrete-time system)**.** *A trajectory at every time is a trajectory on every set.*

Proof. □

Unlike continuous time, a trajectory needs no hypothesis to exist or to be unique: the recursion defines it from any initial time and state for all later times. So no discrete-time solution escapes in finite time, and nothing like forward completeness is ever needed.

Lemma 136 (Trajectories of a discrete-time system exist)**.** *For every input u, initial time k_0 and state x_0 there is a trajectory (definition 132) on $[k_0, \infty)$ with $x(k_0) = x_0$.*

Proof. Define it by the recursion itself. □

Lemma 137 (Trajectories of a discrete-time system are unique)**.** *Two trajectories (definition 132) on $[k_0, \infty)$ under the same input with $x(k_0) = y(k_0)$ satisfy $x(k) = y(k)$ for every $k \geq k_0$.*

Proof. Induction on k. □

Lemma 138 (Trajectories of a time-invariant discrete-time system are iterates)**.** *If $F(k, x, u(k)) = f(x)$ for all k and x, every trajectory (definition 132) on $[k_0, \infty)$ satisfies $x(k_0 + n) = f^n(x(k_0))$ for every $n \in \mathbb{N}$.*

Proof. Induction on n. □

Definition 139 (The unforced discrete-time system)**.** The autonomous system (definition 126) $x^+ = F(k, x, 0)$, $y = h(k, x, 0)$ obtained by holding the input at 0. Stability of a system with input means stability of this system.

Lemma 140 (Trajectories of the unforced discrete-time system)**.** *A sequence is a trajectory of the unforced system (definition 139) on I exactly when it is a trajectory of s under the zero input on I.*

Proof. □

Chapter 4

Contraction theory

A *pairing* replaces the inner product in contraction analysis for norms that come from no inner product. Regular pairings are the pairings for which the norm of a curve can be differentiated through the pairing, which is what contraction arguments use.

4.1 Regular pairings

Every normed space carries a regular pairing, built from the norm alone; on an inner product space the inner product is another.

4.2 What a regular pairing says about the norm

Every regular pairing lies between the left and right derivatives of the norm, and so it computes the derivative of the norm along any curve. This is what makes $\frac{1}{2}\|x\|^2$ usable as a Lyapunov function for every norm.

4.3 The logarithmic norm

With Lumer's property, each basic property of the log norm reduces to a property of a pairing.

On an inner product space the pairing is the inner product, so a log-norm bound is a quadratic-form inequality.

4.4 Weighted norms

A weighted norm is an ordinary norm, so every result above applies to it; the one new fact is how its operator and log norms relate to the unweighted ones.

A weight also carries a regular pairing to the weighted norm.

4.5 Log norms of matrices

The ℓ_1 and ℓ_∞ norms have regular pairings in closed form, and through them a weighted log-norm bound is an inequality for every vector.

For the norm $\sqrt{x^\top P x}$ the log norm is a quadratic-form bound and an eigenvalue; the Euclidean norm is the case $P = I$.

4.6 One-sided Lipschitz maps

The one-sided Lipschitz constant extends the log norm from linear to nonlinear maps. For a continuously differentiable map it is the largest log norm of the Jacobian, which is how contractivity of a nonlinear system is checked in practice.

For maps of $\mathbb{R}^n$ the Jacobian test combines with the matrix log norms above.

Chapter 5

Linear systems

5.1 The linear system objects

Linear systems are a special case of the general systems of the Systems chapter: the continuous-time system with $F(t,x,u) = A(t)x + B(t)u$ and $h(t,x,u) = C(t)x + D(t)u$, and its discrete-time counterpart. The objects below carry only the coefficient data. Each is the input of a constructor into the general system, and the semantics — state equation, output equation, trajectories — are defined through that constructor, so every result about general systems applies to linear ones unchanged.

There are two objects per time domain: a *linear time-varying* (LTV) system with coefficient maps $A(t), B(t), C(t), D(t)$, and a *linear time-invariant* (LTI) system with constant matrices. An LTI system is not a different kind of system: it builds the LTV system with constant coefficients, and the general system it describes is the time-invariant system of the Systems chapter, by definition.

5.1.1 Continuous-time LTV systems

Definition 141 (Continuous-time linear time-varying system)**.** A *continuous-time linear time-varying (LTV) system* with state space E, input space U and output space Y (real normed spaces) is the data of four maps on $\mathbb{R}$ into continuous linear maps,

$$A : \mathbb{R} \to (E \to_L E), \quad B : \mathbb{R} \to (U \to_L E), \quad C : \mathbb{R} \to (E \to_L Y), \quad D : \mathbb{R} \to (U \to_L Y),$$

describing the continuous-time system (definition 76) $\dot{x} = A(t)x + B(t)u$, $y = C(t)x + D(t)u$ (definition 142).

Definition 142 (The system of an LTV system)**.** The continuous-time system (definition 76) on E, U, Y with $F(t,x,u) = A(t)x + B(t)u$ and $h(t,x,u) = C(t)x + D(t)u$.

Definition 143 (State equation of an LTV system)**.** The right-hand side of the state equation, $(t,z) \mapsto A(t)\,z + B(t)\,u(t)$: the vector field (definition 106) of definition 142.

Lemma 144 (The unforced state equation of an LTV system)**.** *At zero input the state equation (definition 143) reduces to* $(t,z) \mapsto A(t)\,z$.

Proof. □

Lemma 145 (The state equation of an LTV system at the origin)**.** *At $z = 0$ the state equation (definition 143) contributes only the input term, $A(t)\,0 + B(t)u(t) = B(t)u(t)$. With the input switched off the origin is therefore an equilibrium of the system.*

Proof. □

Definition 146 (Output equation of an LTV system)**.** The output equation, $(t, z) \mapsto C(t)\,z + D(t)\,u(t)$: the output map (definition 107) of definition 142.

Definition 147 (Trajectory of an LTV system)**.** A map x is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when it is a trajectory (definition 108) of definition 142:

$$\dot{x}(t) = A(t)\,x(t) + B(t)\,u(t), \qquad \forall\, t \in I.$$

Definition 148 (Trajectory for all time of an LTV system)**.** A trajectory (definition 109) for all time of definition 142.

Definition 149 (Carathéodory trajectory of an LTV system)**.** A Carathéodory trajectory (definition 116) of definition 142.

Definition 150 (Time invariance of an LTV system)**.** A continuous-time LTV system is *time-invariant* when each of its four coefficients takes the same value at every pair of times.

Lemma 151 (Time-invariant LTV systems are time-invariant systems)**.** *If an LTV system is time-invariant (definition 150), so is the system it describes (definition 78).*

Proof. □

5.1.2 Continuous-time LTI systems

Definition 152 (Continuous-time linear time-invariant system)**.** A *continuous-time linear time-invariant (LTI) system* is the data of four matrices $A \in \mathbb{K}^{X\times X}$, $B \in \mathbb{K}^{X\times U}$, $C \in \mathbb{K}^{Y\times X}$, $D \in \mathbb{K}^{Y\times U}$, describing the LTV system (definition 141) with these constant coefficients, $\dot{x} = Ax + Bu$, $y = Cx + Du$.

Definition 153 (The LTV system of an LTI system)**.** The LTV system (definition 141) on $\mathbb{R}^X$, $\mathbb{R}^U$, $\mathbb{R}^Y$ whose coefficients are the constant maps $t \mapsto A$, $t \mapsto B$, $t \mapsto C$, $t \mapsto D$, the matrices acting on Euclidean vectors.

Lemma 154 (LTI systems are time-invariant LTV systems)**.** *The LTV system of an LTI system (definition 153) is time-invariant (definition 150).*

Proof. □

Definition 155 (The system of an LTI system)**.** The continuous-time system (definition 76) of the LTV system of s (definition 142, definition 153).

Lemma 156 (An LTI system is a linear time-invariant system)**.** *The system of an LTI system (definition 155) is the time-invariant system (definition 77) with $f(x, u) = Ax + Bu$ and $g(x, u) = Cx + Du$: the two routes from the LTI data to a general system agree.*

Proof. □

5.1.3 Discrete-time LTV systems

Definition 157 (Discrete-time linear time-varying system)**.** A *discrete-time linear time-varying (LTV) system* is the same data as definition 141 with the coefficients indexed by $\mathbb{N}$ instead of $\mathbb{R}$, describing the discrete-time system (definition 121) $x(k+1) = A(k)x(k) + B(k)u(k)$, $y(k) = C(k)x(k) + D(k)u(k)$ (definition 158).

The forward time domain $\mathbb{N}$ rather than $\mathbb{Z}$ matches the discrete solution theory, whose state transition matrix is an ordered product over $[t_0, t)$.

Definition 158 (The system of a discrete-time LTV system)**.** The discrete-time system (definition 121) with $F(k, x, u) = A(k)x + B(k)u$ and $h(k, x, u) = C(k)x + D(k)u$.

Definition 159 (State equation of a discrete-time LTV system)**.** The right-hand side of the state equation, $(k, z) \mapsto A(k)\, z + B(k)\, u(k)$: the next-state map (definition 130) of definition 158.

Lemma 160 (The unforced state equation of a discrete-time LTV system)**.** *At zero input the state equation (definition 159) reduces to $(k, z) \mapsto A(k)\, z$.*

Proof. □

Lemma 161 (The state equation of a discrete-time LTV system at the origin)**.** *At $z = 0$ the state equation (definition 159) contributes only the input term, $A(k)\, 0 + B(k)u(k) = B(k)u(k)$. With the input switched off the origin is therefore an equilibrium of the system.*

Proof. □

Definition 162 (Output equation of a discrete-time LTV system)**.** The output equation, $(k, z) \mapsto C(k)\, z + D(k)\, u(k)$: the output map (definition 131) of definition 158.

Definition 163 (Trajectory of a discrete-time LTV system)**.** A sequence x is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{N}$ when it is a trajectory (definition 132) of definition 158:

$$x(k+1) = A(k)\, x(k) + B(k)\, u(k), \qquad \forall\, k \in I.$$

Definition 164 (Trajectory for all time of a discrete-time LTV system)**.** A trajectory (definition 133) for all time of definition 158.

Definition 165 (Time invariance of a discrete-time LTV system)**.** A discrete-time LTV system is *time-invariant* when each of its four coefficients takes the same value at every pair of times.

Lemma 166 (Time-invariant discrete-time LTV systems are time-invariant systems)**.** *If a discrete-time LTV system is time-invariant (definition 165), so is the system it describes (definition 123).*

Proof. □

5.1.4 Discrete-time LTI systems

Definition 167 (Discrete-time linear time-invariant system)**.** A *discrete-time linear time-invariant (LTI) system* is the data of four matrices A, B, C, D, describing the discrete-time LTV system (definition 157) with these constant coefficients, $x(k+1) = Ax(k) + Bu(k)$, $y(k) = Cx(k) + Du(k)$.

Definition 168 (The LTV system of a discrete-time LTI system)**.** The discrete-time LTV system (definition 157) whose coefficients are the constant maps $k \mapsto A$, $k \mapsto B$, $k \mapsto C$, $k \mapsto D$.

Lemma 169 (Discrete-time LTI systems are time-invariant LTV systems)**.** *The LTV system of a discrete-time LTI system (definition 168) is time-invariant (definition 165).*

Proof. □

Lemma 170 (Time-invariant discrete-time LTV systems come from LTI systems)**.** *A time-invariant discrete-time LTV system (definition 165) is the LTV system of some LTI system. The converse of lemma 169.*

Proof. Read the coefficients off at $k = 0$. □

Definition 171 (The system of a discrete-time LTI system)**.** The discrete-time system (definition 121) of the LTV system of s (definition 158, definition 168).

Lemma 172 (A discrete-time LTI system is a linear time-invariant system)**.** *The system of a discrete-time LTI system (definition 171) is the time-invariant system (definition 122) with $f(x, u) = Ax + Bu$ and $g(x, u) = Cx + Du$: the two routes from the LTI data to a general system agree.*

Proof. □

5.2 Peano–Baker series and the state transition matrix

For a genuinely time-varying coefficient $A(t)$ in a real Banach algebra — matrices under any submultiplicative norm, or the bounded operators on a Banach space — this section constructs the state transition matrix $\Phi(t, t_0)$ as the Peano–Baker series and proves Theorem 5.1: Φ solves $\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0)$ with $\Phi(t_0, t_0) = 1$, and consequently $x(t) := \Phi(t, t_0)x_0$ solves $\dot{x} = A(t)x$, $x(t_0) = x_0$.

Standing hypothesis. Throughout this section A is continuous. The term-wise estimates additionally assume a bound $\|A(s)\| \le M$ on the segment between the two times named; the results about Φ itself need none, since A is bounded on every compact interval.

5.2.1 Uniqueness

5.2.2 Semigroup, invertibility and composition

Uniqueness is what makes these available: two expressions that solve the same matrix initial value problem must agree, and each identity below is an instance of that.

The matrix-level forms below, with the times ordered, predate the general results above and are being retired.

5.2.3 Variation of constants

With a forcing term $B(t)u(t)$, the solution acquires a convolution against Φ. The difficulty is that the elapsed time appears both as the upper limit of integration and inside the integrand; the proof factors the two apart rather than differentiating Φ in its second argument.

5.2.4 The forced response as a trajectory

The solution theory above is stated on the coefficient maps A and B, which is where the analysis lives. The results below restate it for the system object of definition 141: the variation-of-constants formula is a *trajectory* in the sense of definition 148, which the stability theory quantifies over.

Definition 173 (Forced response)**.** The *forced response* of $\dot{x} = A(t)x + B(t)u(t)$ from the state x_0 at time t_0, given by the variation-of-constants formula

$$x(t) = \Phi(t, t_0)\, x_0 + \int_{t_0}^{t} \Phi(t, \tau)\, B(\tau)\, u(\tau)\, \mathrm{d}\tau,$$

with Φ the state transition operator (definition 176).

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Theorem 5.2.

Theorem 174 (The forced response is a trajectory)**.** *Let s be a continuous-time linear system with continuous coefficients A and B, and let u be a continuous input. Then the forced response (definition 173) from any state x_0 at any time t_0 is a trajectory of s (definition 148): it satisfies the state equation at every $t \in \mathbb{R}$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Theorem 5.2.*

Proof. By the semigroup property (theorem 177), $x(t) = \Phi(t, t_0)\, w(t)$ with $w(t) = x_0 + \int_{t_0}^{t} \Phi(t_0, \tau) B(\tau) u(\tau)\, d\tau$. Differentiate the product: $\Phi(\cdot, t_0)$ by **??**, and w by the fundamental theorem of calculus. □

Theorem 175 (Trajectories are forced responses)**.** *Let A, B and u be continuous and let x be a trajectory of s under u on $[t_0, t_1]$ (definition 147). Then x is the forced response (definition 173) from $x(t_0)$ there.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Theorem 5.2.*

Proof. x minus the forced response is a trajectory of the unforced system (theorem 174) starting at 0, so it is $\Phi(t, t_0)\, 0 = 0$ (theorem 180). □

Definition 176 (State transition operator of an LTV system)**.** The state transition matrix $\Phi(t, t_0)$ (**??**) of the state operator A, an operator on the state space.

Theorem 177 (Semigroup property of the state transition operator)**.** *For A continuous, $\Phi(t, \sigma)\, \Phi(\sigma, \tau)\, x = \Phi(t, \tau)\, x$ for all t, σ, τ and every state x.*

Proof. **??** in the algebra of operators. □

Theorem 178 (The homogeneous response is a trajectory)**.** *For a continuous-time linear system with continuous A, the homogeneous response $t \mapsto \Phi(t, t_0)\, x_0$ from any state x_0 at any time t_0 is a trajectory (definition 148) under zero input.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Theorem 5.1.*

Proof. Apply **??** to the fixed state x_0. □

Theorem 179 (The homogeneous response is a trajectory on the forward ray)**.** *The homogeneous response $t \mapsto \Phi(t, t_0)\, x_0$ is a trajectory of the unforced system (definition 147) on $[t_0, \infty)$.*

Proof. theorem 178, restricted to the ray by lemma 111. □

Theorem 180 (Trajectories of the unforced system are homogeneous responses)**.** *Let s have continuous A and let φ be a trajectory of the unforced system on $[t_0, t_1]$ (definition 147). Then $\varphi(t) = \Phi(t, t_0)\,\varphi(t_0)$ for every $t \in [t_0, t_1]$ (definition 176).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Theorem 5.1.*

Proof. $r \mapsto \Phi(t_0, r)\,\varphi(r)$ has derivative $-\Phi(t_0, r)A(r)\varphi(r) + \Phi(t_0, r)A(r)\varphi(r) = 0$ (**??**), so it is constant on $[t_0, t_1]$, equal to $\varphi(t_0)$. Apply $\Phi(t, t_0)$ and use theorem 177. □

The log norm of $A(t)$, for any norm on the state space, bounds the growth of solutions.

Theorem 181 (Linear systems are forward complete)**.** *If A is continuous, the unforced system (definition 86) of definition 142 is forward complete (definition 92).*

Proof. A trajectory on $[t_0, t_1]$ agrees there with the homogeneous response from its initial state (theorem 180), which is a trajectory on $[t_0, \infty)$ (theorem 179). □

Theorem 182 (Linear systems have unique solutions)**.** *If A is continuous, the system of definition 142 has unique solutions (definition 93) under zero input.*

Proof. Both trajectories are the homogeneous response from their common initial state (theorem 180). □

Theorem 183 (The state transition map of a linear system)**.** *If A is continuous, then under zero input the state transition map (definition 99) of the system of definition 142 is*

$$\varphi(t; t_0, x_0) = \Phi(t, t_0)\,x_0, \qquad t \geq t_0,$$

with Φ the state transition operator (definition 176).

Proof. The homogeneous response is a trajectory on $[t_0, t]$ (theorem 179) and solutions are unique (theorem 182), so it is given by the state transition map (lemma 101). □

5.2.5 The time-invariant case: the matrix exponential

For a constant state matrix the Peano–Baker series sums to the matrix exponential, which is Hespanha's (6.1)–(6.2). The point of proving it is economy: every property of Φ established above for a time-varying A then holds for $e^{A(t-t_0)}$ by specialization, with no separate LTI theory. The two exponential facts that need no solution theory at all — the semigroup law (P6.3) and the finite-polynomial form via Cayley–Hamilton (P6.5) — are proved as statements about exp and A alone.

Theorem 184 (The LTI state transition operator is $e^{A(t-t_0)}$)**.** *For an LTI system, $\Phi(t, t_0) = e^{A(t-t_0)}$, acting on Euclidean states.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.2).*

Proof. **??** in the algebra of operators, and exp commutes with the continuous algebra map from matrices to operators. □

Theorem 185 (Uniqueness of the LTI state response)**.** *Let x be a trajectory of the unforced LTI system on $[t_0, t_1]$. Then $x(t) = e^{A(t-t_0)}x(t_0)$ for every t in that interval.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, Property P6.1.*

Proof. theorem 180 at the constant state matrix, rewritten by theorem 184. □

5.2.6 The forced response of a time-invariant system

For a time-invariant system the state transition matrix is the matrix exponential, so the variation-of-constants formula collapses to the closed form every textbook writes.

Lemma 186 (Forced response of a time-invariant system)**.** *For an LTI system the forced response (definition 173) is*

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-\tau)}\,B\,u(\tau)\,\mathrm{d}\tau.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6.*

Proof. theorem 184, applied once to the homogeneous term and once under the integral sign. □

Theorem 187 (The LTI response is a trajectory)**.** *For an LTI system (definition 152) with continuous input u, the closed form*

$$t \mapsto e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-\tau)}\,B\,u(\tau)\,\mathrm{d}\tau$$

is a trajectory (definition 148) of that system.

Proof. theorem 174 at the LTV system of the LTI system (definition 153), whose coefficient maps are constant and so continuous, rewritten by lemma 186. □

5.3 The matrix exponential and complexification

5.3.1 Complexification

Spectral arguments must happen over $\mathbb{C}$, because $\mathbb{R}$ is not algebraically closed, while the systems of interest have real coefficients. Entrywise complexification is the bridge, and these record what it commutes with. The one with content is the transfer of positive definiteness: a real positive-definite matrix remains positive definite against *complex* vectors, which is what lets a Lyapunov certificate be evaluated against an eigenvector.

Definition 188 (Complexification of a real matrix)**.** For a real matrix A, its *complexification* $A_{\mathbb{C}}$ is obtained by applying $\mathbb{R} \hookrightarrow \mathbb{C}$ to each entry.

Lemma 189 (Positive definiteness survives complexification)**.** *If a real matrix P is positive definite, so is its complexification $P_{\mathbb{C}}$, as a Hermitian matrix over $\mathbb{C}$.*

Proof. Write a complex vector as $v = a + ib$ with a, b real. Then $v^* P_{\mathbb{C}} v = a^{\mathsf{T}} P a + b^{\mathsf{T}} P b$: the imaginary part cancels because P is symmetric, and the real part is positive because $v \neq 0$ forces a or b to be nonzero. □

Over a field the spectrum of a matrix is its set of eigenvalues, which is how a condition on eigenpairs meets the spectral radius.

Lemma 190 (The spectrum of a matrix is its set of eigenvalues)**.** *For a square matrix M over a field K, $\mu \in \sigma(M)$ if and only if $Mv = \mu v$ for some $v \neq 0$.*

Proof. □

5.3.2 Spectral facts about the exponential

These say nothing about trajectories, initial times or state transition matrices — they are facts about exp and A alone, and are proved as such. The complexification bridge is what lets spectral arguments, which must happen over $\mathbb{C}$, yield bounds on a real exponential; P6.5 collapses the exponential series to a finite polynomial in A. The eigenpair lemma is the half of spectral mapping that is pure power-series algebra, and is what turns an eigenvalue condition on A into a statement about the growth of e^{At}.

Lemma 191 (Complexification commutes with the matrix exponential)**.** *Entrywise complexification commutes with the matrix exponential: for a real matrix A,*

$$\overline{\exp(A)} = \exp(\overline{A}),$$

where $\overline{\ }$ denotes applying $\mathbb{R} \hookrightarrow \mathbb{C}$ entrywise. This is the bridge used to transfer spectral facts, which live over $\mathbb{C}$, to norm bounds on a real exponential.

Proof. Entrywise complexification is a continuous ring homomorphism, and exp commutes with any such. □

Lemma 192 (The exponential acts on an eigenvector by exponentiating its eigenvalue)**.** *Let M be a square matrix and let (μ, v) be an eigenpair of M, so that $Mv = \mu v$. Then*

$$e^M v = e^\mu\, v.$$

Reference: standard power-series functional calculus for the exponential.

Proof. Induction gives $M^k v = \mu^k v$ for every k. The map $N \mapsto Nv$ is linear and continuous, so it commutes with the sum defining $e^M = \sum_k (k!)^{-1} M^k$, giving $e^M v = \sum_k (k!)^{-1} \mu^k v = e^\mu v$. □

Lemma 193 (A scalar multiple scales the eigenvalues)**.** *If $Av = \mu v$ then $(cA)v = (c\mu)v$.*

Proof. □

Lemma 194 (Powers power the eigenvalues)**.** *If $Av = \mu v$ then $A^k v = \mu^k v$ for every k.*

Proof. □

Lemma 195 (A scalar shift shifts the eigenvalues)**.** *For a square matrix A over a commutative ring and scalars α, μ, a vector v satisfies $(A + \alpha I)v = (\mu + \alpha)v$ if and only if $Av = \mu v$.*

Proof. □

Lemma 196 (The kernel form of an eigenpair)**.** $(\mu I - A)v = 0$ *if and only if* $Av = \mu v$.

Proof. □

Lemma 197 (Negative spectrum gives a contractive exponential)**.** *If every eigenpair (μ, v) of the complexification of a real matrix A, with $v \neq 0$, has $\mathrm{Re}(\mu) < 0$, then $r(e^{A_{\mathbb{C}}}) < 1$.*

Reference: standard spectral mapping for the matrix exponential.

Proof. Reverse spectral mapping: every spectral value of $e^{A_{\mathbb{C}}}$ is e^μ for an eigenvalue μ of $A_{\mathbb{C}}$, and $|e^\mu| = e^{\mathrm{Re}\,\mu} < 1$. □

Theorem 198 (Semigroup property of the matrix exponential)**.** *Hespanha, P6.3. For every* $t, \tau \in \mathbb{R}$,

$$e^{At}e^{A\tau} = e^{A(t+\tau)}.$$

Proof. tA and τA commute, being scalar multiples of the same matrix, so the exponential of the sum factors. The general $e^X e^Y = e^{X+Y}$ needs commutativity and is false without it; here it is free. □

Theorem 199 (The matrix exponential as a finite polynomial)**.** *Hespanha, P6.5. There are scalar functions* $\alpha_0, \dots, \alpha_{n-1} : \mathbb{R} \to \mathbb{R}$ *with*

$$e^{At} = \sum_{i=0}^{n-1} \alpha_i(t)\, A^i \qquad \textit{for every } t.$$

The matrix exponential — an infinite series in A *— is a* finite *polynomial in* A *of degree less than* n*, at the cost of coefficients that are transcendental in* t*.*

Proof. Cayley–Hamilton reduces every power A^k to a combination of $A^0, \dots, A^{n-1}$, say with coefficients $\bar{a}_i(k)$; substituting into the exponential series and exchanging the finite sum over i with the infinite sum over k exhibits $\alpha_i(t) = \sum_k t^k \bar{a}_i(k)/k!$ as a witness.

The exchange needs those series to converge, which holds because the reduced coefficients grow at most geometrically, $|\bar{a}_i(k)| \le CM^k$, dominating each α_i by a scalar exponential series. That growth bound is the crux of the proof and is established by iteration, not by spectral theory: "multiply by X, reduce modulo the characteristic polynomial" is a fixed linear operation on the n-dimensional space of polynomials of degree $< n$, so one step scales the ℓ^1 size of the coefficient vector by at most a constant factor. No eigenvalues and no Jordan form appear anywhere. □

5.4 The discrete-time state transition matrix

For $x(t+1) = A(t)x(t)$ the state transition matrix is an ordered product rather than a series, and the whole theory becomes finite: every proof below is an induction, with no convergence, no differentiation and no Gronwall estimate. Two differences from the continuous case are substantive rather than merely technical. There is no standing bound on A to carry, since a finite product always converges; and Φ need not be invertible, so the semigroup property genuinely requires $\tau \le s \le t$ instead of holding in any order.

Definition 200 (Discrete state transition matrix)**.** The discrete-time *state transition matrix* $\Phi(t, t_0)$:

$$\Phi(t, t_0) := \begin{cases} A(t-1)A(t-2)\cdots A(t_0+1)A(t_0) & t > t_0, \\ I & t \le t_0. \end{cases}$$

Formally, the ordered product of A over $[t_0, t)$, which for $t \le t_0$ is the empty product I. The system runs forward only, so the $t < t_0$ branch carries no meaning; it is there to make the definition total.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, equation (5.11).

Lemma 201 (The discrete state transition matrix at the initial time)**.** $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Property P5.5.*

Proof. The defining product is over the empty range $[t_0, t_0)$. □

Lemma 202 (One-step recursion)**.** *For $t \geq t_0$, $\Phi(t+1, t_0) = A(t)\,\Phi(t, t_0)$. With lemma 201 this is the existence half of Hespanha's P5.5: the discrete state transition matrix solves the matrix recursion.*

Proof. Peel the leading factor $A(t)$ off the defining product. □

Theorem 203 (Discrete state transition matrix: uniqueness)**.** *Hespanha, P5.5. If $Z(t_0) = I$ and $Z(t+1) = A(t)Z(t)$ for all $t \geq t_0$, then $Z(t) = \Phi(t, t_0)$ for all $t \geq t_0$.*

Proof. Induction on t from t_0: the base case is lemma 201 and the step chains the recursion against lemma 202. No analytic content at all, unlike the continuous-time counterpart — the recursion pins each value outright. □

Theorem 204 (Uniqueness of the discrete state response)**.** *If $z(t_0) = x_0$ and $z(t+1) = A(t)z(t)$ for all $t \geq t_0$, then $z(t) = \Phi(t, t_0)x_0$ for all $t \geq t_0$: the vector form of theorem 203.*

Proof. □

Theorem 205 (Columns of the discrete state transition matrix)**.** *Hespanha, P5.6. For each i, the i-th column of $\Phi(t, t_0)$ is the unique solution of $z(t+1) = A(t)z(t)$ with $z(t_0) = e_i$, for $t \geq t_0$.*

Proof. theorem 204 at $x_0 = e_i$, using $\Phi(t, t_0)e_i = (\Phi(t, t_0))_{\cdot,i}$. □

Theorem 206 (Semigroup property, discrete)**.** *Hespanha, P5.7. For $\tau \leq s \leq t$,*

$$\Phi(t, s)\,\Phi(s, \tau) = \Phi(t, \tau).$$

Unlike the continuous-time case there is no inverse: Φ is a product of matrices that need not be nonsingular, so the ordering $\tau \leq s \leq t$ is a genuine restriction here rather than an artifact.

Proof. Induction on t from s, chaining lemma 202 at both s and τ. □

5.4.1 Discrete variation of constants

Lemma 207 (Discrete variation of constants at the initial time)**.** *The discrete variation-of-constants formula*

$$x(t) := \Phi(t, t_0)x_0 + \sum_{\tau=t_0}^{t-1} \Phi(t, \tau+1)B(\tau)u(\tau)$$

takes the value x_0 at $t = t_0$: the forcing sum is over the empty range and $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Section 5.3 (discrete variation of constants).*

Proof. □

Theorem 208 (Discrete variation of constants: existence)**.** *For $t \geq t_0$, the formula of lemma 207 satisfies the forced recursion*

$$x(t+1) = A(t)\,x(t) + B(t)\,u(t).$$

Proof. Peel the top term off the sum defining $x(t+1)$ and fold $A(t)\Phi(t,\tau+1)$ into $\Phi(t+1,\tau+1)$ termwise by lemma 202, applied once more to the leading $\Phi(t,t_0)$ term. The peeled term is exactly $B(t)u(t)$, since $\Phi(t+1,t+1)=I$. □

Theorem 209 (Discrete variation of constants: uniqueness)**.** *Any two solutions of* $z(t+1)=A(t)z(t)+B(t)u(t)$ *with the same value at* t_0 *agree for all* $t\geq t_0$*.*

Proof. Induction on t from t_0. Unlike theorem 175, no Lipschitz condition or Gronwall estimate is needed: the recursion determines each successive value outright, so uniqueness is immediate. □

5.4.2 The discrete responses as trajectories

The variation-of-constants formula and its unforced part, named, and read on the system object (definition 157): they are exactly the trajectories of the system from a given initial state, which is the form in which the solution theory reaches the stability theory.

Definition 210 (Forced response, discrete time)**.** The *forced response* of $x(t+1)=A(t)x(t)+B(t)u(t)$ from the state x_0 at time t_0,

$$x(t)=\Phi(t,t_0)\,x_0+\sum_{\tau=t_0}^{t-1}\Phi(t,\tau+1)\,B(\tau)\,u(\tau),$$

with Φ the discrete state transition matrix (definition 200). It solves the recursion forward from t_0 only.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Section 5.3.

Definition 211 (Homogeneous response, discrete time)**.** The *homogeneous response* of $x(t+1)=A(t)x(t)$ from the state x_0 at time t_0, $x(t)=\Phi(t,t_0)\,x_0$: the forced response (definition 210) with the input switched off.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Section 5.3.

Lemma 212 (The discrete forced response at the initial time)**.** *The forced response (definition 210) takes the value* x_0 *at* $t=t_0$*.*

Proof. lemma 207. □

Lemma 213 (The discrete forced response solves the recursion)**.** *For* $t\geq t_0$ *the forced response (definition 210) satisfies* $x(t+1)=A(t)x(t)+B(t)u(t)$*.*

Proof. theorem 208. □

Lemma 214 (The discrete homogeneous response at the initial time)**.** *The homogeneous response (definition 211) takes the value* x_0 *at* $t=t_0$*.*

Proof. $\Phi(t_0,t_0)=I$ (lemma 201). □

Lemma 215 (The discrete homogeneous response solves the unforced recursion)**.** *For* $t\geq t_0$ *the homogeneous response (definition 211) satisfies* $x(t+1)=A(t)x(t)$*.*

Proof. $\Phi(t+1,t_0)=A(t)\Phi(t,t_0)$ (lemma 202). □

Theorem 216 (The discrete state transition matrix of an LTI system)**.** *For a constant state matrix A, the discrete state transition matrix (definition 200) is $\Phi(t, t_0) = A^{t-t_0}$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Section 5.3.*

Proof. $\Phi(t, t_0)$ is the ordered product of $t - t_0$ factors, all equal to A. □

Theorem 217 (The discrete forced response is a trajectory)**.** *For a discrete-time LTV system s and input u, the forced response (definition 210) from x_0 at t_0 is a trajectory of s under u on $[t_0, \infty)$ (definition 163).*

Proof. lemma 213, which is the trajectory condition on the nose. □

Theorem 218 (Discrete trajectories are forced responses)**.** *If x is a trajectory of s under u on $[t_0, \infty)$ (definition 163), then $x(t)$ is the forced response (definition 210) from $x(t_0)$ at t_0, for every $t \geq t_0$.*

Proof. Both satisfy the forced recursion from the same initial value, so they agree by theorem 209. □

Theorem 219 (The discrete homogeneous response is a trajectory)**.** *The homogeneous response (definition 211) from x_0 at t_0 is a trajectory of the unforced system on $[t_0, \infty)$ (definition 163).*

Proof. lemma 215, after reducing the unforced state equation by lemma 160. □

Theorem 220 (Discrete unforced trajectories are homogeneous responses)**.** *If x is a trajectory of the unforced system on $[t_0, \infty)$ (definition 163), then $x(t) = \Phi(t, t_0)\, x(t_0)$ for every $t \geq t_0$.*

Proof. theorem 204, after reducing the unforced state equation by lemma 160. □

Chapter 6

Comparison functions

Comparison functions are the vocabulary of stability estimates: a class $\mathcal{K}$ function bounds how far a trajectory can go, a class $\mathcal{L}$ function how fast it decays, and a class $\mathcal{KL}$ function both at once. This chapter builds them, and the tools that turn a cruder bound into one of them.

6.1 Class $\mathcal{K}$ and $\mathcal{K}_\infty$ functions

Definition 221 (Class $\mathcal{K}$ function)**.** A *class* $\mathcal{K}$ function on $[0, a)$ is a continuous strictly increasing map $\alpha : [0, a) \to [0, b)$ with $\alpha(0) = 0$. All class K functions have an inverse $\alpha^{-1} : [0, b) \to [0, a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}$). Differences:

- Khalil writes the codomain as $[0, \infty)$; here the bound b is tracked in the type.
- The inverse is bundled as data rather than derived.

Definition 222 (Class $\mathcal{K}_\infty$ function)**.** A *class* $\mathcal{K}_\infty$ function is a continuous strictly increasing map $\alpha : [0, \infty) \to [0, \infty)$ with $\alpha(0) = 0$ and $\alpha(r) \to \infty$ as $r \to \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}_\infty$). The inverse is bundled as data rather than derived.

6.1.1 Building class $\mathcal{K}$ functions

A recognition criterion and the closure properties. The restatements of the definition itself (strict monotonicity as an iff, the two inverse equations, and so on) are left to the Lean API; they say nothing the definition above does not.

Definition 223 (Class $\mathcal{K}$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$ and $f(a) = b$ and be continuous and strictly increasing on $[0, a]$, with $a, b > 0$. Then f is a class $\mathcal{K}$.

Definition 224 (Restriction of a class $\mathcal{K}$ function)**.** If α is class $\mathcal{K}$ on $[0, a) \to [0, b)$ and $0 < c < a$, then α restricted to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, \alpha(c))$.

Definition 225 (Restriction to a prescribed codomain)**.** definition 224 with the codomain bound named: if $\alpha(c) = e$, the restriction of α to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, e)$.

Lemma 226 (Applying the inverse of a class $\mathcal{K}$ function)**.** *If α is class $\mathcal{K}$ on $[0,a) \to [0,b)$, $x \in [0,a)$ and $y \in [0,b)$, then $x \le \alpha^{-1}(y)$ if and only if $\alpha(x) \le y$.*

Proof. □

Definition 227 (The composite $\alpha_3 \circ \alpha_2^{-1}$)**.** If α_2, α_3 are class $\mathcal{K}$ on $[0,r)$ and $0 < c < r$, then $\alpha_3 \circ \alpha_2^{-1}$ is class $\mathcal{K}$ on $[0, \alpha_2(c)) \to [0, \alpha_3(c))$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), proofs of Theorems 4.9 and 4.18.

Definition 228 (Inverse of a class $\mathcal{K}$ function)**.** Class $\mathcal{K}$ is closed under inversion. The inverse of a class K function is also class K.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, first bullet. Khalil's domain $[0, \alpha(a))$ is the codomain bound b carried by the type.

Definition 229 (Composition of class $\mathcal{K}$ functions)**.** Class $\mathcal{K}$ is closed under composition: if α is class $\mathcal{K}$ on $[0,a) \to [0,b)$ and β is class $\mathcal{K}$ on $[0,b) \to [0,c)$, then $\beta \circ \alpha$ is class $\mathcal{K}$ on $[0,a) \to [0,c)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, third bullet.

6.1.2 Building class $\mathcal{K}_\infty$ functions

The same recognition criterion and closure properties on the unbounded domain, together with the closure properties that need the global domain (sum, positive scaling, minimum), the power functions as the standard examples, and the restriction back to class $\mathcal{K}$.

Definition 230 (Class $\mathcal{K}_\infty$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$, be continuous and strictly increasing on $[0, \infty)$, and have $f(r) \to \infty$ as $r \to \infty$. Then f is a class $\mathcal{K}_\infty$ function, its inverse being the inverse of f on $[0, \infty)$.

Definition 231 (Inverse of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under inversion: if α is class $\mathcal{K}_\infty$, so is α^{-1}.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, second bullet.

Definition 232 (Composition of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under composition: if α and β are class $\mathcal{K}_\infty$, so is $\beta \circ \alpha$, with inverse $\alpha^{-1} \circ \beta^{-1}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fourth bullet.

Definition 233 (Sum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise addition: if α and β are class $\mathcal{K}_\infty$, so is $\alpha + \beta$.

Definition 234 (Class $\mathcal{K}_\infty$ plus a nondecreasing function)**.** If α is class $\mathcal{K}_\infty$ and f is continuous and nondecreasing on $[0, \infty)$ with $f(0) = 0$, then $\alpha + f$ is class $\mathcal{K}_\infty$.

Lemma 235 (Weak triangle inequality)**.** *If f is nondecreasing and nonnegative on $[0, \infty)$, then $f(a+b) \le f(2a) + f(2b)$ for all $a, b \ge 0$.*

Reference: Sontag, Input to state stability: basic concepts and results *(2008).*

Proof. $a + b \le 2\max(a,b)$, so $f(a+b)$ is at most whichever of $f(2a)$, $f(2b)$ is larger, and the other is nonnegative. □

Definition 236 (Positive multiple of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under multiplication by a positive constant: if α is class $\mathcal{K}_\infty$ and $c > 0$, so is $c\alpha$.

Definition 237 (Minimum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise minimum: if α and β are class $\mathcal{K}_\infty$, so is $\min(\alpha, \beta)$.

Definition 238 (Powers of a class $\mathcal{K}_\infty$ function)**.** For every $p > 0$, the map $r \mapsto r^p$ is class $\mathcal{K}_\infty$.

Definition 239 (A class $\mathcal{K}_\infty$ function restricts to class $\mathcal{K}$)**.** Every class $\mathcal{K}_\infty$ function α restricts, for each $a > 0$, to a class $\mathcal{K}$ function on $[0, a) \to [0, \alpha(a))$.

Theorem 240 (The integral of a class $\mathcal{K}_\infty$ function)**.** *For a class $\mathcal{K}_\infty$ function α (definition 222), $r \mapsto \int_0^r \alpha(s)\, ds$ is class $\mathcal{K}_\infty$, with derivative α on $[0, \infty)$.*

Proof. Its derivative α is positive on $(0, \infty)$, so it is strictly increasing; since α is increasing, $\int_0^r \alpha \geq \int_0^1 \alpha + (r-1)\alpha(1) \to \infty$. □

Lemma 241 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** *Let $\omega : \mathbb{R} \to \mathbb{R}$ satisfy $\omega(0) = 0$, be monotone on $[0, \infty)$, and be continuous at 0 from the right. Then there is a class $\mathcal{K}_\infty$ function α with $\omega(r) \leq \alpha(r)$ for all $r \geq 0$.*

Proof. lemma 274 supplies a continuous, strictly increasing f with $f(0) = 0$, $f(r) \to \infty$ and $\omega \leq f$ on $[0, \infty)$; by definition 230 that f is class $\mathcal{K}_\infty$. □

6.2 Inverses of antitone functions

A function W that is continuous and strictly antitone on $(0, \infty)$, blows up at 0^+ and vanishes at $+\infty$ surjects onto $(0, \infty)$, and its right inverse has the same shape. This is the mechanism behind the class $\mathcal{KL}$ construction of the next section: W is a mean-value bound and its inverse supplies the decay rate.

Lemma 242 (Surjectivity onto $(0, \infty)$ from the two boundary limits)**.** *Let $W : \mathbb{R} \to \mathbb{R}$ be continuous on $(0, \infty)$ with $W(\eta) \to +\infty$ as $\eta \to 0^+$ and $W(\eta) \to 0$ as $\eta \to +\infty$. Then for every $s > 0$ there exists $c > 0$ with $W(c) = s$.*

Proof. □

Lemma 243 (The right inverse inverts)**.** *Under the surjectivity conditions of lemma 242, $W\big(\mathrm{invFunOn}\, W\, (0, \infty)\, s\big) = s$ for all $s > 0$.*

Proof. □

Lemma 244 (Positivity of the right inverse)**.** *Under the surjectivity conditions of lemma 242,* $\mathrm{invFunOn}\, W\, (0, \infty)\, s > 0$ *for all $s > 0$.*

Proof. □

Lemma 245 (The inverse of a strictly antitone function is strictly antitone)**.** *If W is strictly antitone on $(0, \infty)$, then* $\mathrm{invFunOn}\, W\, (0, \infty)$ *is strictly antitone on $(0, \infty)$.*

Proof. □

Lemma 246 (The inverse vanishes at $+\infty$)**.** *Under the conditions of lemma 242 and with W strictly antitone,* $\mathrm{invFunOn}\, W\, (0, \infty)\, s \to 0$ *as $s \to +\infty$.*

Proof. □

Lemma 247 (Continuity of the inverse)**.** *Under the conditions of lemma 242 and with W strictly antitone,* invFunOn $W\,(0,\infty)$ *is continuous on* $(0,\infty)$*.*

Proof. □

Lemma 248 (The inverse blows up at 0^+)**.** *Under the conditions of lemma 242 and with W strictly antitone,* invFunOn $W\,(0,\infty)\, s \to +\infty$ *as* $s \to 0^+$*.*

Proof. □

6.3 Class $\mathcal{KL}$ functions and the Osgood construction

Definition 249 (Class $\mathcal{L}$ function)**.** A *class* $\mathcal{L}$ function on $[0,\infty)$ is continuous, strictly positive, and strictly decreasing, and satisfies $\sigma(s) \to 0$ as $s \to \infty$. It carries the time decay of an asymptotic stability estimate: the product of a class $\mathcal{K}$ function of the initial deviation with a class $\mathcal{L}$ function of elapsed time is a class $\mathcal{KL}$ bound.

Definition 250 (Singular class $\mathcal{L}$ function)**.** A *singular class* $\mathcal{L}$ function on $(0,\infty)$ is continuous, strictly positive and antitone, with $U(s) \to 0$ as $s \to \infty$ and $U(s) \to \infty$ as $s \to 0^+$. It differs from class $\mathcal{L}$ in two ways: the domain excludes 0, where the function is instead required to blow up, and the decrease is only required to be weak. It arises as the inverse of the sliding-window function in the class $\mathcal{KL}$ characterization of asymptotic stability.

Definition 251 (Class $\mathcal{KL}$ function)**.** A *class* $\mathcal{KL}$ function on $[0,a)\times[0,\infty)$ is continuous, class $\mathcal{K}$ in the first argument, and for each fixed $r>0$ is non-increasing and tends to 0 as $s\to\infty$. It arises as the bound $\|x(t)\| \le \beta(\|x_0\|,t)$ in asymptotic stability estimates.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3. Difference: Khalil says $\beta(\cdot,s)$ is class $\mathcal{K}$; here that is spelled out as separate fields, so no class $\mathcal{K}$ instance is required for each s. Khalil also asks $\beta(r,\cdot)$ to be decreasing; non-increasing loses nothing, since $\beta(r,s)+\alpha(r)e^{-s}$, with α class $\mathcal{K}$, is a strictly decreasing majorant.

Definition 252 (Global class $\mathcal{KL}$ function)**.** A *global class* $\mathcal{KL}$ function is a map $\beta : [0,\infty)\times[0,\infty)\to\mathbb{R}$, jointly continuous, that is class $\mathcal{K}$ in the first argument (strictly increasing, zero at zero) and, for each fixed $r\ge 0$, is antitone and tends to 0 as $s\to\infty$. As in Khalil, $\beta(\cdot,s)$ need not be unbounded.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3 with $a=\infty$.

Lemma 253 (Radially unbounded majorant of a class $\mathcal{KL}$ function)**.** *For every global class $\mathcal{KL}$ function β (definition 252) there is a global class $\mathcal{KL}$ function β' with $\beta(r,s)\le\beta'(r,s)$ for $r,s\ge 0$ and $\beta'(r,0)\to\infty$ as $r\to\infty$.*

Proof. $\beta'(r,s)=\beta(r,s)+re^{-s}$. □

6.3.1 Building class $\mathcal{KL}$ functions

Two constructions and the two composition rules, in the bounded and the global setting. The product $\alpha(r)\gamma(s)$ of a class $\mathcal{K}$ and a class $\mathcal{L}$ function is the easy one; the Sontag-style cap is the one that does real work.

Definition 254 (Class $\mathcal{KL}$ from a class $\mathcal{K}$ times a class $\mathcal{L}$)**.** If α is class $\mathcal{K}$ on $[0, a)$ and γ is class $\mathcal{L}$, then $\beta(r, s) := \alpha(r)\gamma(s)$ is class $\mathcal{KL}$ on $[0, a)$.

The cap is built in the form $\min(\alpha(r), \sqrt{\alpha(r)\,U(s)})$, so the decay of that particular expression is recorded first.

Lemma 255 (Decay of a geometric-mean cap)**.** *Let $c \geq 0$ and let $U : \mathbb{R} \to \mathbb{R}$ satisfy $U(s) \to 0$ as $s \to +\infty$. Then $\min(c, \sqrt{c\,U(s)}) \to 0$ as $s \to +\infty$.*

Proof. □

Theorem 256 (Class $\mathcal{KL}$ from a singular class $\mathcal{L}$ cap)**.** *Let α be class $\mathcal{K}$ on $[0, a)$ and let U be a singular class $\mathcal{L}$ function (positive, antitone, $U(s) \to 0$ as $s \to \infty$ and $U(s) \to \infty$ as $s \to 0^+$). Then*

$$\beta(r, s) := \begin{cases} \alpha(r) & s = 0, \\ \min(\alpha(r), \sqrt{\alpha(r)U(s)}) & s > 0, \end{cases}$$

is class $\mathcal{KL}$ on $[0, a)$.

Proof. Every field but continuity is immediate from the corresponding property of α and U, the decay in s being lemma 255. Continuity is checked in two regimes. For $s > 0$ the min branch is a composition of continuous maps. At $(r, 0)$, pick a buffer radius $r_1 \in (r, a)$; since $U(s) \to \infty$ as $s \to 0^+$, on a neighbourhood we have $\alpha(r_1) < U(s)$, hence $\alpha(r') \leq U(s)$ and so $\alpha(r') \leq \sqrt{\alpha(r')U(s)}$ — the minimum is locally just α, which is continuous. □

Definition 257 (Post-composition with a class $\mathcal{K}$ function)**.** Let β be class $\mathcal{KL}$ on $[0, a)$ and α be class $\mathcal{K}$ on $[0, b)$, and suppose $\beta(r, s) < b$ for every $r \in [0, a)$ and $s \geq 0$. Then $(r, s) \mapsto \alpha(\beta(r, s))$ is class $\mathcal{KL}$ on $[0, a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the outer α_1). The range hypothesis is additional: α_1 is defined only on $[0, b)$, so β's values must land there, which Khalil leaves implicit.

Definition 258 (Post-composition with a class $\mathcal{K}_\infty$ function)**.** If β is class $\mathcal{KL}$ on $[0, a)$ and α is class $\mathcal{K}_\infty$, then $(r, s) \mapsto \alpha(\beta(r, s))$ is class $\mathcal{KL}$ on $[0, a)$. This is definition 257 with the range hypothesis discharged by α being defined on all of $[0, \infty)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet, with α_1 of class $\mathcal{K}_\infty$; no range hypothesis is then needed.

Definition 259 (Pre-composition in the radius argument)**.** If β is class $\mathcal{KL}$ on $[0, b)$ and α is class $\mathcal{K}$ on $[0, a) \to [0, b)$, then $(r, s) \mapsto \beta(\alpha(r), s)$ is class $\mathcal{KL}$ on $[0, a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the inner α_2).

Definition 260 (Restriction of a class $\mathcal{KL}$ function)**.** If β is class $\mathcal{KL}$ on $[0, a) \times [0, \infty)$ and $0 < a' \leq a$, then β is class $\mathcal{KL}$ on $[0, a') \times [0, \infty)$.

Definition 261 (Global class $\mathcal{KL}$ from a product)**.** If α is class $\mathcal{K}_\infty$ and γ is class $\mathcal{L}$, then $\beta(r,s) := \alpha(r)\gamma(s)$ is global class $\mathcal{KL}$.

Definition 262 (Post-composition, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \alpha(\beta(r,s))$ is global class $\mathcal{KL}$. No range hypothesis is needed, since α is defined on all of $[0,\infty)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the outer α_1), in the global case $a = \infty$.

Definition 263 (Pre-composition in the radius argument, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \beta(\alpha(r),s)$ is global class $\mathcal{KL}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the inner α_2), in the global case $a = \infty$.

Theorem 264 (Composition with a nonincreasing radius)**.** *Let β be global class $\mathcal{KL}$ and let $a : [0,\infty)^2 \to \mathbb{R}$ be continuous, with $a(0,s) = 0$, strictly increasing in r and nonincreasing in s. Then $(r,s) \mapsto \beta(a(r,s),s)$ is global class $\mathcal{KL}$.*

Proof. Each property of β transfers, using $a \geq 0$; for the limit, $0 \leq \beta(a(r,s),s) \leq \beta(a(r,0),s) \to 0$. □

Definition 265 (Sum of global class $\mathcal{KL}$ functions)**.** If β and β' are global class $\mathcal{KL}$, so is $\beta + \beta'$.

Definition 266 (Positive multiple of a global class $\mathcal{KL}$ function)**.** If β is global class $\mathcal{KL}$ and $c > 0$, then $c\beta$ is global class $\mathcal{KL}$.

Definition 267 (Rescaling time in a global class $\mathcal{KL}$ function)**.** If β is global class $\mathcal{KL}$ and $c > 0$, then $(r,s) \mapsto \beta(r, s/c)$ is global class $\mathcal{KL}$.

Theorem 268 (Capping a class $\mathcal{KL}$ function by a class $\mathcal{K}$ function)**.** *If β is class $\mathcal{KL}$ on $[0,a)$ and α is class $\mathcal{K}$ on $[0,a)$, then $(r,s) \mapsto \min(\beta(r,s),\alpha(r))$ is class $\mathcal{KL}$ on $[0,a)$.*

Capping a class $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class. Its use is to recover a range condition: a bound obtained only as a majorant — from a smoothing construction, say — carries no upper control, and capping restores it without disturbing the bound, since the minimum still dominates whatever β dominated.

Proof. Each field is inherited. The minimum of two strictly increasing functions of r is strictly increasing; α does not depend on s, so antitonicity in s survives the minimum; and $\min(\beta(r,s),\alpha(r)) \to \min(0,\alpha(r)) = 0$ as $s \to \infty$ because $\alpha \geq 0$. □

Definition 269 (Capping a global class $\mathcal{KL}$ function)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \min(\beta(r,s),\alpha(r))$ is global class $\mathcal{KL}$ — capping a $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class.

6.3.2 Smoothing a bound into a class $\mathcal{KL}$ function

A candidate bound that has the right qualitative behaviour — monotone in the initial deviation, decaying in time — need not itself be class $\mathcal{KL}$: it may fail continuity or strict monotonicity. These two results replace such a candidate by a genuine class $\mathcal{KL}$ function dominating it, which is how the stability characterizations produce their bounds. The first lemma below is the continuity criterion the construction rests on.

Lemma 270 (Joint continuity from separate continuity and monotonicity)**.** *Let $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be nondecreasing in its first argument, nonincreasing in its second, and continuous in each argument separately. Then f is jointly continuous.*

Proof. Near (r_0, s_0), monotonicity gives $f(r_-, s_+) \le f(r, s) \le f(r_+, s_-)$ whenever $r_- < r < r_+$ and $s_- < s < s_+$. Given $u > f(r_0, s_0)$, continuity of $f(r_0, \cdot)$ gives $s_- < s_0$ with $f(r_0, s_-) < u$, and then continuity of $f(\cdot, s_-)$ gives $r_+ > r_0$ with $f(r_+, s_-) < u$; so $f < u$ on the neighbourhood $(-\infty, r_+) \times (s_-, \infty)$. The lower bound is symmetric. □

Lemma 271 (Class $\mathcal{KL}$ majorant)**.** *Let $\psi : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ satisfy, for all $r, s \ge 0$:*

1. *$\psi(r, s) \ge 0$;*
2. *$\psi(0, s) = 0$;*
3. *$r \mapsto \psi(r, s)$ is monotone on $[0, \infty)$;*
4. *$s \mapsto \psi(r, s)$ is antitone on $[0, \infty)$;*
5. *$\psi(r, s) \to 0$ as $s \to \infty$;*
6. *$r \mapsto \psi(r, 0)$ is continuous at 0 within $[0, \infty)$.*

Then there is a global class $\mathcal{KL}$ function β with $\psi(r, s) \le \beta(r, s)$ for all $r, s \ge 0$.

Proof. Clamp both arguments of ψ at 0 and average it over a window in each variable: first $\psi_1(r, s) = \int_0^1 \psi(r, s - v)\, dv$, over $[s-1, s]$, then $\psi_2(r, s) = \int_1^2 \psi_1(ru, s)\, du$, over $[r, 2r]$. Each window sits on the side that makes the average dominate, so $\psi \le \psi_1 \le \psi_2$, and ψ_2 is still nondecreasing in r and nonincreasing in s.

- $\psi_1(r, \cdot)$ is a difference of primitives of $\psi(r, \cdot)$, so continuous; $\psi_2(r, \cdot)$ is then continuous by dominated convergence, the integrands being bounded by $\psi(2r, 0)$.
- For $r > 0$, $\psi_2(r, s) = r^{-1} \int_r^{2r} \psi_1(\cdot, s)$ is a quotient of primitives, so continuous; at $r \le 0$, $0 \le \psi_2(r, s) \le \psi(2 \max(r, 0), 0) \to \psi(0, 0) = 0$ by continuity of $\psi(\cdot, 0)$ at 0.

By lemma 270, ψ_2 is jointly continuous. Also $\psi_2(r, s) \le \psi(2r, s - 1) \to 0$. Finally $\beta(r, s) = \psi_2(r, s) + re^{-s}$ is class $\mathcal{KL}$, the second term making it strictly increasing in r. □

Lemma 272 (A class $\mathcal{KL}$ function above a size bound and a decay bound)**.** *Let α be class $\mathcal{K}_\infty$ and let U be a family of functions such that*

1. *$U(r, s) > 0$ for all $r > 0$ and $s > 0$;*
2. *$U(r, \cdot)$ is antitone on $(0, \infty)$ for each $r > 0$;*
3. *$U(r, s) \to 0$ as $s \to \infty$, for each $r > 0$;*
4. *$r \mapsto U(r + 1, s)$ is monotone on $[0, \infty)$ for each $s > 0$.*

Then there is a global class $\mathcal{KL}$ function β with $\alpha(r) \le \beta(r, 0)$ and $\min\{\alpha(r), U(r+1, s)\} \le \beta(r, s)$ for all $r \ge 0$ and $s > 0$.

Proof. Let $\psi(r,0) = \alpha(r)$ and $\psi(r,s) = \min\{\alpha(r), U(r+1,s)\}$ for $s > 0$; the radius is shifted by 1 so that U is only evaluated at a positive radius. Each hypothesis of lemma 271 holds for ψ: monotonicity in r from that of α and of $r \mapsto U(r+1,s)$, antitonicity in s from that of $U(r+1,\cdot)$, decay since $0 \le \psi \le U$, and continuity at $r = 0$ because $\psi(\cdot,0) = \alpha$. The resulting β dominates ψ. □

6.4 Class $\mathcal{K}$ bounds for positive definite functions

A Lyapunov function is positive definite: a pointwise condition, awkward to combine with estimates. The theorem below converts it into monotone bounds in the single variable $\|x\|$, after which every Lyapunov argument can be run in the comparison-function calculus. It rests on two smoothing results, which replace a monotone bound by a continuous strictly increasing one.

Lemma 273 (Class $\mathcal{K}$ minorant of a positive monotone function)**.** *Let $r > 0$ and let $\psi : \mathbb{R} \to \mathbb{R}$ satisfy $\psi(0) = 0$, $\psi(s) > 0$ for $s \in (0, r]$, and ψ nondecreasing on $[0, r]$. Then there are $b > 0$ and a function f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $f(s) \le \psi(s)$ for all $s \in [0, r]$.*

That is: a positive nondecreasing function admits a class $\mathcal{K}$ minorant on any bounded interval. This is what turns a positive-definite V into the lower comparison bound of theorem 276.

Proof. Take $f(s) = r^{-1} \int_0^s \psi$ (with the argument of ψ clamped to $[0, r]$, so that it is monotone, hence integrable, on all of $\mathbb{R}$). It is continuous as a primitive, and $f(0) = 0$. For $s < s'$ in $[0, r]$, $f(s') - f(s) = r^{-1} \int_s^{s'} \psi > 0$ since $\psi > 0$ on (s, s'), so f is strictly increasing and $b = f(r) > 0$. Finally, ψ is nondecreasing, so $\int_0^s \psi \le s\,\psi(s) \le r\,\psi(s)$, i.e. $f(s) \le \psi(s)$. □

Lemma 274 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ satisfy $\varphi(0) = 0$, be monotone on $[0, \infty)$, and be continuous at 0 from the right. Then there is an f with $f(0) = 0$, continuous and strictly increasing on $[0, \infty)$, $f(s) \to \infty$ as $s \to \infty$, and $\varphi(s) \le f(s)$ for all $s \ge 0$.*

That is: a nondecreasing function admits a class $\mathcal{K}_\infty$ majorant. The unbounded domain is what distinguishes this from lemma 275, which it implies.

Continuity at 0 is necessary: the unit step has no continuous majorant vanishing at 0.

Proof. Take $f(s) = s + g(s)$ with $g(s) = \int_1^2 \varphi(su)\,du$, the average of φ over $[s, 2s]$ (with φ clamped to $\varphi(\max(t, 0))$ so that it is monotone, hence integrable, on all of $\mathbb{R}$). Monotonicity of φ gives $\varphi(s) \le g(s) \le \varphi(2s)$ and that g is nondecreasing, so f is strictly increasing, $f \ge \varphi$, and $f(s) \ge s \to \infty$. For $s > 0$, $g(s) = s^{-1}(\Phi(2s) - \Phi(s))$ with Φ the (continuous) primitive of φ, so g is continuous there; at 0, $0 \le g(s) \le \varphi(2s) \to 0$ by continuity of φ at 0. □

Lemma 275 (Class $\mathcal{K}$ majorant on a bounded interval)**.** *Let $r > 0$ and let φ satisfy $\varphi(0) = 0$, be monotone on $[0, r]$, and be continuous at 0 from the right. Then there are $b > 0$ and an f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $\varphi(s) \le f(s)$ for all $s \in [0, r]$.*

Proof. Clamp φ to $\varphi_{\text{ext}}(s) = \varphi(\min(s, r))$, which is monotone on all of $[0, \infty)$, still vanishes at 0 and agrees with φ near 0 (so is still continuous there), and apply lemma 274. Restricting the resulting majorant to $[0, r]$ gives f; $b = f(r) > 0$ because f is strictly increasing from $f(0) = 0$. □

Theorem 276 (Class $\mathcal{K}$ sandwich bounds)**.** *Let $W : \mathbb{R}^n \to \mathbb{R}$ be positive definite (definition 29) about x_{eq} on a set containing $\overline{B}(x_{\text{eq}}, r)$. Then there are class $\mathcal{K}$ functions α_1, α_2 on $[0, r)$ with*

$$\alpha_1(\|x - x_{\text{eq}}\|) \;\le\; W(x) \;\le\; \alpha_2(\|x - x_{\text{eq}}\|) \qquad \forall\, \|x - x_{\text{eq}}\| < r.$$

The conclusion holds on the open *ball: α_1 and α_2 are class $\mathcal{K}$ on $[0, r)$, so neither is defined at $\|x - x_{\text{eq}}\| = r$.*

This is what lets a Lyapunov argument be run entirely in terms of comparison functions: positive definiteness is a pointwise condition, and this converts it into monotone bounds in the single variable $\|x - x_{\text{eq}}\|$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.3 (proved in his Appendix C.4). Differences:*

- *The bound holds on the open ball, with α_1, α_2 class $\mathcal{K}$ on $[0, r)$; Khalil states it on $[0, r]$.*
- *The radially unbounded $\mathcal{K}_\infty$ case is not derived here.*

Proof. Translate to the origin, $V(y) = W(y + x_{\text{eq}})$. For the lower bound take $\psi(s) = \inf\{V(y) : s \le \|y\| \le r\}$, the infimum of V over the annulus of inner radius s: it vanishes at 0, is positive for $s > 0$ by compactness and positive definiteness, and is nondecreasing. For the upper bound take $\phi(s) = \sup\{V(y) : \|y\| \le s\}$, similarly nondecreasing and vanishing at 0. Neither is continuous or strictly monotone, so smooth them: lemma 273 gives a class $\mathcal{K}$ minorant of ψ and lemma 275 a class $\mathcal{K}$ majorant of ϕ. Since $\psi(\|y\|) \le V(y) \le \phi(\|y\|)$ pointwise, these sandwich V. □

6.5 The decay system and the comparison bound

A class $\mathcal{K}$ decay rate produces a class $\mathcal{KL}$ bound. This is the step that turns a Lyapunov estimate — which controls only the *rate* of decrease — into an asymptotic stability statement, and it is where the comparison system $\dot{y} = -\alpha(y)$ enters: a Lyapunov function along a trajectory is a subsolution of it.

Definition 277 (The decay system of a class $\mathcal{K}$ function)**.** For α class $\mathcal{K}$ on $[0, a)$ (definition 221), the *decay system* is the scalar time-invariant system

$$\dot{y} = -\alpha(y),$$

as a continuous-time system (definition 76) with no input. Its meaningful states are $y \in [0, a)$.

Theorem 278 (The decay system has a class $\mathcal{KL}$ solution operator)**.** *Let α be class $\mathcal{K}$ on $[0, a)$ and at most linear near the origin, $\alpha(x) \le Lx$ on $(0, \text{base}]$. Then there is a class $\mathcal{KL}$ function σ (definition 251) with $\sigma(y_0, 0) = y_0$ such that for every $y_0 \in [0, a)$ and every t_0, $t \mapsto \sigma(y_0, t - t_0)$ is a trajectory on $[t_0, \infty)$ (definition 108) of the decay system $\dot{y} = -\alpha(y)$ (definition 277).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.4. Differences:*

- *Khalil assumes α locally Lipschitz; here the hypothesis is the linear bound $\alpha(x) \le Lx$ near the origin, which is what the construction uses.*
- *Uniqueness of the solution is not asserted.*

Proof. **Osgood's construction.** Let $\eta(y) = -\int_{\text{base}}^{y} \mathrm{d}x/\alpha(x)$ be the time for the solution started at base to reach y. Since $\alpha > 0$ on $(0, a)$, η is continuous and strictly decreasing there; and since $\alpha(x) \le Lx$ near 0, $\eta(y) \ge (\log \text{base} - \log y)/L \to +\infty$ as $y \to 0^+$, so the origin is not reached in finite time. Setting $\sigma(r, s) := \eta^{-1}(\eta(r) + s)$, extended by $\sigma(0, s) = 0$, the class $\mathcal{KL}$ conditions follow from monotonicity and continuity of η^{-1}, and differentiating $\eta(\sigma) = \eta(r) + s$ gives $\dot\sigma = -\alpha(\sigma)$ for $s > 0$. At $s = 0$ the one-sided derivative is the limit of these, $-\alpha(r)$, by continuity of σ and α. □

Lemma 279 (Lipschitz class $\mathcal{K}$ minorant)**.** *Let α be a class $\mathcal{K}$ function on $[0, a)$. Then there are $b' > 0$, $L > 0$ and a class $\mathcal{K}$ function $\beta : [0, a) \to [0, b')$ with $\beta(x) \le \alpha(x)$ for $x \in [0, a)$, $\beta(x) \le Lx$ for $x \ge 0$, and β L-Lipschitz on all of $\mathbb{R}$.*

The range of β cannot in general be the range $[0, b)$ of α: $\alpha(x) = 1 - \sqrt{1 - x}$ on $[0, 1)$ approaches 1 more slowly than any Lipschitz function can.

Proof. Extend α to a nondecreasing $g : \mathbb{R} \to [0, b]$ by 0 on the left and b from a on, and set $\beta(x) = a^{-1} \int_0^x g$, so $b' = \beta(a)$. Since $0 \le g \le b$, β is (b/a)-Lipschitz, and with $\beta(0) = 0$ this gives $\beta(x) \le (b/a)\, x$. Since $g > 0$ on $(0, \infty)$, β is strictly increasing, hence class $\mathcal{K}$ by definition 223. Finally g is nondecreasing, so for $x \in [0, a)$, $\int_0^x g \le x\, \alpha(x) \le a\, \alpha(x)$, i.e. $\beta(x) \le \alpha(x)$. □

Theorem 280 (Subsolutions of the decay system decay)**.** *Let α be class $\mathcal{K}$ on $[0, a)$. Then there is a class $\mathcal{KL}$ function σ with $\sigma(r, 0) = r$ such that: whenever v is continuous on $[t_0, t]$, takes values in $[0, a)$ there, and is a Dini subsolution of the decay system (definition 277),*

$$D^+ v(s) \;\le\; -\alpha(v(s)),$$

it obeys $v(t) \le \sigma(v(t_0),\, t - t_0)$.

A differential inequality *with a class $\mathcal{K}$ decay rate therefore yields a class $\mathcal{KL}$ bound — which is what converts a Lyapunov decay estimate into an asymptotic stability statement. No hypothesis is placed on α beyond being class $\mathcal{K}$: the linear-growth condition the construction needs is obtained internally by passing to a Lipschitz minorant.*

Proof. Replace α by a class $\mathcal{K}$ minorant $\beta \le \alpha$ that is Lipschitz near the origin; this only weakens the Dini hypothesis, and supplies the linear bound $\beta(x) \le Lx$ that theorem 278 requires. That theorem gives a class $\mathcal{KL}$ σ whose sections solve $\dot y = -\beta(y)$ with $\sigma(r, 0) = r$. Since v is a Dini subsolution of the same equation and starts at $\sigma(v(t_0), 0)$, theorem 58 bounds v by it — the perturbed solutions that lemma requires coming from theorem 54, whose hypotheses hold because β is globally Lipschitz (lemma 279). □

Theorem 281 (Subsolutions above a level decay to it)**.** *Let α be class $\mathcal{K}$ on $[0, a)$, and σ the class $\mathcal{KL}$ function of theorem 280. Let v be continuous on $[t_0, t]$ with values in $[0, a)$, with right derivative v' on $[t_0, t)$ satisfying $v'(s) \le -\alpha(v(s))$ whenever $v(s) > \eta$. Then*

$$v(t) \le \max\{\sigma(v(t_0), t - t_0),\ \eta\}.$$

Proof. If $v > \eta$ on all of $[t_0, t)$, v is a subsolution of the decay system there and theorem 280 applies. Otherwise $v(\tau) \le \eta$ for some $\tau \in [t_0, t)$. Wherever $v > \eta$, $v' \le -\alpha(v) \le 0$, so by lemma 28 on $[\tau, t]$, $v(t) \le \max\{v(\tau), \eta\} = \eta$. □

6.6 Massera's lemma

A class $\mathcal{K}$ function with class $\mathcal{K}$ derivative can flatten any decay rate enough to make a Lyapunov integral along the solutions converge, together with its derivative in the initial state.

Lemma 282 (Massera's lemma)**.** *Let $g : [0,\infty) \to \mathbb{R}$ be positive, continuous and strictly decreasing with $g(t) \to 0$, and $h : [0,\infty) \to \mathbb{R}$ positive, continuous and nondecreasing. There is a class $\mathcal{K}_\infty$ function G (definition 222) whose derivative G' is class $\mathcal{K}_\infty$, and constants $k_1, k_2 > 0$, such that for every continuous u with $0 \le u(t) \le g(t)$ the integrals below converge and*

$$\int_0^\infty G(u(t))\,dt \le k_1, \qquad \int_0^\infty G'(u(t))\,h(t)\,dt \le k_2.$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma C.1. Differences: G and G' are class $\mathcal{K}_\infty$ rather than class $\mathcal{K}$ on $[0,\infty)$, since only their values on $[0, g(0)]$ matter; convergence of the integrals is stated.*

Proof. Let $\eta(t) = g(t) + 1/t$: continuous and strictly decreasing on $(0,\infty)$, with $\eta \to \infty$ at 0^+, $\eta \to 0$ at ∞, and $\eta > g$. Its inverse $\theta = \eta^{-1}$ (lemma 247) satisfies $\theta(s) \ge t$ whenever $0 < s \le g(t)$. Let

$$G'(s) = \frac{e^{-\theta(s)}}{h(\theta(s))} + \max(s - g(0), 0), \qquad G(r) = \int_0^r G'(s)\,ds,$$

so that $G'(s) \le e^{-t}/h(t)$ for $0 \le s \le g(t)$. Then $G'(u(t))\,h(t) \le e^{-t}$ and $G(u(t)) \le u(t)\,e^{-t}/h(t) \le (g(0)/h(0))\,e^{-t}$, so $k_1 = g(0)/h(0)$ and $k_2 = 1$. Khalil's piecewise-linear η is replaced by $g(t) + 1/t$, which lies above g for every $t > 0$, so no splitting at t_1 is needed. □

Chapter 7

Stability of continuous-time systems

Stability notions for an equilibrium of an autonomous continuous-time system $\dot{x} = f(t, x)$ (definition 76); a system with input is first unforced (definition 86). Each notion is defined once, for time-varying systems, with the initial time $t_0 \geq 0$ explicit; a time-invariant system is a special case, and that its notions may be read with $t_0 = 0$ is a theorem rather than a second definition.

Bounds — stability, uniform convergence times, exponential estimates — quantify over trajectories on segments $[t_0, t_1]$, so that a solution escaping in finite time counts against stability instead of being silently excluded. Limits quantify over trajectories on rays $[t_0, \infty)$.

7.1 Definitions

Definition 283 (Stability)**.** The equilibrium x_{eq} of an autonomous continuous-time system $\dot{x} = f(t, x)$ is *stable* when for every $\varepsilon > 0$ and $t_0 \geq 0$ there is $\delta = \delta(\varepsilon, t_0) > 0$ such that every trajectory on a segment $[t_0, t_1]$ with $\|x(t_0) - x_{\text{eq}}\| < \delta$ satisfies $\|x(t) - x_{\text{eq}}\| < \varepsilon$ for all $t \in [t_0, t_1]$.

Quantifying over segments rather than over solutions on $[t_0, \infty)$ means a solution that escapes in finite time counts against stability instead of being silently excluded.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*stable*).

Definition 284 (Instability)**.** The equilibrium is *unstable* when it is not stable (definition 283).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 285 (Uniform stability)**.** The equilibrium is *uniformly stable* when δ in definition 283 can be chosen independently of t_0: $\delta = \delta(\varepsilon)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly stable*).

Definition 286 (Global uniform stability)**.** The equilibrium is *globally uniformly stable* when it is uniformly stable (definition 285) with a choice of $\delta(\varepsilon)$ satisfying $\delta(\varepsilon) \to \infty$ as $\varepsilon \to \infty$. In particular every trajectory is bounded, so none escapes in finite time.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 287 (Local asymptotic stability)**.** The equilibrium is *locally asymptotically stable* when it is stable (definition 283) and for every $t_0 \geq 0$ there is $c = c(t_0) > 0$ such that the system is forward complete from the ball of radius c about x_{eq} at t_0 (definition 91) and every trajectory on $[t_0, \infty)$ starting in that ball satisfies $x(t) \to x_{\text{eq}}$. Forward completeness keeps solutions that escape in finite time from being silently excluded.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*asymptotically stable*).

Definition 288 (Global asymptotic stability)**.** The equilibrium is *globally asymptotically stable* when it is stable (definition 283), the system is forward complete from every state at every $t_0 \geq 0$ (definition 91), and every trajectory on $[t_0, \infty)$ satisfies $x(t) \to x_{\text{eq}}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Section 4.1.

Definition 289 (Uniform convergence time)**.** Trajectories starting within c of x_{eq} have a *uniform convergence time*: for every $\eta > 0$ there is $T = T(\eta) > 0$ such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_{\text{eq}}\| < c$ satisfies $\|x(t) - x_{\text{eq}}\| < \eta$ for $t \in [t_0 + T, t_1]$.

Definition 290 (Uniform asymptotic stability)**.** The equilibrium is *uniformly asymptotically stable* when it is uniformly stable (definition 285) and there is $c > 0$ from which convergence has a uniform convergence time (definition 289).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly asymptotically stable*).

Definition 291 (Global uniform asymptotic stability)**.** The equilibrium is *globally uniformly asymptotically stable* when it is globally uniformly stable (definition 286) and every $c > 0$ admits a uniform convergence time (definition 289).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 292 (Local exponential stability)**.** The equilibrium is *locally exponentially stable* when there are $c, k, \gamma > 0$ such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_{\text{eq}}\| < c$ satisfies

$$\|x(t) - x_{\text{eq}}\| \leq k \, \|x(t_0) - x_{\text{eq}}\| \, e^{-\gamma(t - t_0)} \qquad \forall\, t \in [t_0, t_1].$$

The constants do not depend on t_0: exponential stability is uniform, as in the reference.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5.

Definition 293 (Global exponential stability)**.** The equilibrium is *globally exponentially stable* when the estimate of definition 292 holds for every initial state, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5.

Boundedness is a property of the system near a point that need not be an equilibrium: trajectories stay, or end up and stay, within a prescribed distance of it.

Definition 294 (Uniform boundedness)**.** Trajectories are *uniformly bounded* around x_c when there is $c > 0$ such that for every $0 < a < c$ there is $\beta = \beta(a) > 0$ with

$$\|x(t_0) - x_c\| \leq a \;\Rightarrow\; \|x(t) - x_c\| \leq \beta \qquad \forall\, t \in [t_0, t_1]$$

for every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$. The point x_c need not be an equilibrium. The bound applies to every segment a solution exists on, so finite escape violates it.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 295 (Global uniform boundedness)**.** Trajectories are *globally uniformly bounded* around x_c when the bound of definition 294 holds for every $a > 0$, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 296 (Uniform ultimate boundedness)**.** Trajectories are *uniformly ultimately bounded* with ultimate bound b around x_c when there is $c > 0$ such that for every $t_0 \geq 0$ the system is forward complete from the ball of radius c about x_c (definition 91), and for every $0 < a < c$ there is $T = T(a, b) \geq 0$ with

$$\|x(t_0) - x_c\| \leq a \;\Rightarrow\; \|x(t) - x_c\| \leq b \qquad \forall\, t \geq t_0 + T$$

for every trajectory on $[t_0, \infty)$. The point x_c need not be an equilibrium. Khalil leaves forward completeness implicit; without it, a solution escaping before $t_0 + T$ would never be tested.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 297 (Global uniform ultimate boundedness)**.** Trajectories are *globally uniformly ultimately bounded* with ultimate bound b around x_c when the system is forward complete from every state at every $t_0 \geq 0$ (definition 91) and the ultimate bound of definition 296 holds for every $a > 0$, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

7.2 Implications

Lemma 298 (Instability from a fixed escape radius)**.** *Fix $\varepsilon > 0$. If for every $\delta > 0$ some trajectory on a segment $[0, T]$ starts within δ of x_{eq} and reaches distance at least ε from it, then x_{eq} is unstable (definition 284).*

Proof. Stability at $t_0 = 0$ with tolerance ε gives a δ; the trajectory for that δ contradicts it. □

Lemma 299 (Uniform stability implies stability)**.** *A uniformly stable equilibrium (definition 285) is stable (definition 283).*

Proof. □

Lemma 300 (Global asymptotic stability implies local asymptotic stability)**.** *A globally asymptotically stable equilibrium (definition 288) is locally asymptotically stable (definition 287).*

Proof. □

Lemma 301 (Uniform asymptotic stability implies local asymptotic stability)**.** *A uniformly asymptotically stable equilibrium (definition 290) of a forward-complete system (definition 92) is locally asymptotically stable (definition 287). Forward completeness is needed: bounds on solution segments do not by themselves make the segments extend.*

Proof. Uniform stability gives stability; a uniform convergence time gives convergence of every trajectory on a ray, with the same radius c for every t_0. □

Lemma 302 (Global uniform asymptotic stability implies global asymptotic stability)**.** *A globally uniformly asymptotically stable equilibrium (definition 291) is globally asymptotically stable (definition 288), given forward completeness (definition 92).*

Proof. Every trajectory starts in some ball, and every ball has a uniform convergence time. □

Lemma 303 (Exponential stability implies uniform stability)**.** *A locally exponentially stable equilibrium (definition 292) is uniformly stable (definition 285).*

Proof. The estimate bounds the distance by k times the initial distance, so $\delta = \min(c, \varepsilon/k)$ works for every t_0. □

Lemma 304 (Exponential stability implies uniform asymptotic stability)**.** *A locally exponentially stable equilibrium (definition 292) is uniformly asymptotically stable (definition 290).*

Proof. Uniform stability is lemma 303. From the ball of radius c the distance after time T is below $kc\,e^{-\gamma T}$, which is below any $\eta > 0$ for T large, independently of t_0. □

Lemma 305 (Global exponential stability implies local exponential stability)**.** *A globally exponentially stable equilibrium (definition 293) is locally exponentially stable (definition 292).*

Proof. □

Lemma 306 (Global exponential stability implies global uniform asymptotic stability)**.** *A globally exponentially stable equilibrium (definition 293) is globally uniformly asymptotically stable (definition 291).*

Proof. The estimate bounds the distance by k times the initial distance, so $\delta(\varepsilon) = \varepsilon/k$ witnesses global uniform stability. For each radius c the estimate is locally exponential from that ball, which gives a uniform convergence time by lemma 304. □

Lemma 307 (Exponential stability under a change of coordinates)**.** *Let $e : E \to E'$ be continuous linear with $\|x\| \le a\|ex\|$ and $\|ex\| \le b\|x\|$, $a, b > 0$, and let $F'(t, x'_{\mathrm{eq}} + e(x - x_{\mathrm{eq}})) = e(F(t, x))$. If x'_{eq} is locally exponentially stable for s' (definition 292), so is x_{eq} for s.*

Proof. If x is a trajectory of s, then $x'_{\mathrm{eq}} + e(x - x_{\mathrm{eq}})$ is one of s'. Its estimate, with radius c/b and constant abk, is one for x. □

7.3 Comparison-function characterizations

The uniform notions are equivalent to comparison-function bounds on the trajectories: uniform stability to a class $\mathcal{K}$ bound, uniform asymptotic stability to a class $\mathcal{KL}$ bound. The hard direction is the decay in elapsed time. A uniform convergence time says that for each tolerance η there is a delay after which every trajectory from a ball is inside the η-ball. Read the other way round it bounds how far a trajectory can still be after a given elapsed time, but only through a function of η that may be discontinuous. The results below regularize it into a decay envelope, a function of elapsed time alone that is continuous, positive and vanishing at $+\infty$: the time-decay half of a class $\mathcal{KL}$ bound, whose other half is a uniform class $\mathcal{K}$ bound.

Definition 308 (Uniform class $\mathcal{K}$ bound)**.** The trajectories of s *have the uniform class $\mathcal{K}$ bound* $\alpha : [0, a) \to [0, b)$ about x_{eq} when

$$\|x(t) - x_{\mathrm{eq}}\| \le \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$$

for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \in [t_0, t_1]$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), (4.19).

Definition 309 (Uniform class $\mathcal{K}_\infty$ bound)**.** The trajectories of s *have the uniform class* $\mathcal{K}_\infty$ *bound* α about x_{eq} when $\|x(t) - x_{\mathrm{eq}}\| \le \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$ for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and every $t \in [t_0, t_1]$, with no restriction on the initial state.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), (4.20).

Lemma 310 (Local uniform decay envelope)**.** *Suppose the trajectories of s have a uniform convergence time from the ball of radius c (definition 289) and a uniform class $\mathcal{K}$ bound α defined on $[0, a_\alpha)$ (definition 308). Let $0 < a \le c$ with $a < a_\alpha$. Then there is a singular class $\mathcal{L}$ function U (definition 250) such that*

$$\|x(t) - x_{\mathrm{eq}}\| \le U(t - t_0)$$

for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \in (t_0, t_1]$.

The bound depends on the elapsed time alone: neither on t_0, nor on the trajectory, nor on where in the a-ball it started.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5.*

Proof. Let $\bar{T}(\eta, r)$ be the least delay after which every trajectory starting within r of x_{eq} lies inside the η-ball; it is finite by hypothesis and antitone in η, but possibly discontinuous. Smooth it by the sliding average

$$W(r, \eta) = \frac{2}{\eta} \int_{\eta/2}^{\eta} \left[\bar{T}(s, r) + \frac{r}{\eta} \right] ds,$$

which equals $\frac{2}{\eta} \int_{\eta/2}^{\eta} \bar{T}(s, r)\, ds + r/\eta$. The averaged term is antitone in η and the r/η term is strictly antitone, so $W(r, \cdot)$ is continuous and strictly antitone on $(0, \infty)$; it dominates r/η, hence tends to $+\infty$ as $\eta \to 0^+$, and $\bar{T}(\cdot, r)$ vanishes beyond $\alpha(r)$, hence $W(r, \eta) \to 0$ as $\eta \to \infty$. Inverting it (lemma 244, lemma 245, lemma 246, lemma 247, lemma 248) gives $U = \mathrm{invFunOn}\, W(r, \cdot)\, (0, \infty)$, singular class $\mathcal{L}$. Finally $W(r, U(s)) = s$ and $W \ge \bar{T} + r/\eta$ give $\bar{T}(U(s), r) < s$, so s is itself a valid convergence time to the $U(s)$-ball, which is the claimed bound. □

Lemma 311 (Global uniform decay envelope)**.** *Suppose the trajectories of s have a uniform convergence time from every ball (definition 289) and a uniform class $\mathcal{K}_\infty$ bound (definition 309). Then there is a family $U : (0, \infty) \times (0, \infty) \to \mathbb{R}$ with*

1. *$U(r, s) > 0$ for all $r > 0$ and $s > 0$;*
2. *$U(r, \cdot)$ antitone on $(0, \infty)$ for each $r > 0$;*
3. *$U(r, s) \to 0$ as $s \to \infty$, for each $r > 0$;*
4. *$r \mapsto U(r + 1, s)$ monotone on $[0, \infty)$ for each $s > 0$;*
5. *$\|x(t) - x_{\mathrm{eq}}\| \le U(r, t - t_0)$ for every $r > 0$, every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < r$, and every $t \in (t_0, t_1]$.*

Conditions (1)–(4) are exactly the hypotheses of **??***; (5) is the decay bound. Unlike the local case (lemma 310) no single envelope can serve every initial state, so the radius remains a parameter.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5 (global case).*

Proof. Every radius is finite, so at radius r the global hypotheses restrict to the local ones (a uniform convergence time from the r-ball, and the class $\mathcal{K}_\infty$ bound read as a class $\mathcal{K}$ bound on $[0, r+1)$). Take $U(r,\cdot)$ to be the envelope of lemma 310 at that radius, namely the inverse of $W(r,\cdot)$. Conditions (1)–(3) and (5) are then that lemma's conclusions, and (4) holds because W is monotone in r — a larger ball has larger optimal convergence times — so inverting reverses nothing. □

Lemma 312 (Uniform convergence from a class $\mathcal{KL}$ bound)**.** *Let β be class $\mathcal{KL}$ on $[0,a)\times[0,\infty)$ (definition 251), and suppose every trajectory on a segment $[t_0,t_1]$, $t_0 \geq 0$, with $\|x(t_0)-x_{\text{eq}}\| < a$ satisfies $\|x(t)-x_{\text{eq}}\| \leq \beta(\|x(t_0)-x_{\text{eq}}\|, t-t_0)$ on the segment. Then for every $c < a$, trajectories from the ball of radius c have a uniform convergence time (definition 289).*

Proof. For $c < 0$ no trajectory starts in the ball. Otherwise every deviation is at most $\beta(c, t-t_0)$, by monotonicity of β in its first argument, and $\beta(c,\cdot) \to 0$. □

Lemma 313 (Class $\mathcal{K}$ characterization of stability)**.** *The equilibrium x_{eq} is stable (definition 283) if and only if for each $t_0 \geq 0$ there is a class $\mathcal{K}$ function α_{t_0} on $[0, a_{t_0})$ with $\|x(t)-x_{\text{eq}}\| \leq \alpha_{t_0}(\|x(t_0)-x_{\text{eq}}\|)$ for every trajectory on a segment $[t_0,t_1]$ with $\|x(t_0)-x_{\text{eq}}\| < a_{t_0}$, and every $t \in [t_0,t_1]$. A single α for all t_0 is uniform stability (lemma 314).*

Proof. The argument of lemma 314 with the initial time fixed. □

Lemma 314 (Class $\mathcal{K}$ characterization of uniform stability)**.** *The equilibrium x_{eq} is uniformly stable (definition 285) if and only if the trajectories have a uniform class $\mathcal{K}$ bound (definition 308): there is a class $\mathcal{K}$ function α on $[0,a)$ with*

$$\|x(t)-x_{\text{eq}}\| \leq \alpha(\|x(t_0)-x_{\text{eq}}\|)$$

for every trajectory on a segment $[t_0,t_1]$ with $t_0 \geq 0$ and $\|x(t_0)-x_{\text{eq}}\| < a$, and every $t \in [t_0,t_1]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, first half.*

Proof. Forward: the worst deviation $\omega(r)$ reachable from the r-ball is nondecreasing, vanishes at 0 and is continuous there by uniform stability; a continuous strictly increasing majorant (lemma 275) is the class $\mathcal{K}$ bound. Backward: α is continuous at 0 with $\alpha(0) = 0$, so a small radius δ has $\alpha(\delta) < \varepsilon$. □

Lemma 315 (Class $\mathcal{KL}$ characterization of uniform asymptotic stability)**.** *The equilibrium x_{eq} is uniformly asymptotically stable (definition 290) if and only if there are a class $\mathcal{KL}$ function β on $[0,a)\times[0,\infty)$ (definition 251) with*

$$\|x(t)-x_{\text{eq}}\| \leq \beta(\|x(t_0)-x_{\text{eq}}\|, t-t_0)$$

for every trajectory on a segment $[t_0,t_1]$ with $t_0 \geq 0$ and $\|x(t_0)-x_{\text{eq}}\| < a$, and every $t \in [t_0,t_1]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, second half.*

Proof. Forward: uniform stability gives a class $\mathcal{K}$ bound α (lemma 314) and, with the uniform convergence time, a decay envelope U (lemma 310); $\beta = \min(\alpha, \sqrt{\alpha\, U})$ is class $\mathcal{KL}$ (theorem 256) and bounds the trajectories. Backward: $\beta(\cdot,0)$ is continuous at 0, which gives uniform stability, and lemma 312 gives the convergence time. □

Lemma 316 (Class $\mathcal{KL}$ characterization, global case)**.** *The equilibrium x_{eq} is globally uniformly asymptotically stable (definition 291) if and only if there is a $\beta : [0,\infty) \times [0,\infty) \to \mathbb{R}$ that is*

1. *zero at the origin: $\beta(0,s) = 0$ for $s \geq 0$;*
2. *jointly continuous on $[0,\infty) \times [0,\infty)$;*
3. *strictly increasing in r for each fixed $s \geq 0$;*
4. *radially unbounded: $\beta(r,0) \to \infty$ as $r \to \infty$;*
5. *antitone in s for each fixed $r \geq 0$, with $\beta(r,s) \to 0$ as $s \to \infty$;*

such that $\|x(t) - x_{\mathrm{eq}}\| \leq \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ for every trajectory on a segment $[t_0, t_1]$ with $t_0 \geq 0$ and every $t \in [t_0, t_1]$, with no restriction on the initial state.

Condition (4) is what makes the characterization global*: without it β is class $\mathcal{KL}$ on a bounded radius and the statement collapses to the local one (lemma 315).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, global case.*

Proof. Forward: global uniform stability gives a class $\mathcal{K}_\infty$ bound, the convergence times from every ball a family of decay envelopes (lemma 311), and **??** combines them. Backward: $\alpha = \beta(\cdot, 0)$ is class $\mathcal{K}_\infty$ and $\delta = \alpha^{-1}$ gives global uniform stability; from the r-ball every deviation is at most $\beta(r,\cdot) \to 0$. □

The same characterizations hold with the bounds on solutions over $[t_0,\infty)$. A bound on every segment restricts to those solutions, so that direction needs nothing more. The converse needs every segment from the ball to extend to such a solution: forward completeness from some ball about x_{eq} locally, and from every state globally.

Lemma 317 (Class $\mathcal{K}$ characterization for forward-complete systems)**.** *Let the system be forward complete from some ball $B(x_{\mathrm{eq}}, c)$ at every $t_0 \geq 0$ (definition 91). Then x_{eq} is uniformly stable if and only if there is a class $\mathcal{K}$ function α on $[0,a)$ with $\|x(t) - x_{\mathrm{eq}}\| \leq \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$ for every solution x on $[t_0,\infty)$ with $t_0 \geq 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 314, with the bound moved between segments and solutions on $[t_0,\infty)$. Forward: restrict a solution to $[t_0, t]$. Converse: shrink the radius to $a' = \min(c, a/2)$, restricting α (definition 224); a segment starting within a' extends by completeness to a solution on $[t_0,\infty)$, which the bound covers. □

Lemma 318 (Class $\mathcal{KL}$ characterization for forward-complete systems)**.** *Let the system be forward complete from some ball $B(x_{\mathrm{eq}}, c)$ at every $t_0 \geq 0$ (definition 91). Then x_{eq} is uniformly asymptotically stable if and only if there is a class $\mathcal{KL}$ function β on $[0,a) \times [0,\infty)$ with $\|x(t) - x_{\mathrm{eq}}\| \leq \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ for every solution x on $[t_0,\infty)$ with $t_0 \geq 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 315, with the bound moved between segments and solutions on $[t_0,\infty)$ as in lemma 317, shrinking the radius to $\min(c, a)$ (definition 260). □

Lemma 319 (Class $\mathcal{KL}$ characterization for forward-complete systems, global case)**.** *Let the system be forward complete (definition 92). Then x_{eq} is globally uniformly asymptotically stable if and only if there is a β with the properties of lemma 316 and $\|x(t) - x_{\mathrm{eq}}\| \leq \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ for every solution x on $[t_0, \infty)$ with $t_0 \geq 0$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 316, with the bound moved between segments and solutions on $[t_0, \infty)$ as in lemma 317. □

7.4 Lyapunov functions

A Lyapunov function is a certificate: its existence implies a stability notion. The certificates are time-varying, $V(t, x)$, and named by what they certify. What separates them is which bounds hold, not whether time appears: a positive definite (definition 29) lower bound gives stability, and a positive definite upper bound as well (decrescence) is what makes the conclusion uniform. A time-invariant $V(x)$ is its own lower and upper bound, so it is a uniform certificate as soon as it decreases along the system.

Definition 320 (Lyapunov function)**.** A function $V : \mathbb{R} \times E \to \mathbb{R}$ is a *Lyapunov function* on a neighbourhood D of x_{eq} when, for all $t \geq 0$ and $x \in D$, V is differentiable at (t, x) and

$$V(t, x_{\mathrm{eq}}) = 0, \qquad W_1(x) \leq V(t, x), \qquad \dot{V}(t, x) \leq 0,$$

for some W_1 positive definite on D (definition 29), with $\dot{V}$ the derivative along the system (definition 115). It certifies stability (definition 283).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.8.

Definition 321 (Uniform Lyapunov function)**.** A Lyapunov function (definition 320) that is also *decrescent*: $V(t, x) \leq W_2(x)$ for all $t \geq 0$, $x \in D$, for some W_2 positive definite on D. The upper bound is what makes the conclusion uniform: it certifies uniform stability (definition 285).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.8.

Definition 322 (Strict uniform Lyapunov function)**.** A uniform Lyapunov function (definition 321) with $\dot{V}(t, x) \leq -W_3(x)$ for all $t \geq 0$, $x \in D$, for some W_3 positive definite on D. It certifies uniform asymptotic stability (definition 290).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.9.

Definition 323 (Exponential Lyapunov function)**.** A function $V : \mathbb{R} \times E \to \mathbb{R}$ is an *exponential Lyapunov function* with exponent $a > 0$ on a neighbourhood D of x_{eq} when there are $k_1, k_2, k_3 > 0$ with, for all $t \geq 0$ and $x \in D$, V differentiable at (t, x) and

$$k_1 \|x - x_{\mathrm{eq}}\|^a \leq V(t, x) \leq k_2 \|x - x_{\mathrm{eq}}\|^a, \qquad \dot{V}(t, x) \leq -k_3 \|x - x_{\mathrm{eq}}\|^a.$$

It certifies local exponential stability (definition 292).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.10.

Lemma 324 (Time-invariant Lyapunov functions are uniform)**.** *Let $V : E \to \mathbb{R}$ be positive definite on a neighbourhood D of x_{eq} (definition 29) and differentiable at its points, with $\nabla V(x) \cdot f(t, x) \leq 0$ for all $t \geq 0$, $x \in D$. Then $(t, x) \mapsto V(x)$ is a uniform Lyapunov function (definition 321).*

Proof. Take $W_1 = W_2 = V$. The derivative along the system of a function of x alone is its directional derivative along the vector field. □

Lemma 325 (Strict time-invariant Lyapunov functions are strict uniform)**.** *Let $V : E \to \mathbb{R}$ be positive definite on a neighbourhood D of x_{eq} and differentiable at its points, with $\nabla V(x) \cdot f(t,x) \leq -W_3(x)$ for all $t \geq 0$, $x \in D$, for some W_3 positive definite on D. Then $(t,x) \mapsto V(x)$ is a strict uniform Lyapunov function (definition 322).*

Proof. As for lemma 324; $-W_3 \leq 0$ gives the non-strict decrease. □

Lemma 326 (Strict decrease suffices for a time-invariant system)**.** *Let the state equation be $\dot{x} = f(x)$ with f continuous and $f(x_{\text{eq}}) = 0$, and let $V : E \to \mathbb{R}$ be C^1 and positive definite on a neighbourhood D of x_{eq} with $\nabla V(x) \cdot f(x) < 0$ for $x \in D \setminus \{x_{\text{eq}}\}$. Then $(t,x) \mapsto V(x)$ is a strict uniform Lyapunov function (definition 322).*

Proof. Take $W_3 = -\nabla V \cdot f$: continuous since V is C^1 and f is continuous, zero at x_{eq} since $f(x_{\text{eq}}) = 0$, and positive elsewhere on D; apply lemma 325. □

7.5 Lyapunov's direct method

For a time-varying V the derivative along the system acquires an explicit time term, and positive definiteness of $V(t,\cdot)$ on its own buys nothing uniform in t_0: V could flatten as $t \to \infty$. Sandwiching V between two positive-definite functions of the state alone is what restores uniformity. Only trajectories on segments $[t_0, t_1]$ are used, so no solution is assumed to exist for all time.

Theorem 327 (Lyapunov's uniform stability theorem)**.** *If V is a uniform Lyapunov function for x_{eq} (definition 321) on a system with state space $\mathbb{R}^n$, then x_{eq} is uniformly stable (definition 285).*

The sandwich $W_1(x) \leq V(t,x) \leq W_2(x)$ is what makes the conclusion uniform in t_0: W_1 and W_2 do not depend on t, so neither does the $\delta(\varepsilon)$ extracted from them.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.8.*

Proof. Given ε, work on a radius $r \leq \varepsilon$ whose closed ball lies in the region of the certificate. Bound W_1 below and W_2 above by class $\mathcal{K}$ functions α_1, α_2 on it (theorem 276), fix a level d below both ranges, and take $\delta = \alpha_2^{-1}(d)$: starting within δ of x_{eq} forces $V(t_0, x(t_0)) < d$. To leave the r-ball the trajectory would first cross a sphere on which $W_1 > d$; up to that time V has not increased (lemma 66), so it is still below d, a contradiction. □

Theorem 328 (Lyapunov's uniform asymptotic stability theorem)**.** *If V is a strict uniform Lyapunov function for x_{eq} (definition 322) on a system with state space $\mathbb{R}^n$, then x_{eq} is uniformly asymptotically stable (definition 290).*

A strictly negative $\dot{V}$ alone would not do: W_3 must be bounded away from zero on each annulus, which is exactly what positive definiteness of a function of x alone buys.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.9. Khalil's two "moreover" clauses are not part of this conclusion: the explicit class $\mathcal{KL}$ bound, which the proof constructs, and the global case.*

Proof. Uniform stability is theorem 327. For convergence, take class $\mathcal{K}$ bounds $\alpha_1, \alpha_2, \alpha_3$ for W_1, W_2, W_3 on a ball in the region of the certificate, and the trapping radius c of that proof. Along

a trajectory from the c-ball, $\dot V \le -W_3(x) \le -\alpha_3(\|x - x_{\text{eq}}\|) \le -\alpha_3(\alpha_2^{-1}(V))$, so $v(t) = V(t, x(t))$ satisfies $D^+v \le -\alpha(v)$ with $\alpha = \alpha_3 \circ \alpha_2^{-1}$ of class $\mathcal{K}$. theorem 280 gives a class $\mathcal{KL}$ σ with $v(t) \le \sigma(v(t_0), t - t_0)$; unwinding the sandwich,

$$\|x(t) - x_{\text{eq}}\| \le \alpha_1^{-1}(\sigma(\alpha_2(\|x(t_0) - x_{\text{eq}}\|), t - t_0)),$$

a class $\mathcal{KL}$ bound. Conclude by lemma 312. □

In the global theorem compact sublevel sets of V are the hypothesis; for a continuous V they follow from radial unboundedness.

Lemma 329 (Sublevel sets of a radially unbounded function are compact)**.** *Let E be a proper normed space. If $V : E \to \mathbb{R}$ is continuous and radially unbounded ($V(x) \to \infty$ as $\|x\| \to \infty$), then every sublevel set $\{x \mid V(x) \le c\}$ is compact.*

Proof. Closed: it is $V^{-1}((-\infty, c])$, with V continuous. Bounded: radial unboundedness gives a compact K off which $V > c$, so the sublevel set lies in K. Compact: closed and bounded in a proper space. □

Theorem 330 (Lyapunov's global asymptotic stability theorem)**.** *Let the state equation be $\dot x = f(x)$ with f continuous, on $\mathbb{R}^n$. Let $V : \mathbb{R}^n \to \mathbb{R}$ be C^1, positive definite about x_{eq} on all of $\mathbb{R}^n$ (definition 29), with compact sublevel sets $\{V \le c\}$ (equivalently, radially unbounded) and $\nabla V(x) \cdot f(x) < 0$ for $x \ne x_{\text{eq}}$. Then x_{eq} is globally uniformly asymptotically stable (definition 291).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.2.*

Proof. V does not increase along trajectories: $\nabla V \cdot f \le 0$ everywhere, since $\nabla V(x_{\text{eq}}) = 0$ at the minimum. So a trajectory starting below every value V takes outside the ε-ball never leaves it. Take $\delta(\varepsilon)$ the supremum of the radii $\rho \le \varepsilon$ whose closed ball lies strictly below those values: it is positive, since V is bounded below by a positive constant outside the ε-ball and small near x_{eq}, and it tends to ∞, since V is bounded on each ball and large far out. This is global uniform stability.

For convergence from the c-ball, trajectories stay in the compact set $\{V \le M\}$, $M = \max_{\overline{B}(x_{\text{eq}}, c)} V$, on which, outside the $\delta(\eta)$-ball, $\nabla V \cdot f \le -\gamma < 0$. So V falls at rate γ while the trajectory is outside, and since $V \ge 0$ it enters the $\delta(\eta)$-ball by time $t_0 + M/\gamma + 1$; from then on it stays in the η-ball. □

Theorem 331 (Lyapunov's exponential stability theorem)**.** *If V is an exponential Lyapunov function for x_{eq} (definition 323), then x_{eq} is locally exponentially stable (definition 292): near x_{eq},*

$$\|x(t) - x_{\text{eq}}\| \le \left(\frac{k_2}{k_1}\right)^{1/a} \|x(t_0) - x_{\text{eq}}\|\, e^{-(k_3/(ak_2))(t - t_0)}.$$

The state space is any normed space.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.10.*

Proof. The bounds give $\dot V \le -(k_3/k_2)V$, so $e^{(k_3/k_2)t}V(t, x(t))$ does not increase and $V(t, x(t)) \le V(t_0, x(t_0))\, e^{-(k_3/k_2)(t - t_0)}$ while the trajectory stays in D; the sandwich and an a-th root give the estimate. To stay in D, start within $r/(K + 1)$ of x_{eq}, where $\overline{B}(x_{\text{eq}}, r) \subseteq D$ and $K = (k_2/k_1)^{1/a}$: at the first time the trajectory reached the sphere of radius r the estimate would put it strictly inside. □

Theorem 332 (Lyapunov's global exponential stability theorem)**.** *If V is an exponential Lyapunov function for x_{eq} on the whole space (definition 323 with $D = E$), then x_{eq} is globally exponentially stable (definition 293).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.10, global case.*

Proof. The estimate of theorem 331 holds on every trajectory segment, since every segment stays in $D = E$. □

7.6 Converse Lyapunov theorems

Exponential stability has a quadratic Lyapunov function, built by integrating the squared distance along solutions over a fixed horizon. Its gradient is of the order of the distance to the equilibrium, which is what lets it absorb perturbations of the vector field.

Theorem 333 (Converse Lyapunov theorem for global exponential stability)**.** *Let x_{eq} be an equilibrium of $\dot{x} = f(t, x)$ on $\mathbb{R}^n$, with f continuously differentiable on $\mathbb{R} \times \mathbb{R}^n$ and $f(t, \cdot)$ L-Lipschitz for every $t \in \mathbb{R}$ (for C^1 f, equivalently $\|\partial f/\partial x\| \le L$). If x_{eq} is globally exponentially stable (definition 293), there is an exponential Lyapunov function V with exponent 2 on $\mathbb{R}^n$ (definition 323) and $c_4 > 0$ with $\|\partial V/\partial x\| \le c_4\|x - x_{\text{eq}}\|$ for all $t \ge 0$ and $x \in \mathbb{R}^n$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.14, global case. Differences:*

- *The Lipschitz bound replaces Khalil's equivalent Jacobian bound, and is asked for all $t \in \mathbb{R}$: differentiability of V at $t = 0$ involves solutions from negative initial times.*
- *The constants are $c_1 = \delta e^{-2L\delta}$, $c_2 = \delta k^2$ and $c_4 = 2k\delta e^{L\delta}$, cruder than Khalil's.*

Proof. Solutions $\phi(\tau; t, x)$ exist from every (t, x), since f is globally Lipschitz in x. Let $V(t, x) = \int_0^\delta \|\phi(t + \sigma; t, x) - x_{\text{eq}}\|^2 \, d\sigma$ with $\delta = \ln(2k^2 + 1)/(2\gamma)$. The exponential estimate and the lower bound $\|\phi(\tau) - x_{\text{eq}}\|^2 \ge e^{-2L(\tau - t)}\|x - x_{\text{eq}}\|^2$ sandwich V. V is differentiable: appending time to the state, solutions are differentiable in their initial data (**??**) and locally Lipschitz in it (**??**), so the integral can be differentiated under the integral sign. Along a solution, uniqueness gives $V = \int_h^{h+\delta} \|\phi(t + \sigma) - x_{\text{eq}}\|^2 d\sigma$, so $\dot{V} = \|\phi(t + \delta) - x_{\text{eq}}\|^2 - \|x - x_{\text{eq}}\|^2 \le (k^2 e^{-2\gamma\delta} - 1)\|x - x_{\text{eq}}\|^2 \le -\frac{1}{2}\|x - x_{\text{eq}}\|^2$. Finally, solutions from nearby states separate at most like $e^{L\sigma}$, which makes $V(t, \cdot)$ locally Lipschitz with constant $\approx 2k\delta e^{L\delta}\|x - x_{\text{eq}}\|$. □

Theorem 334 (Converse Lyapunov theorem for exponential stability)**.** *Let x_{eq} be an equilibrium of $\dot{x} = f(t, x)$ on $\mathbb{R}^n$, $f(t, x_{\text{eq}}) = 0$ for all t, with f continuously differentiable on $\mathbb{R} \times D$, $D = \{x : \|x - x_{\text{eq}}\| < r\}$, and $f(t, \cdot)$ L-Lipschitz on D for all t (for C^1 f, equivalently $\|\partial f/\partial x\| \le L$ there). Let $k, \gamma, r_0 > 0$ with $r_0 < r/k$, $D_0 = \{x : \|x - x_{\text{eq}}\| < r_0\}$, and suppose every trajectory on a segment $[t_0, t_1]$, $t_0 \ge 0$, with $x(t_0) \in D_0$ satisfies*

$$\|x(t) - x_{\text{eq}}\| \le k\,\|x(t_0) - x_{\text{eq}}\|\,e^{-\gamma(t - t_0)} \qquad \forall\, t \in [t_0, t_1].$$

Then there is an exponential Lyapunov function V with exponent 2 on D_0 (definition 323), that is, with

$$c_1\|x - x_{\text{eq}}\|^2 \le V(t, x) \le c_2\|x - x_{\text{eq}}\|^2, \qquad \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t, x) \le -c_3\|x - x_{\text{eq}}\|^2$$

for $t \ge 0$ and $x \in D_0$, and $c_4 > 0$ with $\|\partial V/\partial x\| \le c_4\|x - x_{\text{eq}}\|$ there.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.14. Differences:*

- *f is C^1 and Lipschitz for all $t \in \mathbb{R}$, not only $t \geq 0$: differentiability of V at $t = 0$ involves solutions from negative initial times.*
- *The Lipschitz bound replaces Khalil's equivalent Jacobian bound.*
- *The estimate is assumed on trajectory segments.*

Proof. Cut the field off outside the ball: with χ a smooth bump equal to 1 on the ball of radius ρ_1, $kr_0 < \rho_1 < r$, and vanishing near the sphere of radius r, $\tilde{f} = \chi f$ is C^1 and globally Lipschitz in x, uniformly in t, and agrees with f on the ρ_1-ball. A solution of $\tilde{f}$ from D_0 never reaches the ρ_1-sphere: up to the first time it would, it is a solution of f, so Khalil's estimate keeps it within $kr_0 < \rho_1$. It is therefore a solution of f, and obeys the estimate for all time. The construction of theorem 333, run for $\tilde{f}$ from D_0, gives V; on D_0, where $\tilde{f} = f$, its derivative along $\tilde{f}$ is its derivative along f. □

7.7 Ultimate boundedness

A Lyapunov-like function whose derivative is negative only outside a ball does not certify stability, but it does bound the trajectories: they decay like a class $\mathcal{KL}$ function until they reach a neighbourhood of the ball, and stay there.

Theorem 335 (Ultimate boundedness theorem)**.** *Let the state space be $\mathbb{R}^n$, $\alpha_1 : [0, r) \to [0, b_1)$ and $\alpha_2 : [0, r) \to [0, b_2)$ class $\mathcal{K}$, and $V : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}$ differentiable at (t, x) and*

$$\alpha_1(\|x - x_c\|) \leq V(t, x) \leq \alpha_2(\|x - x_c\|), \qquad \dot{V}(t, x) \leq -W_3(x) \quad \text{for } \mu \leq \|x - x_c\|$$

whenever $t \geq 0$ and $\|x - x_c\| < r$, where W_3 is positive definite about x_c on the closed r-ball (definition 29) and $0 \leq \mu < c$, where $c < r$ and $\alpha_2(c) < b_1$. Then there is a class $\mathcal{KL}$ function β on $[0, c)$, depending only on α_1, α_2, W_3 and c — not on the system, V or μ — such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_c\| < c$ satisfies

$$\|x(t) - x_c\| \leq \max\{\beta(\|x(t_0) - x_c\|, t - t_0),\ \alpha_1^{-1}(\alpha_2(\mu))\} \qquad \forall\, t \in [t_0, t_1].$$

Khalil's estimates (4.42) and (4.43) are the two branches of the maximum.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.18. Differences:*

- *The sandwich holds on an open ball; Khalil's level $\alpha_1(r)$ becomes b_1, and the initial states form the open c-ball.*
- *$\mu = 0$ is allowed.*
- *$\beta(\rho, s) \leq \alpha_1^{-1}(\alpha_2(\rho))$ is part of the conclusion. The global case is theorem 338.*

Proof. Let $v(t) = V(t, x(t))$, $\eta = \alpha_2(\mu)$ and $d = \alpha_2(c)$. If $v > \eta$ then $\alpha_2(\|x - x_c\|) \geq v > \alpha_2(\mu)$, so $\|x - x_c\| > \mu$ and $\dot{v} \leq -W_3 \leq 0$. The trajectory therefore never reaches the sphere $\alpha_1(\|x - x_c\|) = d$: up to the first time it did, v could only fall while above η, so it would be at most $\max\{v(t_0), \eta\} < d$ (lemma 28), while on the sphere $v \geq d$.

Bound W_3 below by a class $\mathcal{K}$ α_3 (theorem 276). Above η, $\dot{v} \leq -\alpha_3(\|x - x_c\|) \leq -\alpha_3(\alpha_2^{-1}(v))$, so theorem 281 gives $v(t) \leq \max\{\sigma(\alpha_2(\|x(t_0) - x_c\|), t - t_0), \eta\}$. Apply the increasing α_1^{-1} and set $\beta(\rho, s) = \alpha_1^{-1}(\sigma(\alpha_2(\rho), s))$. □

Corollary 336 (Uniform boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 335, the trajectories are uniformly bounded around x_c (definition 294).*

Proof. From the c-ball, start within $a < c$: the estimate gives $\|x(t) - x_c\| \le \max\{\beta(a,0), \alpha_1^{-1}(\alpha_2(\mu))\}$, as β is increasing in its first argument and decreasing in its second. □

Corollary 337 (Uniform ultimate boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 335 with $\mu > 0$, if the system is forward complete from the c-ball about x_c at every $t_0 \ge 0$ (definition 91), its trajectories are uniformly ultimately bounded around x_c with ultimate bound $\alpha_1^{-1}(\alpha_2(\mu))$ (definition 296).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.18, estimate (4.43).*

Proof. The ultimate bound $b = \alpha_1^{-1}(\alpha_2(\mu))$ is positive, and $\beta(a,s) \to 0$, so $\beta(a,s) \le b$ for $s \ge T(a)$. Restrict a trajectory on $[t_0, \infty)$ to $[t_0, t]$ and apply the estimate. □

Theorem 338 (Ultimate boundedness theorem, global case)**.** *Let the state space be $\mathbb{R}^n$, V differentiable, α_1, α_2 class $\mathcal{K}_\infty$, W_3 positive definite about x_c on $\mathbb{R}^n$ (definition 29) and $\mu \ge 0$, with*

$$\alpha_1(\|x - x_c\|) \le V(t,x) \le \alpha_2(\|x - x_c\|), \qquad \dot V(t,x) \le -W_3(x) \quad \text{for } \mu \le \|x - x_c\|$$

for all $t \ge 0$ and $x \in \mathbb{R}^n$. Then there is a global class $\mathcal{KL}$ function β (definition 252), depending only on α_1, α_2 and W_3 — not on the system, V or μ — such that every trajectory on a segment $[t_0, t_1]$, $t_0 \ge 0$, satisfies

$$\|x(t) - x_c\| \le \max\{\beta(\|x(t_0) - x_c\|, t - t_0),\ \alpha_1^{-1}(\alpha_2(\mu))\} \qquad \forall\, t \in [t_0, t_1].$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.18, global case. Khalil's proof chooses one class $\mathcal{K}$ lower bound for W_3 on all of $\mathbb{R}^n$, which need not exist: W_3 may tend to 0 at infinity.*

Proof. For each $k \in \mathbb{N}$, apply theorem 335 with initial radius $c_k = k + 1$, on a ball large enough that $\alpha_2(c_k) < \alpha_1(R_k)$. If $\mu \ge \|x(t_0) - x_c\|$, the bound of β_k at an index with $c_k > \mu$ is at most $\alpha_1^{-1}(\alpha_2(\mu))$, so the trajectory never leaves the ultimate bound. It gives a class $\mathcal{KL}$ β_k with $\beta_k(\rho, s) \le \alpha_1^{-1}(\alpha_2(\rho))$. The sums $U(r,s) = \sum_{k \le \lceil r \rceil} \beta_k(k,s)$ are positive, decreasing to 0 in s, and increasing in r, and a trajectory from the r-ball that is outside the ultimate bound at time t is within $\beta_{\lceil r \rceil}(r, t - t_0) \le U(r+1, t - t_0)$ and within $\alpha(r) = \alpha_1^{-1}(\alpha_2(r))$ of x_c. **??** turns the class $\mathcal{K}_\infty$ bound α and the family U into a global class $\mathcal{KL}$ β above $\min\{\alpha(r), \sqrt{\alpha(r)\, U(r+1, s)}\}$. □

Corollary 339 (Global uniform boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 338, the trajectories are globally uniformly bounded around x_c (definition 295).*

Proof. Starting within a, the estimate gives $\|x(t) - x_c\| \le \max\{\beta(a,0), \alpha_1^{-1}(\alpha_2(\mu))\}$. □

Corollary 340 (Global uniform ultimate boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 338 with $\mu > 0$, if the system is forward complete from every state at every $t_0 \ge 0$ (definition 91), its trajectories are globally uniformly ultimately bounded around x_c with ultimate bound $\alpha_1^{-1}(\alpha_2(\mu))$ (definition 297).*

Proof. The ultimate bound b is positive and $\beta(a,s) \to 0$, so $\beta(a,s) \le b$ for $s \ge T(a)$; apply the estimate on $[t_0, t]$. □

7.8 Input-to-state stability

Input-to-state stability bounds the state by a decaying function of the initial condition plus a gain of the size of the input. An ISS-Lyapunov function certifies it through the global ultimate boundedness theorem, applied once per input: its class $\mathcal{KL}$ function does not depend on the input.

Definition 341 (Input-to-state stability)**.** A continuous-time system with input is *input-to-state stable* about x_{eq} when there are a global class $\mathcal{KL}$ function β (definition 252) and a class $\mathcal{K}_\infty$ function γ (definition 222) such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, under any input u satisfies

$$\|x(t) - x_{\text{eq}}\| \leq \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0) + \gamma(M) \qquad \forall\, t \in [t_0, t_1]$$

whenever $\|u(\tau)\| \leq M$ for all $\tau \in [t_0, t]$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.7. Differences:

- Any bound M on the input replaces its supremum, which is equivalent.
- γ is class $\mathcal{K}_\infty$, as in Sontag, rather than class $\mathcal{K}$; a class $\mathcal{K}$ gain γ is dominated by $\gamma + \text{id}$.
- Solutions are not required to exist for all time.

Theorem 342 (Input-to-state stability implies global uniform asymptotic stability)**.** *If s is input-to-state stable about x_{eq} (definition 341), then x_{eq} is a globally uniformly asymptotically stable equilibrium of the unforced system $\dot{x} = F(t, x, 0)$ (definition 291).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Section 4.9.*

Proof. With $u \equiv 0$ and $M = 0$, $\gamma(M) = 0$ and the estimate is $\|x(t) - x_{\text{eq}}\| \leq \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0)$. The bound $\beta(r, s) + re^{-s}$ is again global class $\mathcal{KL}$, now also radially unbounded, and lemma 316 concludes. □

Theorem 343 (Input-to-state estimate from an ISS-Lyapunov function)**.** *Let the state space be $\mathbb{R}^n$, V differentiable, α_1, α_2 class $\mathcal{K}_\infty$, W_3 positive definite about x_c on $\mathbb{R}^n$ (definition 29), and $\rho : [0, \infty) \to [0, \infty)$ nondecreasing, with*

$$\alpha_1(\|x - x_c\|) \leq V(t, x) \leq \alpha_2(\|x - x_c\|), \qquad \dot{V}(t, x, u) \leq -W_3(x) \quad \text{for } \|x - x_c\| \geq \rho(\|u\|)$$

for all $t \geq 0$, x and u. Then there is a global class $\mathcal{KL}$ function β such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, satisfies

$$\|x(t) - x_c\| \leq \beta(\|x(t_0) - x_c\|, t - t_0) + \alpha_1^{-1}(\alpha_2(\rho(M)))$$

for $t \in [t_0, t_1]$, whenever $\|u(\tau)\| \leq M$ for all $\tau \in [t_0, t]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.19, whose ρ is class $\mathcal{K}$; only its monotonicity is used.*

Proof. Fix the input and t. Freeze the input, clamped to $[t_0, t]$, into the system: $\dot{x} = F(\tau, x, u(\tau))$ is an unforced system with the same trajectories on $[t_0, t]$, and since $\|u(\tau)\| \leq M$ and ρ is nondecreasing, $\dot{V} \leq -W_3$ wherever $\|x - x_c\| \geq \rho(M)$. Apply theorem 338 with $\mu = \rho(M)$; its β does not depend on the system or on μ, so it is the same for every input. Finally $\max\{a, b\} \leq a + b$ for $a, b \geq 0$. □

Theorem 344 (ISS-Lyapunov theorem)**.** *Under the hypotheses of theorem 343 with ρ class $\mathcal{K}_\infty$, the system is input-to-state stable about x_c (definition 341) with gain $\gamma = \alpha_1^{-1} \circ \alpha_2 \circ \rho$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.19, with ρ of class $\mathcal{K}$; a class $\mathcal{K}$ ρ can be replaced by $\rho + \mathrm{id}$, which only weakens the hypothesis.*

Proof. theorem 343, with $\alpha_1^{-1} \circ \alpha_2 \circ \rho$ class $\mathcal{K}_\infty$ (definition 231, definition 232). □

7.9 LaSalle's invariance principle

When $\nabla V \cdot f \leq 0$ only, V may stall, and Lyapunov's theorems give stability but not convergence. LaSalle's principle recovers convergence for a time-invariant system: a trajectory confined to a compact set converges to its ω-limit set, on which V is constant, and forward invariance of that set turns constancy of V into $\nabla V \cdot f = 0$.

Lemma 345 (The ω-limit set is forward invariant)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, and let φ be a trajectory on $[t_0, \infty)$. Then its ω-limit set $\omega(\varphi)$ (definition 70) is forward invariant (definition 114).*

Neither boundedness of φ nor compactness of an enclosing set is needed; those hypotheses of Khalil's Lemma 4.1 are what make $\omega(\varphi)$ nonempty and compact, not what make it invariant.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.1.*

Proof. Let ψ be a trajectory on $[a, b]$ with $p = \psi(a) \in \omega(\varphi)$, and fix $t \in [a, b]$. Given $\varepsilon > 0$, put $C = e^{K(b-a)}$ and pick, by the definition of the ω-limit set, arbitrarily large r with $\|\varphi(r) - p\| < \varepsilon/C$. The time-shifted curve $u(\cdot) = \varphi(\cdot + r - a)$ solves the same time-invariant equation on $[a, b]$ with $u(a) = \varphi(r)$, so Grönwall gives

$$\|\varphi(r + t - a) - \psi(t)\| = \|u(t) - \psi(t)\| \leq \|\varphi(r) - p\|\, e^{K(t-a)} < \varepsilon.$$

Since r may be taken arbitrarily large, so may $r + t - a$; hence φ is frequently within ε of $\psi(t)$, i.e. $\psi(t) \in \omega(\varphi)$.

Khalil argues instead with the flow map and the semigroup identity, which presumes a globally defined flow; comparing the two solutions directly needs only uniqueness, in its quantitative Grönwall form. □

Lemma 346 ($\dot{V}$ vanishes on the ω-limit set)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, Ω compact and forward invariant, $V \in C^1$ with $\nabla V \cdot f \leq 0$ on Ω, and φ a trajectory on $[t_0, \infty)$ with $\varphi(t_0) \in \Omega$. Then $\nabla V(y) \cdot f(y) = 0$ for every $y \in \omega(\varphi)$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), proof of Theorem 4.4.*

Proof. $V(\varphi(t))$ is nonincreasing and bounded below on the compact Ω, so it converges to some L, and $V \equiv L$ on $\omega(\varphi)$ (lemma 73). Through $y \in \omega(\varphi)$ take a solution ψ on $[0, 1]$ (Picard–Lindelöf, theorem 53); it stays in $\omega(\varphi)$ by lemma 345, so $V \circ \psi \equiv L$ and its derivative at 0, which is $\nabla V(y) \cdot f(y)$, vanishes. □

Theorem 347 (LaSalle's invariance principle)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, Ω compact and forward invariant (definition 114), and $V \in C^1$ with $\nabla V \cdot f \leq 0$ on Ω. Put*

$$E = \{x \in \Omega : \nabla V(x) \cdot f(x) = 0\},$$

and let M contain every forward invariant subset of E. Then every trajectory φ on $[t_0, \infty)$ with $\varphi(t_0) \in \Omega$ satisfies $\varphi(t) \to M$ as $t \to \infty$.

Khalil takes M to be the largest invariant *subset of E. Here M must absorb every* forward *invariant subset, a larger family, so the hypothesis on M is correspondingly stronger: backward invariance of $\omega(\varphi)$ would need solutions through its points to extend backwards in time.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.4.*

Proof. The ω-limit set lies in Ω, which is closed and contains the trajectory (lemma 74), and $\nabla V \cdot f$ vanishes on it (lemma 346), so $\omega(\varphi) \subseteq E$. It is forward invariant (lemma 345), so $\omega(\varphi) \subseteq M$. A trajectory confined to a compact set converges to every set containing its ω-limit set (lemma 75). □

Theorem 348 (Barbashin's theorem)**.** *Let $\dot{x} = f(x)$ on $\mathbb{R}^n$ with f Lipschitz, and $V \in C^1$ positive definite on a neighbourhood D of x_{eq} (definition 29) with $\nabla V \cdot f \leq 0$ on D. Suppose that for some $c > 0$ the sublevel set $\Omega_c = \{V \leq c\}$ is contained in D, compact and forward invariant, and that $\{x_{\text{eq}}\}$ contains every forward invariant subset of $\{x \in \Omega_c : \nabla V(x) \cdot f(x) = 0\}$. Then x_{eq} is locally asymptotically stable (definition 287).*

The hypothesis on E is the one a user of LaSalle discharges: solve $\nabla V \cdot f = 0$ and check that no solution can stay there but the equilibrium.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.1.*

Proof. Stability: V is a uniform Lyapunov function (lemma 324, theorem 327). Forward completeness: f is Lipschitz (theorem 95). Convergence: near x_{eq}, $V < c$, so a trajectory starting there starts in Ω_c, and theorem 347 with $M = \{x_{\text{eq}}\}$ applies. □

Theorem 349 (Krasovskii's theorem)**.** *Let $\dot{x} = f(x)$ on $\mathbb{R}^n$ with f Lipschitz, and $V \in C^1$ positive definite on $\mathbb{R}^n$, with compact sublevel sets (equivalently, radially unbounded) and $\nabla V \cdot f \leq 0$ everywhere. Suppose that for every c, $\{x_{\text{eq}}\}$ contains every forward invariant subset of $\{x : V(x) \leq c,\ \nabla V(x) \cdot f(x) = 0\}$. Then x_{eq} is globally asymptotically stable (definition 288).*

Unlike Barbashin's theorem (theorem 348) no forward invariant set is assumed: every sublevel set is compact, and $\nabla V \cdot f \leq 0$ keeps a trajectory in the one through its initial state.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.2.*

Proof. Stability and forward completeness as in Barbashin's theorem. The sublevel set $\{V \leq V(x(t_0))\}$ is compact and, since V does not increase along trajectories, forward invariant; apply theorem 347 with $M = \{x_{\text{eq}}\}$. □

7.10 Instability

Instability is the failure of stability, and the simplest witness is a trajectory that leaves a fixed neighbourhood from arbitrarily close initial states (lemma 298). Chetaev's theorem produces such trajectories from a function that grows along the system near the equilibrium.

The geometric criterion uses a positive certificate with positive derivative in an open region and vanishing on its boundary, with these conditions local to a closed ball.

Definition 350 (Chetaev function)**.** Let s be an input-free continuous-time system. A Chetaev certificate consists of an open region D, a point $x_{\text{eq}} \in \partial D$, a radius $\rho > 0$, and a globally C^1 function V. For $\|x - x_{\text{eq}}\| \leq \rho$, $V(x) > 0$ in D and $V(x) = 0$ on ∂D; in D the derivative $\nabla V(x) \cdot F(t, x, ())$ is strictly positive for every $t \geq 0$.

Reference: Khalil, *Nonlinear Systems*, 3rd ed., Theorem 4.3, p. 125.

Theorem 351 (Geometric Chetaev instability criterion)**.** *For an input-free system $\dot{x} = f(x)$ with f globally C^1, a Chetaev certificate at x_{eq} (definition 350) implies instability (definition 284).*

Reference: Khalil, Nonlinear Systems*, 3rd ed., Theorem 4.3, p. 125.*

Proof. Choose a positive seed near the boundary point and continue a cutoff curve. Boundary vanishing retains it in the region. The Lie derivative has a positive minimum on its compact superlevel, forcing growth beyond the ball's certificate ceiling. □

The exponential criterion is its corollary on the full positive set $D = \{V > 0\}$.

Theorem 352 (Exponential Chetaev instability criterion)**.** *Suppose the vector field f and certificate V are C^1, V is quadratically bounded on a ball, its positive set accumulates at the base point, and its Lie derivative satisfies $\dot{V} \geq 2\alpha V$ there for some $\alpha > 0$. Then the base point is unstable (definition 284) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems*, 3rd ed. (Prentice Hall, 2002), Theorem 4.3, p. 125, for the method; this quantitative variant is not its statement.*

Proof. Take $D = \{V > 0\}$. The quadratic bound gives $V(x_{\text{eq}}) = 0$; the nearby positive seeds put x_{eq} in its closure. Continuity makes $V = 0$ on ∂D, and $\dot{V} \geq 2\alpha V > 0$ in D within the ball. Apply theorem 351. □

Chapter 8

Stability of discrete-time systems

The notions of the previous chapter for an autonomous discrete-time system $x^+ = f(k, x)$ (definition 121), under the same names. Discrete time is simpler in one respect: trajectories always exist and are unique (lemma 136, lemma 137), so every notion quantifies over trajectories on $[k_0, \infty)$ and no forward completeness is needed anywhere. Exponential estimates are geometric, ρ^{k-k_0} with $0 \leq \rho < 1$, in place of $e^{-\gamma(t-t_0)}$.

8.1 Definitions

Definition 353 (Stability of a discrete-time system)**.** The equilibrium x_{eq} of an autonomous discrete-time system $x^+ = f(k, x)$ is *stable* when for every $\varepsilon > 0$ and $k_0 \in \mathbb{N}$ there is $\delta = \delta(\varepsilon, k_0) > 0$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < \delta$ satisfies $\|x(k) - x_{\text{eq}}\| < \varepsilon$ for all $k \geq k_0$.

The counterpart of definition 283. Trajectories on rays suffice: a discrete-time trajectory exists from every state for all later times (lemma 136).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4; Jungers and van de Wouw (2026), Definition 2.

Definition 354 (Instability of a discrete-time system)**.** The equilibrium is *unstable* when it is not stable (definition 353).

Definition 355 (Uniform stability of a discrete-time system)**.** The equilibrium is *uniformly stable* when δ in definition 353 can be chosen independently of k_0.

Definition 356 (Global uniform stability of a discrete-time system)**.** The equilibrium is *globally uniformly stable* when it is uniformly stable (definition 355) with a choice of $\delta(\varepsilon)$ satisfying $\delta(\varepsilon) \to \infty$ as $\varepsilon \to \infty$.

Definition 357 (Local asymptotic stability of a discrete-time system)**.** The equilibrium is *locally asymptotically stable* when it is stable (definition 353) and for every k_0 there is $c = c(k_0) > 0$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < c$ satisfies $x(k) \to x_{\text{eq}}$. Unlike definition 287 no forward completeness is asked for: discrete-time trajectories always exist.

Definition 358 (Global asymptotic stability of a discrete-time system)**.** The equilibrium is *globally asymptotically stable* when it is stable (definition 353) and every trajectory on $[k_0, \infty)$ satisfies $x(k) \to x_{\text{eq}}$.

Definition 359 (Uniform convergence time, discrete time)**.** Trajectories starting within c of x_{eq} have a *uniform convergence time*: for every $\eta > 0$ there is $T = T(\eta) \in \mathbb{N}$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < c$ satisfies $\|x(k) - x_{\text{eq}}\| < \eta$ for $k \geq k_0 + T$.

Definition 360 (Uniform asymptotic stability of a discrete-time system)**.** The equilibrium is *uniformly asymptotically stable* when it is uniformly stable (definition 355) and there is $c > 0$ from which convergence has a uniform convergence time (definition 359).

Definition 361 (Global uniform asymptotic stability of a discrete-time system)**.** The equilibrium is *globally uniformly asymptotically stable* when it is globally uniformly stable (definition 356) and every $c > 0$ admits a uniform convergence time (definition 359).

Definition 362 (Local exponential stability of a discrete-time system)**.** The equilibrium is *locally exponentially stable*, or *geometrically stable*, when there are $c, C > 0$ and $0 \leq \rho < 1$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < c$ satisfies

$$\|x(k) - x_{\text{eq}}\| \leq C \, \|x(k_0) - x_{\text{eq}}\| \, \rho^{k-k_0} \qquad \forall \, k \geq k_0.$$

The counterpart of definition 292, with $\rho = e^{-\gamma}$.

Reference: Jungers and van de Wouw (2026), Definition 2, estimate (6).

Definition 363 (Global exponential stability of a discrete-time system)**.** The equilibrium is *globally exponentially stable* when the estimate of definition 362 holds for every initial state, with no radius c.

Boundedness is a property of the system near a point that need not be an equilibrium: trajectories stay, or end up and stay, within a prescribed distance of it.

Definition 364 (Uniform boundedness of a discrete-time system)**.** Trajectories are *uniformly bounded* around x_c when there is $c > 0$ such that for every $0 < a < c$ there is $\beta = \beta(a) > 0$ with

$$\|x(k_0) - x_c\| \leq a \;\Rightarrow\; \|x(k) - x_c\| \leq \beta \qquad \forall \, k \geq k_0$$

for every trajectory on $[k_0, \infty)$. The counterpart of definition 294.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 365 (Global uniform boundedness of a discrete-time system)**.** Trajectories are *globally uniformly bounded* around x_c when the bound of definition 364 holds for every $a > 0$, with no radius c.

Definition 366 (Uniform ultimate boundedness of a discrete-time system)**.** Trajectories are *uniformly ultimately bounded* with ultimate bound b around x_c when there is $c > 0$ such that for every $0 < a < c$ there is $T = T(a, b) \in \mathbb{N}$ with

$$\|x(k_0) - x_c\| \leq a \;\Rightarrow\; \|x(k) - x_c\| \leq b \qquad \forall \, k \geq k_0 + T$$

for every trajectory on $[k_0, \infty)$. The point x_c need not be an equilibrium.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6; Jungers and van de Wouw (2026), Definition 3.

Definition 367 (Global uniform ultimate boundedness of a discrete-time system)**.** Trajectories are *globally uniformly ultimately bounded* with ultimate bound b around x_c when definition 366 holds for every $a > 0$, with no radius c.

8.2 Implications

Lemma 368 (A stable equilibrium is a fixed point)**.** *If x_{eq} is stable (definition 353), then $f(k, x_{\text{eq}}) = x_{\text{eq}}$ for every k.*

Proof. The trajectory from x_{eq} at time k stays within every $\varepsilon > 0$ of x_{eq}, so its value at $k+1$, which is $f(k, x_{\text{eq}})$, is at distance 0. □

Lemma 369 (Instability from a fixed escape radius, discrete time)**.** *Fix $\varepsilon > 0$ and k_0. If for every $\delta > 0$ some trajectory on $[k_0, \infty)$ starts within δ of x_{eq} and reaches distance at least ε from it, then x_{eq} is unstable (definition 354).*

Proof. □

Lemma 370 (Uniform stability implies stability, discrete time)**.** *A uniformly stable equilibrium (definition 355) is stable (definition 353).*

Proof. □

Lemma 371 (Global asymptotic stability implies local, discrete time)**.** *A globally asymptotically stable equilibrium (definition 358) is locally asymptotically stable (definition 357).*

Proof. □

Lemma 372 (Uniform asymptotic stability implies local, discrete time)**.** *A uniformly asymptotically stable equilibrium (definition 360) is locally asymptotically stable (definition 357). Unlike lemma 301, no forward completeness is needed.*

Proof. □

Lemma 373 (Global uniform asymptotic stability implies global, discrete time)**.** *A globally uniformly asymptotically stable equilibrium (definition 361) is globally asymptotically stable (definition 358).*

Proof. □

Lemma 374 (Exponential stability implies uniform stability, discrete time)**.** *A locally exponentially stable equilibrium (definition 362) is uniformly stable (definition 355).*

Proof. The estimate bounds the distance by C times the initial distance, so $\delta = \min(c, \varepsilon/C)$ works for every k_0. □

Lemma 375 (Exponential stability implies uniform asymptotic stability, discrete time)**.** *A locally exponentially stable equilibrium (definition 362) is uniformly asymptotically stable (definition 360).*

Proof. Uniform stability is lemma 374. From the ball of radius c the distance after T steps is below $Cc\rho^T$, which is below any $\eta > 0$ for T large, independently of k_0. □

Lemma 376 (Global exponential stability implies local, discrete time)**.** *A globally exponentially stable equilibrium (definition 363) is locally exponentially stable (definition 362).*

Proof. □

Lemma 377 (Global exponential stability implies global uniform, discrete time)**.** *A globally exponentially stable equilibrium (definition 363) is globally uniformly asymptotically stable (definition 361).*

Proof. $\delta(\varepsilon) = \varepsilon/C$ witnesses global uniform stability, and from every ball the estimate gives a uniform convergence time as in lemma 375. □

8.3 Lyapunov's direct method

For a time-invariant system $x^+ = f(x)$ the decrease of a Lyapunov function along the system is the difference $\Delta V(x) = V(f(x)) - V(x)$. Unlike continuous time, the theorems need continuity of f for a reason of their own: a trajectory can jump over a sphere in one step, so the first-exit argument of continuous time is replaced by keeping one step from a small ball inside the region where V is controlled.

Theorem 378 (Lyapunov's stability theorem, discrete time)**.** *Let E be proper, and let the system be $x^+ = f(x)$ with $f(x_{\mathrm{eq}}) = x_{\mathrm{eq}}$ and f continuous at x_{eq}. Let $\mathcal{D}$ be a neighbourhood of x_{eq} and V positive definite on $\mathcal{D}$ about x_{eq} (definition 29), with*

$$\Delta V(x) = V(f(x)) - V(x) \leq 0 \qquad \forall\, x \in \mathcal{D} \quad \{x_{\mathrm{eq}}\}.$$

Then x_{eq} is uniformly stable (definition 355).

Continuity of f at x_{eq} is not among the hypotheses of the reference, but it cannot be dropped: on $\mathbb{R}$, $V(x) = x^2/(1+x^4)$ and $f(x) = 1/x^2$ for $0 < |x| < 1$ ($f(x) = x$ otherwise) satisfy everything else, yet $x_0 = 10^{-3}$ jumps to $x_1 = 10^6$.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 1.

Proof. Given ε, shrink it so that the closed ball of radius ε' lies in $\mathcal{D}$, and choose $s \leq \varepsilon'$ so that f maps the ball of radius s into the ball of radius ε'. On the compact annulus $s \leq \|x - x_{\mathrm{eq}}\| \leq \varepsilon'$, V has a positive minimum m; choose $\delta < s$ with $V < m$ on the ball of radius δ. By induction, a trajectory with $\|x(k_0) - x_{\mathrm{eq}}\| < \delta$ keeps $\|x(k) - x_{\mathrm{eq}}\| < s$ and $V(x(k)) < m$: the next state is within ε' by the choice of s, has $V(x(k+1)) \leq V(x(k)) < m$, and so cannot lie in the annulus. Nothing depends on k_0, since a trajectory from k_0 is the iterate sequence of $x(k_0)$ (lemma 138). □

Theorem 379 (Lyapunov's theorem for local asymptotic stability, discrete time)**.** *Let E be proper, and let the system be $x^+ = f(x)$ with $f(x_{\mathrm{eq}}) = x_{\mathrm{eq}}$. Let $\mathcal{D}$ be a neighbourhood of x_{eq} on which f is continuous, and V positive definite on $\mathcal{D}$ about x_{eq} (definition 29), with*

$$\Delta V(x) = V(f(x)) - V(x) < 0 \qquad \forall\, x \in \mathcal{D} \quad \{x_{\mathrm{eq}}\}.$$

Then x_{eq} is locally asymptotically stable (definition 357).

Continuity of f on $\mathcal{D}$ is not among the hypotheses of the reference, but it cannot be dropped. With $V(x) = |x|$ on $\mathbb{R}$, let f be odd with $f(0) = 0$ and $f(x) = (2^{-n-1} + x)/2$ on each interval $(2^{-n-1}, 2^{-n}]$: then $\Delta V < 0$ away from 0, but a trajectory starting in $(2^{-n-1}, 2^{-n}]$ converges to 2^{-n-1}, at every scale.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 2.

Proof. Stability is theorem 378. For attractivity, choose a closed ball $K \subseteq \mathcal{D}$ around x_{eq} and, by stability, a δ whose trajectories stay in K. Along such a trajectory $V(x(k))$ is nonincreasing and bounded below, so it converges to some L. A cluster point y of the trajectory, which exists in the compact set K, has $V(y) = L$; by continuity of f at y, $x(k+1) = f(x(k))$ clusters at $f(y)$, so $V(f(y)) = L$ too. Then $\Delta V(y) = 0$, so $y = x_{\text{eq}}$. A sequence in a compact set whose only cluster point is x_{eq} converges to it. □

Theorem 380 (Lyapunov's theorem for global asymptotic stability, discrete time)**.** *Let E be proper, and let the system be $x^+ = f(x)$ with f continuous and $f(x_{\text{eq}}) = x_{\text{eq}}$. Let V be positive definite on E about x_{eq} (definition 29), with compact sublevel sets $\{V \leq c\}$ (equivalently, for continuous V, radially unbounded: lemma 329), and*

$$\Delta V(x) = V(f(x)) - V(x) < 0 \qquad \forall\, x \neq x_{\text{eq}}.$$

Then x_{eq} is globally asymptotically stable (definition 358).

Continuity of f is not among the hypotheses of the reference but cannot be dropped: with $V(x) = |x|$ on $\mathbb{R}$, the map $f(x) = x/2$ for $|x| \leq 1$, $f(x) = \operatorname{sign}(x)(1+|x|)/2$ otherwise, has $\Delta V < 0$ away from 0, yet the trajectory from 2 converges to 1.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 3.

Proof. Stability is theorem 378 with $\mathcal{D} = E$. From any initial state, $V(x(k)) \leq V(x(k_0))$, so the trajectory stays in the compact sublevel set $\{V \leq V(x(k_0))\}$, and the ω-limit argument of theorem 379 applies there. □

Theorem 381 (Lyapunov's theorem for local exponential stability, discrete time)**.** *Let the system be $x^+ = f(x)$ with $f(x_{\text{eq}}) = x_{\text{eq}}$ and f continuous at x_{eq}, and let $\mathcal{D}$ be a neighbourhood of x_{eq}. Let $0 < c_1 \leq c_2$, $p > 0$ and $0 < c_3 < 1$, and let $V : E \to \mathbb{R}$ satisfy, for every $x \in \mathcal{D}$,*

$$c_1\|x - x_{\text{eq}}\|^p \leq V(x) \leq c_2\|x - x_{\text{eq}}\|^p, \qquad \Delta V(x) \leq -c_3 V(x).$$

Then x_{eq} is locally exponentially stable (definition 362).

Continuity of f at x_{eq} is not among the hypotheses of the reference, but without it a trajectory can leave $\mathcal{D}$ in a single step.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 4.

Proof. Choose $r_0 > 0$ so that the ball $B(x_{\text{eq}}, r_0)$ lies in $\mathcal{D}$ and f maps it into $\mathcal{D}$, and let $C = (c_2/c_1)^{1/p} \geq 1$ and $\delta = r_0/(2C)$. By induction, a trajectory starting within δ stays in $B(x_{\text{eq}}, r_0)$ and satisfies $V(x(k)) \leq (1-c_3)^{k-k_0} V(x(k_0))$: the decrease applies at $x(k) \in \mathcal{D}$, and $x(k+1) = f(x(k)) \in \mathcal{D}$, so the sandwich gives $\|x(k+1) - x_{\text{eq}}\| \leq C\delta = r_0/2$. The geometric bound with $\rho = (1-c_3)^{1/p}$ then follows as in theorem 382. □

Theorem 382 (Lyapunov's theorem for global exponential stability, discrete time)**.** *Let the system be $x^+ = f(x)$, let $0 < c_1 \leq c_2$, $p > 0$ and $0 < c_3 < 1$, and let $V : E \to \mathbb{R}$ satisfy, for every x,*

$$c_1\|x - x_{\text{eq}}\|^p \leq V(x) \leq c_2\|x - x_{\text{eq}}\|^p, \qquad \Delta V(x) = V(f(x)) - V(x) \leq -c_3 V(x).$$

Then x_{eq} is globally exponentially stable (definition 363), with $C = (c_2/c_1)^{1/p}$ and $\rho = (1-c_3)^{1/p}$. No continuity of f is needed.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 4, on the whole space. They ask for $p \geq 1$; $p > 0$ suffices.

Proof. The decrease condition is $V(f(x)) \leq (1-c_3)V(x)$, so by induction $V(x(k)) \leq (1-c_3)^{k-k_0}V(x(k_0))$. Sandwiching,

$$c_1\|x(k) - x_{\text{eq}}\|^p \leq (1-c_3)^{k-k_0}c_2\|x(k_0) - x_{\text{eq}}\|^p,$$

and taking p-th roots gives the geometric bound. □

Chapter 9

Stability of linear systems

The stability notions of a linear system are those of definition 283 and the definitions after it, applied to the unforced system (definition 86): there is no separate notion of stability for linear systems. What linearity adds is that each notion becomes a condition on the state transition matrix, and for constant A a condition on its eigenvalues.

9.1 Linear time-varying systems

Hespanha states Lyapunov stability for the continuous-time linear time-varying system $\dot{x} = A(t)x$ as four conditions on its homogeneous state response $x(t) = \Phi(t, t_0)x_0$. None of them is taken as a definition here: each is proved *equivalent* to the corresponding stability predicate of the unforced system, so the linear track and the nonlinear track share one notion of stability rather than two.

Two things stand between the textbook conditions and the predicates, and both are settled in this section. The predicates quantify over trajectories where the textbook quantifies over initial conditions, which the trajectory characterization below closes in both directions; and the predicates bound the response from a *ball* where the textbook bounds it from every initial state, which homogeneity closes. The second is why a linear system has no nontrivial region of attraction, recorded as lemma 389.

9.1.1 Hespanha's Definition 8.1

Lemma 383 (Stability as a bound on the state transition matrix)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\,x,$$

*with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (**??**).*

The origin is a stable equilibrium (definition 283) of this system if and only if for every $t_0 \geq 0$ there is C with

$$\|\Phi(t, t_0)\,x\| \leq C\,\|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1).*

Proof. ($\Rightarrow$) Apply stability at $\varepsilon = 1$ to get $\delta(1, t_0)$. Every response from the ball of radius δ stays inside the unit ball, and scaling turns that into the bound with $C = 2/\delta$.

($\Leftarrow$) Given ε, take $\delta = \varepsilon/(C+1)$. A trajectory is its own response (theorem 180), so $\|\varphi(t)\| \leq C\|\varphi(t_0)\| < \varepsilon$. □

Theorem 384 (Lyapunov stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

*with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (**??**).*

The origin is a stable equilibrium (definition 283) of this system if and only if, for every $t_0 \geq 0$ and every initial condition x_0, the homogeneous state response $x(t) = \Phi(t, t_0)\, x_0$ is bounded on $[t_0, \infty)$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1).*

Proof. The state-transition form is lemma 383. A bound on Φ gives the response bound $C\|x_0\|$; conversely a family of operators bounded at every point of a Banach space is norm-bounded, by the uniform boundedness principle. □

Theorem 385 (Uniform stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

*with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (**??**).*

The origin is a uniformly stable equilibrium (definition 285) of this system if and only if there is a single C with

$$\|\Phi(t, t_0)\, x\| \leq C\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1), with the margin independent of t_0.*

Proof. Identical to lemma 383, with the margin δ and the bound C both carried outside the quantifier on t_0. □

Theorem 386 (Asymptotic stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

*with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (**??**).*

The origin is an asymptotically stable equilibrium (definition 287) of this system if and only if, for every $t_0 \geq 0$ and every x_0, the homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is bounded on $[t_0, \infty)$ and tends to 0 as $t \to \infty$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(2).*

Proof. The stability conjunct is theorem 384. For attractivity: an arbitrary x_0 is rescaled into the attractivity ball, and the limit scales back, since the response is linear in the initial state; conversely every trajectory is a response (theorem 180), so convergence of the responses is convergence of the trajectories, from any radius. □

Theorem 387 (Global exponential stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

*with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (**??**).*

The origin is a globally exponentially stable equilibrium (definition 293) of this system if and only if there are $k, \gamma > 0$ with

$$\|\Phi(t, t_0)\, x\| \leq k\, e^{-\gamma(t-t_0)}\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(3).*

Proof. Both directions are the trajectory characterization plus homogeneity: the exponential estimate along every trajectory is the estimate applied to the initial state, and conversely the estimate on any ball scales to the estimate everywhere. □

Theorem 388 (Exponential stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

*with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (**??**).*

The origin is an exponentially stable equilibrium (definition 292) of this system if and only if there are $k, \gamma > 0$ with

$$\|\Phi(t, t_0)\, x\| \leq k\, e^{-\gamma(t-t_0)}\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

The condition is the one in theorem 387: the radius in the local definition carries no information here.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(3).*

Proof. The radius c of the local definition is absorbed by the scaling argument; conversely the estimate is global exponential stability (theorem 387), which is local. □

Lemma 389 (Local exponential stability is global, for a linear system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

*with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (**??**).*

The origin is an exponentially stable equilibrium (definition 292) of this system if and only if it is globally exponentially stable (definition 293).

Proof. Both are equivalent to the same estimate, theorem 388 and theorem 387. □

For a linear system, uniform asymptotic stability already forces exponential decay of the state transition matrix: splitting time into windows of a uniform convergence time, each window contracts by a fixed factor.

Theorem 390 (Uniform asymptotic stability of a linear time-varying system is exponential)**.** *Consider the linear time-varying system $\dot{x} = A(t)\, x$ with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (**??**). The origin is uniformly asymptotically stable (definition 290) if and only if there are $k, \gamma > 0$ with*

$$\|\Phi(t, t_0)\, x\| \leq k\, e^{-\gamma(t-t_0)}\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.11.*

Proof. ($\Leftarrow$) The estimate is exponential stability (theorem 388), which is uniform asymptotic stability. ($\Rightarrow$) Uniform stability bounds $\|\Phi(t,t_0)\| \le C$ (theorem 385), and a uniform convergence time T from the c-ball to the $c/(2e)$-ball gives $\|\Phi(t,t_0)\| \le 1/e$ for $t \ge t_0 + T$, by homogeneity. Splitting $[t_0, t]$ into $N = \lfloor (t-t_0)/T \rfloor$ windows of length T and using $\Phi(t,t_0) = \Phi(t,t_0+T)\,\Phi(t_0+T,t_0)$ gives $\|\Phi(t,t_0)\| \le \max(C,1)\,e^{-N} \le e\max(C,1)\,e^{-(t-t_0)/T}$. □

Corollary 391 (Global uniform asymptotic stability of a linear time-varying system)**.** *Under the hypotheses of theorem 390, the origin is globally uniformly asymptotically stable (definition 291) if and only if $\Phi(t,t_0)$ decays exponentially, uniformly in $t_0 \ge 0$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.11.*

Proof. Global uniform asymptotic stability is uniform asymptotic stability, hence exponential decay (theorem 390); the decay is global exponential stability (theorem 387), which is global uniform asymptotic stability. □

The time-varying form of the Lyapunov equation.

Definition 392 (Differential Lyapunov equation)**.** For operator-valued functions A, P and Q of time on a real Hilbert space, P satisfies the *differential Lyapunov equation* for A with forcing Q when, for every $t \ge 0$, P is differentiable at t and

$$-\dot{P}(t) = P(t)A(t) + A(t)^* P(t) + Q(t).$$

For constant matrices it is the continuous Lyapunov equation (definition 416).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), equation (4.28).

Lemma 393 (Constant solutions of the differential Lyapunov equation)**.** *For constant A, P and Q, the differential Lyapunov equation (definition 392) holds if and only if the continuous Lyapunov equation (definition 416) does.*

Proof. The derivative of a constant is 0, and derivatives are unique, so the differential equation says $0 = -(PA + A^{\mathsf{T}}P + Q)$. □

Definition 394 (Integrand of the Lyapunov integral)**.** For a linear time-varying system with state transition operator Φ (definition 176) and an operator function Q,

$$\Phi(\tau,t)^* Q(\tau)\,\Phi(\tau,t).$$

Reference: Khalil, *Nonlinear Systems* (3rd ed.), proof of Theorem 4.12.

Definition 395 (Lyapunov integral)**.** For a linear time-varying system and an operator function Q,

$$P(t) = \int_t^\infty \Phi(\tau,t)^* Q(\tau)\,\Phi(\tau,t)\,d\tau$$

(definition 394).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), proof of Theorem 4.12.

Theorem 396 (Converse Lyapunov theorem for linear time-varying systems)**.** *Consider $\dot{x} = A(t)x$ with A continuous and $\|A(t)\| \le L$, and let the origin be exponentially stable (definition 292). Let $Q(t)$ be continuous, symmetric, with $\|Q(t)\| \le q$ and $x^{\mathsf{T}}Q(t)x \ge q_1\|x\|^2$, $q_1 > 0$. Then there is a bounded symmetric $P(t)$ satisfying the differential Lyapunov equation (definition 392), and $V(t,x) = x^{\mathsf{T}}P(t)x$ is an exponential Lyapunov function with exponent 2 (definition 323); in particular $c_1\|x\|^2 \le x^{\mathsf{T}}P(t)x \le c_2\|x\|^2$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.12. Differences: the state space carries the sup norm, as everywhere for linear time-varying systems here.*

Proof. Let $P(t) = \int_t^\infty \Phi(\tau,t)^{\mathsf{T}}Q(\tau)\Phi(\tau,t)\,d\tau$, written as $\Phi(0,t)^{\mathsf{T}}R(t)\Phi(0,t)$ with $R(t) = \int_t^\infty \Phi(\tau,0)^{\mathsf{T}}Q(\tau)\Phi(\tau,0)\,d\tau$, which converges by exponential decay of Φ; the same decay bounds $P(t)$. Then $\dot{R}(t) = -\Phi(t,0)^{\mathsf{T}}Q(t)\Phi(t,0)$ and $\partial_t\Phi(0,t) = -\Phi(0,t)A(t)$ (**??**), and the product rule gives $-\dot{P} = PA + A^{\mathsf{T}}P + Q$. Along a solution, $x^{\mathsf{T}}P(t)x = \int_t^\infty (\Phi(\tau,t)x)^{\mathsf{T}}Q(\tau)\Phi(\tau,t)x\,d\tau$: exponential decay bounds it above, the growth bound (**??**) bounds it below on $[t,t+1]$, and its derivative along the solution is $-x^{\mathsf{T}}Q(t)x$. □

Theorem 397 (Instability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\,x,$$

*with A continuous, and write $\Phi(t,t_0)$ for its state transition matrix (**??**).*

The origin is unstable (definition 284) for this system if and only if there are $t_0 \ge 0$ and x_0 whose homogeneous response $t \mapsto \Phi(t,t_0)x_0$ is unbounded on $[t_0,\infty)$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(4).*

Proof. The negation of theorem 384. □

9.2 Linear time-invariant systems

For constant A the state transition matrix is $e^{A(t-t_0)}$ (**??**), and the stability conditions of the previous section collapse to a condition on the eigenvalues of A alone. This is Hespanha's Theorem 8.1.

Clauses (1) and (4) of that theorem — marginal stability and instability — ask that the Jordan blocks at imaginary-axis eigenvalues be 1×1. They are not recorded here: they need a Jordan normal form, which is a project rather than a lemma.

9.2.1 The Hurwitz condition and its rate-indexed refinement

For a real square matrix A, Hurwitz stability is expressed through the complex eigenpairs of its entrywise complexification, over an abstract index type. The rate-indexed predicate records a strict spectral margin. Ordinary Hurwitz stability is the zero-rate case, but is stated separately rather than derived, so that it unfolds directly to $\mathrm{Re}(\mu) < 0$.

Definition 398 (Hurwitz matrix)**.** A real square matrix A is *Hurwitz* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $\mathrm{Re}(\mu) < 0$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1.

Definition 399 (Hurwitz with a prescribed decay rate)**.** A real square matrix A is *Hurwitz with decay rate α* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $\mathrm{Re}(\mu) < -\alpha$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1.

Theorem 400 (Rate-Hurwitz via a spectral shift)**.** *A real matrix A is Hurwitz with decay rate α if and only if the spectrally shifted matrix $A + \alpha I$ is Hurwitz.*

Reference: João P. Hespanha, Linear Systems Theory *(2nd ed.), continuous-time stability criterion. This spectral-shift corollary is proved directly from eigenpairs.*

Proof. A complex eigenvalue μ of A becomes $\mu + \alpha$ after the shift, and $\operatorname{Re}(\mu) < -\alpha$ is equivalent to $\operatorname{Re}(\mu + \alpha) < 0$. □

Theorem 401 (A Hurwitz matrix has a positive spectral margin)**.** *If A is Hurwitz then there is $c > 0$ such that every eigenvalue μ of A satisfies $\operatorname{Re}(\mu) < -c$.*

Proof. The spectrum is compact and nonempty, so Re attains a maximum on it at some μ_0, which is an eigenvalue and hence has $\operatorname{Re}(\mu_0) < 0$. Take $c = -\operatorname{Re}(\mu_0)/2$. □

The exponential decay estimates rest on a general Banach-algebra fact, stated with no matrices in it: an element of spectral radius below one has a contractive power.

Lemma 402 (A contractive power from a spectral radius below one)**.** *Let $\mathbb{A}$ be a nontrivial complete normed algebra over $\mathbb{C}$ and let $a \in \mathbb{A}$ have spectral radius $r(a) < 1$. Then there is an $m > 0$ with $\|a^m\| < 1$.*

Reference: Rudin, Functional Analysis *(Gelfand's spectral-radius formula).*

Proof. Gelfand's formula gives $\|a^m\|^{1/m} \to r(a)$, so $\|a^m\|^{1/m} < 1$ for all large m; pick any such m with $m \neq 0$. Were $\|a^m\| \geq 1$, raising to the positive power $1/m$ would give $\|a^m\|^{1/m} \geq 1$, a contradiction. □

The two predicates are related by a spectral shift (theorem 400), and they agree at rate zero.

Lemma 403 (Hurwitz is rate zero)**.** *A real square matrix A is Hurwitz (definition 398) if and only if it is Hurwitz with decay rate 0 (definition 399).*

Proof. Both unfold to a condition on $\operatorname{Re}\mu$, the first to $\operatorname{Re}\mu < 0$ and the second to $\operatorname{Re}\mu < -0$. □

Lemma 404 (A certified decay rate may be weakened)**.** *If A is Hurwitz with decay rate α (definition 399) and $\beta \leq \alpha$, then A is Hurwitz with decay rate β.*

Proof. Each eigenvalue satisfies $\operatorname{Re}\mu < -\alpha \leq -\beta$. □

Lemma 405 (Zero-dimensional matrices are vacuously Hurwitz)**.** *If the index type is empty then every A is Hurwitz with every decay rate α (definition 399).*

This records the convention chosen for the degenerate case: the predicate quantifies over nonzero eigenvectors, and in dimension zero there are none.

Proof. There is no nonzero vector, so the hypothesis of the predicate is never met. □

Lemma 406 (The scalar example)**.** *For $\alpha < \gamma$, the 1×1 matrix $(-\gamma)$ is Hurwitz with decay rate α (definition 399).*

Included as a concrete check on the sign convention of the definition.

Proof. The only eigenvalue is $-\gamma$, and $\operatorname{Re}(-\gamma) = -\gamma < -\alpha$. □

The Hurwitz condition bounds each eigenvalue away from the imaginary axis one at a time; because the spectrum is compact, that upgrades to a uniform margin (theorem 401). This is what turns a qualitative hypothesis into a quantitative rate, and it is used twice below.

9.2.2 Equivalent analytic conditions

Nothing in this subsection mentions a system: the Hurwitz condition, decay of e^{At}, and an explicit exponential envelope for $\|e^{At}\|$ are shown to be the same condition on the matrix. Two generic facts about exponential envelopes are recorded first.

Lemma 407 (The decay envelope vanishes)**.** *For $\gamma > 0$ and any constants K, t_0,*

$$\lim_{t\to\infty} K\, e^{-\gamma(t-t_0)} = 0.$$

Proof. The affine map $t \mapsto -\gamma(t - t_0)$ tends to $-\infty$ because $-\gamma < 0$; compose with exp and scale by K. □

Lemma 408 (Squeeze against an exponentially decaying envelope)**.** *Let E be a normed space, $f : \mathbb{R} \to E$, and $\gamma > 0$. If eventually*

$$\|f(t)\| \le k\, e^{-\gamma(t-t_0)}\, c,$$

then $f(t) \to 0$ as $t \to \infty$.

Stated over an arbitrary normed space because it is applied to both matrix- and vector-valued quantities.

Proof. The squeeze theorem: $\|f\|$ is bounded below by 0 and above by an envelope that tends to 0 (lemma 407, after reassociating $k \cdot E \cdot c = (kc) \cdot E$). □

Theorem 409 (Exponential decay of the state under a Hurwitz matrix)**.** *If A is Hurwitz (definition 398) then there are $k > 0$ and $\gamma > 0$ with*

$$\|e^{At}x\| \le k\, e^{-\gamma t}\, \|x\|, \qquad \forall\, t \ge 0,\ \forall\, x \in \mathbb{R}^X,$$

in the Euclidean norm. This is the form the Lyapunov stability predicates consume.

Proof. An exponential envelope for e^{At} in any matrix norm, transported to the Euclidean operator norm by continuity of the linear map from matrices to operators. □

Theorem 410 (Decay of the matrix exponential forces the Hurwitz condition)**.** *If $e^{At} \to 0$ as $t \to \infty$ then A is Hurwitz (definition 398).*

Stated from mere convergence rather than from an exponential bound, so that it serves clause (2) and clause (3) of Theorem 8.1 alike.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1.*

Proof. Suppose $\operatorname{Re}\mu \ge 0$ for some eigenpair (μ, v) with $v \neq 0$. Complexification is continuous, so $e_{\mathbb{C}}^{At}v \to 0$ as well. But v is an eigenvector of $tA_{\mathbb{C}}$ with eigenvalue $t\mu$, hence of its exponential (lemma 192), so

$$\|e_{\mathbb{C}}^{At}v\| = |e^{t\mu}|\,\|v\| = e^{t\operatorname{Re}\mu}\|v\| \ge \|v\| > 0$$

for every $t \ge 0$, contradicting convergence to 0. □

Theorem 411 (Pointwise decay of the state forces the Hurwitz condition)**.** *If $e^{At}x \to 0$ as $t \to \infty$ for every real state x, then A is Hurwitz (definition 398).*

This is the composite that both clauses of Theorem 8.1 reach for: the stability predicates speak about states, not about the matrix.

Proof. Feeding the hypothesis the standard basis vectors returns the columns of e^{At}, and convergence in the matrix space is entrywise, so $e^{At} \to 0$; apply theorem 410. □

Theorem 412 (Hurwitz is decay of the matrix exponential)**.** *A real square matrix A is Hurwitz (definition 398) if and only if $e^{At} \to 0$ as $t \to \infty$.*

Proof. Forwards, an exponentially decaying envelope in some matrix norm and lemma 408 squeeze; backwards is theorem 410. □

Theorem 413 (Hurwitz is exponential decay of the matrix exponential)**.** *A real square matrix A is Hurwitz (definition 398) if and only if $e^{\gamma t}e^{At} \to 0$ as $t \to \infty$ for some $\gamma > 0$.*

Recorded separately from theorem 412 because the two are what clauses (2) and (3) of Theorem 8.1 respectively need: bare convergence for asymptotic stability, a rate for exponential stability. No norm appears: the convergence is in the matrix topology, which all matrix norms induce.

Proof. Forwards, an exponential envelope $\|e^{At}\| \le ke^{-\gamma t}$ in some matrix norm gives $e^{(\gamma/2)t}e^{At} \to 0$. Backwards, $e^{At} = e^{-\gamma t} \cdot e^{\gamma t}e^{At} \to 0$, so theorem 412 applies. □

9.2.3 Hespanha's Theorem 8.1

Theorem 414 (Asymptotic stability of a linear time-invariant system)**.** *Consider the time-invariant system $\dot{x} = Ax$ with A a real square matrix.*

The origin is an asymptotically stable equilibrium (definition 287) if and only if A is Hurwitz (definition 398), that is, every eigenvalue of A has strictly negative real part.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1(2).*

Proof. For constant A the state transition matrix is $e^{A(t-t_0)}$ (**??**), so theorem 386 turns the statement into one about $e^{At}x_0$.

If the response converges then A is Hurwitz (theorem 411). Conversely a Hurwitz A gives an exponentially decaying envelope for $\|e^{At}x_0\|$ (**??**); the envelope is bounded by $k\|x_0\|$, which supplies the stability half, and tends to zero (lemma 407), which supplies the attractivity half. □

Theorem 415 (Exponential stability of a linear time-invariant system)**.** *Consider the time-invariant system $\dot{x} = Ax$ with A a real square matrix.*

The origin is an exponentially stable equilibrium (definition 292) if and only if A is Hurwitz (definition 398).

Together with theorem 414 this says the two notions coincide for a time-invariant system — a genuine feature of constant A, not of linearity: for time-varying $A(t)$ they differ, as $\dot{x} = -x/(1+t)$ shows.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1(3).*

Proof. As for theorem 414, but against theorem 388, which already asks for an exponential envelope. The forward direction squeezes $\|e^{At}x\|$ to zero under that envelope and applies theorem 411; the converse is **??** re-anchored from 0 to t_0. □

9.2.4 The Lyapunov equation

Definition 416 (Continuous Lyapunov equation)**.** For real square matrices A, P and Q, the matrix P satisfies the *continuous Lyapunov equation* for A with forcing Q when

$$PA + A^{\mathsf{T}}P = -Q.$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.2.

Definition 417 (Matrix quadratic form)**.** A real matrix P represents the quadratic form $x \mapsto x^{\mathsf{T}}Px$ on Euclidean state space.

Reference: standard quadratic Lyapunov-function construction.

Hespanha's Theorem 8.2 ties the spectral condition to solvability of $PA + A^{\mathsf{T}}P = -Q$. The direction a Lyapunov certificate is *used* in comes first: exhibiting a single positive-definite P proves the spectrum lies in the open left half-plane.

Theorem 418 (A Lyapunov certificate proves the spectrum lies in the open left half-plane)**.** *If P and Q are positive definite and $PM + M^{\mathsf{T}}P = -Q$, then every eigenvalue of M has strictly negative real part.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Complexify the equation and pair it against an eigenvector v of M with eigenvalue μ. The two terms contribute $\mu\, v^*Pv$ and $\bar{\mu}\, v^*Pv$, so the left-hand side is $2\,\mathrm{Re}(\mu)\, v^*Pv$ while the right-hand side is $-v^*Qv$. Both v^*Pv and v^*Qv are strictly positive (lemma 189), so $\mathrm{Re}(\mu) < 0$. □

Theorem 419 (The Lyapunov inequality certifies the Hurwitz condition)**.** *If P is positive definite and $-(PM + M^{\mathsf{T}}P)$ is positive definite, then M is Hurwitz (definition 398).*

This is Hespanha's Theorem 8.2, (5) ⇒ (3). It is stated through positive definiteness of the negated residual rather than through a matrix order relation, so that it needs no Loewner order.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Take the forcing to be minus the residual and apply theorem 418. □

The converse is a construction. One finite block of the integral $\int_0^m e^{tA^{\mathsf{T}}} Q e^{tA}\,\mathrm{d}t$ is summed into a convergent discrete Lyapunov series, with convergence and uniqueness both coming from the contraction $\|e^{Am}\| < 1$. A Hurwitz matrix supplies that contraction, which is theorem 421.

Theorem 420 (The Lyapunov equation from a contracting exponential)**.** *Suppose $\|e^{Am}\| < 1$ for some integer $m > 0$. Then for every positive definite Q there is a positive definite P with $PA + A^{\mathsf{T}}P = -Q$ (definition 416), and P is the unique matrix solution.*

Reference: Khalil, Nonlinear Systems.

Proof. Write $L(P) = PA + A^{\mathsf{T}}P$ for the Lyapunov operator, $C = e^{Am}$, and $T(W) = C^{\mathsf{T}}WC$ for the congruence (Stein) operator.

One block. Let $R = \int_0^m e^{tA^{\mathsf{T}}} Q e^{tA}\,\mathrm{d}t$. Differentiating the integrand two ways and equating the derivatives by uniqueness gives the key identity $L(R) = T(Q) - Q$. R is positive definite because the integrand is, pointwise, and positive definiteness survives the integral.

Summing the blocks. $\|C\| < 1$ makes $\|T\| < 1$, so $\sum_k T^k R$ converges; call it P. It is positive definite, being a convergent sum of positive definite terms. Since A and C commute, L commutes with T, so $L(T^k R) = T^{k+1}Q - T^k Q$ and the series for $L(P)$ telescopes to $-Q$.

Uniqueness. A contraction has only the trivial fixed point, so L is injective; two solutions with the same forcing therefore coincide.

The contraction is taken in the Frobenius norm, the only place in the construction where $\|C^{\mathsf{T}}\| = \|C\|$ is needed. □

Theorem 421 (Lyapunov equation for a Hurwitz matrix)**.** *If a real matrix A is Hurwitz, then for every positive-definite Q there is a positive-definite matrix P satisfying $PA + A^{\mathsf{T}}P = -Q$, and this P is the unique matrix solution.*

omit [DecidableEq X] in Reference: Khalil, Nonlinear Systems*.*

Proof. A contracting integer-time matrix exponential is obtained from the Hurwitz spectrum. Integrating over one time block and summing the resulting discrete Lyapunov series constructs P; contraction proves convergence and uniqueness. □

Lemma 422 (From every forcing to one certificate)**.** *If every positive definite Q admits a positive definite solution of $PM + M^{\mathsf{T}}P = -Q$ (definition 416), then some positive definite P makes $-(PM + M^{\mathsf{T}}P)$ positive definite.*

This is Hespanha's Theorem 8.2, (4) ⇒ (5)*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Instantiate the forcing at $Q = I$; the residual is then $-I$, whose negation is the identity. The edge carries no mathematics beyond the instantiation. □

9.2.5 The rate form

The Lyapunov inequality $PA + A^{\mathsf{T}}P \preceq -2cP$ carries an explicit decay rate: along a trajectory it gives $\dot{V} \le -2cV$ for $V = x^{\mathsf{T}}Px$. It is not one of Hespanha's clauses, but it is the shape every contraction and guaranteed-decay-rate condition is written in, so it is recorded as a sixth equivalent.

Theorem 423 (The rate form certifies the Hurwitz condition)**.** *If P is positive definite, $c > 0$, and $PA + A^{\mathsf{T}}P \preceq -2cP$ — that is, $-(PA + A^{\mathsf{T}}P) - 2cP$ is positive semidefinite — then A is Hurwitz (definition 398).*

Along a trajectory this inequality gives $\dot{V} \le -2cV$ for $V = x^{\mathsf{T}}Px$, so it certifies not merely stability but a decay rate.

Proof. Split $-(PA + A^{\mathsf{T}}P)$ as $(-(PA + A^{\mathsf{T}}P) - 2cP) + 2cP$: positive semidefinite plus positive definite is positive definite, so theorem 419 applies. □

Theorem 424 (A Hurwitz matrix admits a rate certificate)**.** *If A is Hurwitz (definition 398) then there are a positive definite P and a rate $c > 0$ with $PA + A^{\mathsf{T}}P \preceq -2cP$.*

Proof. Solve the Lyapunov equation for the *shifted* matrix. A Hurwitz A has a positive spectral margin c (theorem 401), so $A + cI$ is still Hurwitz (theorem 400). Solving $P(A+cI) + (A+cI)^{\mathsf{T}}P = -I$ (theorem 421) and expanding gives $PA + A^{\mathsf{T}}P = -I - 2cP$, so the residual $-(PA + A^{\mathsf{T}}P) - 2cP$ is *exactly* the identity.

No domination bound between two positive definite matrices is needed: the residual is not merely bounded below, it is I. □

9.2.6 Theorem 8.2

Theorem 425 (Hurwitz is solvability of the Lyapunov equation for every forcing)**.** *A real square matrix A is Hurwitz (definition 398) if and only if for every positive definite Q there is a positive definite P with $PA + A^{\mathsf{T}}P = -Q$ (definition 416), unique among all matrix solutions.*

This is Hespanha's Theorem 8.2, (3) ⇔ (4)*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Forwards is theorem 421. Backwards, instantiate the forcing at $Q = I$ and apply theorem 418. □

Theorem 426 (Hurwitz is the existence of one Lyapunov certificate)**.** *A real square matrix A is Hurwitz (definition 398) if and only if some positive definite P makes $-(PA + A^{\mathsf{T}}P)$ positive definite.*

This is Hespanha's Theorem 8.2, (3) ⇔ (5)*, and the form in which a Lyapunov certificate is actually supplied: one matrix, not a family.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Forwards, theorem 421 produces a solution for each forcing and lemma 422 extracts one certificate from them; backwards is theorem 419. □

Theorem 427 (Hurwitz is the existence of a rate certificate)**.** *A real square matrix A is Hurwitz (definition 398) if and only if there are a positive definite P and a rate $c > 0$ with $PA + A^{\mathsf{T}}P \preceq -2cP$.*

This is clause 6 of theorem 428, which is not in Hespanha's list. It is the rate-carrying refinement of theorem 426, and the shape every contraction and guaranteed-decay-rate condition is written in.

Proof. theorem 424 and theorem 423. □

Theorem 428 (Hespanha's Theorem 8.2)**.** *For the time-invariant system $\dot{x} = Ax$ the following are equivalent.*

1. *The origin is asymptotically stable (definition 287).*
2. *The origin is exponentially stable (definition 292).*
3. *A is Hurwitz (definition 398).*
4. *For every positive definite Q there is a positive definite P with $PA + A^{\mathsf{T}}P = -Q$ (definition 416), and this P is the unique matrix solution.*
5. *There is a positive definite P with $-(PA + A^{\mathsf{T}}P)$ positive definite.*
6. *There are a positive definite P and a rate $c > 0$ with $PA + A^{\mathsf{T}}P \preceq -2cP$.*

Clause 6 is not in Hespanha's list. It is the rate-carrying form of clause 5: along a trajectory it gives $\dot{V} \le -2cV$ for $V = x^{\mathsf{T}}Px$, hence $V(t) \le V(0)e^{-2ct}$, and it is the shape every contraction and guaranteed-decay-rate condition is written in.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Every clause is tied to clause 3 and the cycle is closed by transitivity: 1 ⇔ 3 is theorem 414, 2 ⇔ 3 is theorem 415, 3 ⇔ 4 is theorem 425, 3 ⇔ 5 is theorem 426, and 3 ⇔ 6 is theorem 427. □

9.2.7 Weighted Euclidean log norms

Read through the LMI form of the weighted Euclidean log norm (**??**), a Lyapunov certificate is a norm in which A is contracting.

9.2.8 Quadratic forms along a linear vector field

What makes a solution of the Lyapunov equation a Lyapunov *function*: the forcing matrix is exactly the dissipation rate of the associated quadratic form.

Theorem 429 (A Lyapunov solution makes its quadratic form decrease)**.** *Let P solve $PA+A^\mathsf{T}P = -Q$ (definition 416). Then the derivative of the centered quadratic form $x \mapsto (x-x_{\mathrm{eq}})^\mathsf{T}P(x-x_{\mathrm{eq}})$ (definition 417) along the linear vector field $y \mapsto Ay$ is*

$$-(x-x_{\mathrm{eq}})^\mathsf{T}Q\,(x-x_{\mathrm{eq}}).$$

This is what makes a Lyapunov solution a Lyapunov function*: the forcing Q is exactly the dissipation rate.*

Proof. The derivative of the quadratic form in the direction Ay is $\langle y, Py'\rangle + \langle y', Py\rangle$ with $y' = Ay$, which is the form represented by $PA + A^\mathsf{T}P$. Transporting the equation through `toEuclideanCLM` — under which transpose becomes adjoint — rewrites that as $-Q$. □

Lemma 430 (Identity forcing gives decrease at the squared norm)**.** *For P solving $PA+A^\mathsf{T}P = -I$ (definition 416), the derivative of the centered quadratic form (definition 417) along $y \mapsto Ay$ is $-\|x-x_{\mathrm{eq}}\|^2$.*

Proof. theorem 429 at $Q = I$, where the quadratic form is the squared Euclidean norm. □

9.2.9 A quadratic certificate for an unstable eigenmode

The instability counterpart (theorem 437): an eigenvalue with positive real part yields a real quadratic Chetaev certificate. The construction solves a shifted Lyapunov–Sylvester equation, whose injectivity rests on a nonresonance condition on the spectrum — which is what the lemmas below supply.

Lemma 431 (A nonresonant bilinear form vanishes on a pair of generalized eigenspaces)**.** *Let T be an endomorphism of a complex vector space and B a bilinear form satisfying the Sylvester-type identity*

$$B(x,Ty) + B(Tx,y) = c\,B(x,y) \qquad \forall\, x, y.$$

If $\xi + \nu \neq c$, then B vanishes on $\ker(T-\xi)^k \times \ker(T-\nu)^l$ for all k, l.

Proof. Strong induction on $k+l$. The base cases are the zero generalized eigenspaces, where the vector is 0. Otherwise write $Tx = N_x + \xi x$ and $Ty = N_y + \nu y$ with N_x, N_y one order lower; the induction hypothesis kills $B(N_x, y)$ and $B(x, N_y)$, leaving $(\xi+\nu-c)B(x,y) = 0$, and the coefficient is nonzero by nonresonance. □

Lemma 432 (A nonzero generalized eigenvector witnesses its eigenvalue)**.** *If $x \neq 0$ lies in the maximal generalized eigenspace of T at ξ, then ξ is an eigenvalue of T.*

Proof. Membership gives $x \in \ker(T-\xi)^k$ for some k, necessarily $k \neq 0$ since $x \neq 0$; so that generalized eigenspace is nontrivial, and a nontrivial generalized eigenspace forces an eigenvalue. □

Lemma 433 (A nonresonant bilinear form vanishes identically)**.** *Let V be a finite-dimensional complex vector space, T an endomorphism, and B a bilinear form with $B(x,Ty) + B(Tx,y) = c\,B(x,y)$. If no pair of eigenvalues of T sums to c, then $B = 0$.*

This is the uniqueness half of solvability for the Sylvester-type equation: the associated operator has trivial kernel exactly when the spectrum is nonresonant.

Proof. Over $\mathbb{C}$ the generalized eigenspaces of T span V, so it suffices to check $B(x,y) = 0$ for x, y in generalized eigenspaces. A nonzero such vector witnesses its eigenvalue (lemma 432), so nonresonance applies and lemma 431 gives $B(x,y) = 0$. □

Lemma 434 (Right multiplication acts on the right argument)**.** *For square matrices H, A over a commutative semiring, $(HA)(x,y) = H(x,Ay)$ as bilinear forms.*

Proof. Both sides unfold to $x \cdot (HAy)$ by associativity of matrix-vector multiplication. □

Lemma 435 (Left multiplication by the transpose acts on the left argument)**.** *For square matrices H, A over a commutative semiring, $(A^{\mathsf{T}}H)(x,y) = H(Ax,y)$ as bilinear forms.*

Together with lemma 434 this is what turns the matrix equation $HA + A^{\mathsf{T}}H = cH$ into the Sylvester-type identity on bilinear forms that lemma 433 consumes.

Proof. Move the transpose across the dot product: $x \cdot (A^{\mathsf{T}}Hy) = (Ax) \cdot (Hy)$. □

Lemma 436 (A symmetric matrix represents a symmetric form)**.** *If H is a symmetric real matrix then $H(x,y) = H(y,x)$ as a bilinear form.*

Proof. Move the transpose across the dot product and apply $H^{\mathsf{T}} = H$. □

Theorem 437 (Quadratic instability certificate from an unstable eigenpair)**.** *A complex eigenpair of a real matrix A with $\operatorname{Re}\mu > 0$ yields a Hermitian real matrix H, a shift $\alpha > 0$, and a direction w such that $w^{\mathsf{T}}Hw > 0$ and*

$$HA + A^{\mathsf{T}}H - 2\alpha H$$

is positive definite.

Reference: Hahn, Stability of Motion*; Khalil,* Nonlinear Systems*.*

Proof. Choose a shift avoiding the finitely many pairwise spectral resonances, solve the resulting Lyapunov–Sylvester equation by finite-dimensional injectivity, and extract w from the real or imaginary part of the supplied eigenvector. □

9.3 Discrete-time linear systems

For $x(t+1) = A(t)x(t)$ Hespanha's conditions are again theorems about the stability predicates of the unforced system, now those of definition 353 and the definitions after it. The state transition matrix is a finite product, so no regularity of A is needed, and trajectories always exist, so no forward completeness enters. For constant A the spectral condition is the *Schur* condition: every eigenvalue in the open unit disc.

9.3.1 Hespanha's Definition 8.2

Lemma 438 (Discrete-time stability as a bound on the state transition matrix)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$ with state transition matrix Φ (definition 200), the origin is stable (definition 353) if and only if for every t_0 there is $C \geq 0$ with*

$$\|\Phi(t, t_0)\, x\| \leq C\, \|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(1).*

Proof. ($\Rightarrow$) Stability at $\varepsilon = 1$ gives a margin δ; every response from the δ-ball stays in the unit ball, and homogeneity scales that to the bound with $C = 2/\delta$. ($\Leftarrow$) Given ε, take $\delta = \varepsilon/(C+1)$: a trajectory is its own homogeneous response (theorem 220). □

Theorem 439 (Lyapunov stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is stable (definition 353) if and only if, for every t_0 and every initial condition x_0, the homogeneous response $x(t) = \Phi(t, t_0)\, x_0$ (definition 211) is bounded on $t \geq t_0$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(1).*

Proof. The state-transition form is lemma 438. A bound on Φ bounds each response by $C\|x_0\|$; conversely a family of operators bounded at every point of a Banach space is norm-bounded, by the uniform boundedness principle. □

Theorem 440 (Uniform stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is uniformly stable (definition 355) if and only if there is a single C with $\|\Phi(t, t_0)\, x\| \leq C\, \|x\|$ for all $t \geq t_0$ and all x.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(1), with the margin independent of t_0.*

Proof. As lemma 438, with the margin and the bound both carried outside the quantifier on t_0. □

Theorem 441 (Asymptotic stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is locally asymptotically stable (definition 357) if and only if, for every t_0 and every x_0, the homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is bounded on $t \geq t_0$ and tends to 0.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(2).*

Proof. The stability conjunct is theorem 439. For attractivity an arbitrary x_0 is rescaled into the attractivity ball and the limit scales back; conversely every trajectory is a response (theorem 220). □

Theorem 442 (Global exponential stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is globally exponentially stable (definition 363) if and only if there are $c > 0$ and $0 \leq \rho < 1$ with*

$$\|\Phi(t, t_0)\, x\| \leq c\, \rho^{t-t_0}\, \|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(3).*

Proof. The estimate along every trajectory is the estimate applied to its initial state, since a trajectory is its own homogeneous response; conversely the response from x is a trajectory. □

Theorem 443 (Exponential stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is locally exponentially stable (definition 362) if and only if there are $c > 0$ and $0 \le \rho < 1$ with $\|\Phi(t, t_0)\, x\| \le c\, \rho^{t-t_0}\, \|x\|$ for all $t \ge t_0$ and all x: the condition of theorem 442.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(3).*

Proof. As in the global case; the radius of the local definition is absorbed by the scaling argument, and radius 1 serves in the converse. □

Lemma 444 (Local exponential stability is global, for a discrete-time linear system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is locally exponentially stable (definition 362) if and only if it is globally exponentially stable (definition 363).*

Proof. Both are equivalent to the same estimate: theorem 443 and theorem 442. □

Theorem 445 (Instability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is unstable (definition 354) if and only if there are t_0 and x_0 whose homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is unbounded on $t \ge t_0$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(4).*

Proof. The negation of theorem 439. □

9.3.2 The Schur condition

Nothing in this subsection mentions a system: the Schur condition, decay of A^k, and a geometric envelope for $\|A^k\|$ are the same condition on the matrix. The proof runs through Gelfand's formula, as for the Hurwitz condition, with powers in place of the exponential.

Definition 446 (Schur matrix)**.** A real square matrix A is *Schur* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $|\mu| < 1$. The discrete-time counterpart of definition 398.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.3.

Theorem 447 (A Schur matrix has geometrically decaying powers)**.** *If A is Schur (definition 446) then there are $C > 0$ and $0 \le \rho < 1$ with*

$$\|A^k\| \le C\, \rho^k, \qquad \forall\, k \in \mathbb{N}.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3.*

Proof. The spectral radius of $A_{\mathbb{C}}$ is below 1, so by Gelfand's formula (lemma 402) some power has $\|A^m\| < 1$; fix $\|A^m\| \le c < 1$ with $c > 0$, and let M bound $\|A^r\|$ for $r < m$. Writing $k = r + mq$ with $r < m$, submultiplicativity gives $\|A^k\| \le Mc^q$, and with $\rho = c^{1/m}$, $c^q = \rho^{mq} \le \rho^k/c$. □

Theorem 448 (Geometric decay of the state under a Schur matrix)**.** *If A is Schur (definition 446) then there are $C > 0$ and $0 \le \rho < 1$ with $\|A^k x\| \le C\, \rho^k\, \|x\|$ for every k and x.*

Proof. theorem 447 and $\|Ax\|_\infty \le \|A\|_\infty \|x\|_\infty$. □

Theorem 449 (Decay of the matrix powers forces the Schur condition)**.** *If $A^k \to 0$ as $k \to \infty$ then A is Schur (definition 446).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3.*

Proof. Suppose $|\mu| \geq 1$ for an eigenpair (μ, v) of $A_{\mathbb{C}}$ with $v \neq 0$. Complexification is continuous, so $A_{\mathbb{C}}^k v \to 0$; but $A_{\mathbb{C}}^k v = \mu^k v$ (lemma 194) has norm $|\mu|^k \|v\| \geq \|v\| > 0$. □

Theorem 450 (Pointwise decay of the state forces the Schur condition)**.** *If $A^k x \to 0$ as $k \to \infty$ for every real state x, then A is Schur (definition 446).*

Proof. The standard basis vectors return the columns of A^k, so $A^k \to 0$; apply theorem 449. □

Theorem 451 (Schur is decay of the matrix powers)**.** *A real square matrix A is Schur (definition 446) if and only if $A^k \to 0$ as $k \to \infty$.*

Proof. Forwards, theorem 447 supplies a geometric envelope; backwards is theorem 449. □

Theorem 452 (Schur is a geometric envelope for the matrix powers)**.** *A real square matrix A is Schur (definition 446) if and only if there are $C > 0$ and $0 \leq \rho < 1$ with $\|A^k\| \leq C\rho^k$ for all k.*

Proof. Forwards is theorem 447; backwards, the envelope forces $A^k \to 0$ and theorem 451 applies. □

9.3.3 Hespanha's Theorem 8.3

Clauses (1) and (3) of the theorem, marginal stability and instability, ask that the Jordan blocks at eigenvalues of magnitude one be 1×1; as in continuous time they need a Jordan normal form and are not recorded here.

Theorem 453 (Asymptotic stability of a discrete-time linear time-invariant system)**.** *For the time-invariant system $x^+ = Ax$, the origin is locally asymptotically stable (definition 357) if and only if A is Schur (definition 446): every eigenvalue of A has magnitude strictly smaller than 1.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3(2).*

Proof. For constant A the state transition matrix is A^{t-t_0} (theorem 216), so theorem 441 turns the statement into one about $A^k x_0$. If the responses converge then A is Schur (theorem 450); conversely a Schur A has a geometric envelope $\|A^k x\| \leq C\rho^k \|x\|$ (theorem 448), which bounds the responses and makes them vanish. □

Theorem 454 (Exponential stability of a discrete-time linear time-invariant system)**.** *For the time-invariant system $x^+ = Ax$, the origin is locally exponentially stable (definition 362) if and only if A is Schur (definition 446).*

Together with theorem 453 this says the two notions coincide for a time-invariant system.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3(2).*

Proof. Against theorem 443, which already asks for a geometric envelope $\|A^{t-t_0}x\| \leq c\rho^{t-t_0}\|x\|$. Forwards, at $t_0 = 0$ the envelope makes $A^k x \to 0$, so A is Schur (theorem 450); backwards is theorem 448. □

9.3.4 The discrete Lyapunov equation

The discrete-time counterpart of $PA + A^\mathsf{T}P = -Q$ is $A^\mathsf{T}PA - P = -Q$. Unlike continuous time, the solution needs no integral: it is the convergent series $\sum_k (A^\mathsf{T})^k Q A^k$.

Definition 455 (Discrete Lyapunov equation)**.** For real square matrices A, P and Q, the matrix P satisfies the *discrete-time Lyapunov equation* for A with forcing Q when

$$A^\mathsf{T}PA - P = -Q.$$

Along $x(t+1) = Ax(t)$ the quadratic form $V(x) = x^\mathsf{T}Px$ then changes by $V(Ax) - V(x) = -x^\mathsf{T}Qx$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.4.

Lemma 456 (Decrease of the quadratic form along a discrete LTI system)**.** *If $A^\mathsf{T}PA - P = -Q$ (definition 455), then for every x,*

$$(Ax)^\mathsf{T}P(Ax) - x^\mathsf{T}Px = -x^\mathsf{T}Qx.$$

Proof. $(Ax)^\mathsf{T}P(Ax) = x^\mathsf{T}(A^\mathsf{T}PA)x$, and $A^\mathsf{T}PA = P - Q$. □

Lemma 457 (The transpose of a Schur matrix is Schur)**.** *If A is Schur (definition 446) then so is A^T.*

Proof. A and A^T have the same characteristic polynomial, hence the same spectrum over $\mathbb{C}$, and the spectrum is the set of eigenvalues (lemma 190). □

Theorem 458 (Discrete Lyapunov equation for a Schur matrix)**.** *If A is Schur (definition 446), then for every positive definite Q there is a positive definite P with $A^\mathsf{T}PA - P = -Q$ (definition 455), and P is the unique matrix solution.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Take $P = \sum_k (A^\mathsf{T})^k Q A^k$. It converges because both A and A^T are Schur (lemma 457), so their powers decay geometrically (theorem 447). Shifting the index, $A^\mathsf{T}PA = P - Q$. The first term is Q and the others are positive semidefinite, so P is positive definite (**??**). For uniqueness, a difference W of two solutions satisfies $W = A^\mathsf{T}WA$, hence $W = (A^\mathsf{T})^k W A^k \to 0$. □

Theorem 459 (A discrete Lyapunov certificate proves the spectrum lies in the unit disc)**.** *If P and Q are positive definite and $A^\mathsf{T}PA - P = -Q$, then A is Schur (definition 446).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Complexify the equation and pair it against an eigenvector v with eigenvalue μ: the left-hand side is $(|\mu|^2 - 1)\, v^*Pv$ and the right-hand side is $-v^*Qv$. Both quadratic forms are strictly positive (lemma 189), so $|\mu|^2 < 1$. □

Theorem 460 (The discrete Lyapunov inequality certifies the Schur condition)**.** *If P is positive definite and $-(A^\mathsf{T}PA - P)$ is positive definite, then A is Schur (definition 446).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Take the forcing to be minus the residual and apply theorem 459. □

Theorem 461 (Schur is solvability of the discrete Lyapunov equation for every forcing)**.** *A real square matrix A is Schur (definition 446) if and only if for every positive definite Q there is a positive definite P with $A^\mathsf{T}PA - P = -Q$ (definition 455), unique among all matrix solutions.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Forwards is theorem 458. Backwards, instantiate the forcing at $Q = I$ and apply theorem 459. □

Theorem 462 (Schur is the existence of one discrete Lyapunov certificate)**.** *A real square matrix A is Schur (definition 446) if and only if some positive definite P makes $-(A^\mathsf{T}PA - P)$ positive definite.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Forwards, solve the equation with $Q = I$ (theorem 458); the residual is then $-I$. Backwards is theorem 460. □

9.3.5 Hespanha's Theorem 8.4

Theorem 463 (Hespanha's Theorem 8.4)**.** *For the time-invariant system $x^+ = Ax$ the following are equivalent.*

1. *The origin is asymptotically stable (definition 357).*
2. *The origin is exponentially stable (definition 362).*
3. *A is Schur (definition 446).*
4. *For every positive definite Q there is a positive definite P with $A^\mathsf{T}PA - P = -Q$ (definition 455), and this P is the unique matrix solution.*
5. *There is a positive definite P with $-(A^\mathsf{T}PA - P)$ positive definite.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Every clause is tied to clause 3: $1 \Leftrightarrow 3$ is theorem 453, $2 \Leftrightarrow 3$ is theorem 454, $3 \Leftrightarrow 4$ is theorem 461, and $3 \Leftrightarrow 5$ is theorem 462. □

Chapter 10

Lyapunov's indirect method

Near an equilibrium a nonlinear system behaves like its linearization: if the Jacobian is Hurwitz the equilibrium is locally exponentially stable, and if the Jacobian has an eigenvalue with positive real part it is unstable. Both branches run the same way. A quadratic certificate for the linear system (theorem 421, theorem 437) is a certificate for the nonlinear one as well, once the linearization error is absorbed near the equilibrium.

10.1 The linearization error

Theorem 464 (Uniform linearization error bound)**.** *Let $g : E \to E$ be C^1 on a finite-dimensional space and $C \subseteq E$ compact. For every $\eta > 0$ there is $r > 0$ with*

$$\|g(w) - g(z) - Dg(z)(w - z)\| \leq \eta \|w - z\|$$

for all $z \in C$ and $\|w - z\| < r$.

Proof. Dg is uniformly continuous on the compact thickening of C by 1; take r below its modulus for η and below 1. On the ball of radius r about z, $\|Dg(u) - Dg(z)\| \leq \eta$, and the mean value inequality bounds the remainder. □

Theorem 465 (Quadratic linearization error)**.** *Let f have derivative f' within a convex set s, with f' L-Lipschitz on s. For $x_0, x \in s$,*

$$\|f(x) - f(x_0) - f'(x_0)(x - x_0)\| \leq \tfrac{L}{2} \|x - x_0\|^2.$$

Proof. With $h = x - x_0$, let $g(\tau) = f(x_0 + \tau h) - f(x_0) - \tau f'(x_0)h$ on $[0, 1]$. Then $g(0) = 0$ and $\|g'(\tau)\| = \|(f'(x_0 + \tau h) - f'(x_0))h\| \leq L\tau\|h\|^2$, the derivative of $B(\tau) = \frac{L}{2}\tau^2\|h\|^2$. The norm comparison theorem gives $\|g(1)\| \leq B(1)$. □

Theorem 466 (Gradient of a quadratic form)**.** *For a bounded operator M on a real inner product space, $\|\partial_x \langle x, Mx\rangle\| \leq 2\|M\|\|x\|$.*

Proof. The derivative is $z \mapsto \langle x, Mz\rangle + \langle z, Mx\rangle$; apply Cauchy–Schwarz to each term. □

Lemma 467 (Absorbing the linearization remainder)**.** *Let f be C^1 with $Df(x_{\mathrm{eq}}) = A$, let M be a matrix, and let $c > 0$. Then there is $r > 0$ such that for every x with $\|x - x_{\mathrm{eq}}\| < r$,*

$$\left| D\, q_M(x) \left[f(x) - f(x_{\mathrm{eq}}) - A(x - x_{\mathrm{eq}}) \right] \right| \le c \, \|x - x_{\mathrm{eq}}\|^2,$$

where q_M is the centred quadratic form $q_M(x) = \langle M(x - x_{\mathrm{eq}}),\, x - x_{\mathrm{eq}} \rangle$.

The bracket is the first-order remainder of f at x_{eq}. The point is that c is arbitrary*: near enough to the equilibrium the remainder perturbs $\dot{q}_M$ by less than any prescribed multiple of $\|x - x_{\mathrm{eq}}\|^2$, which is the order of $\dot{q}_M$ itself. This is why the linearization decides stability.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7 (the quadratic-Lyapunov proof of Lyapunov's indirect method).*

Proof. Write $y = x - x_{\mathrm{eq}}$ and e for the remainder. Differentiability of f at x_{eq} gives, for any $\eta > 0$, a radius within which $\|e\| \le \eta \|y\|$. The derivative of a centred quadratic form is bounded by $|Dq_M(x)[e]| \le 2\|M\| \, \|y\| \, \|e\|$, so the product is at most $2\|M\|\eta \, \|y\|^2$. Choosing $\eta = c/(4(\|M\| + 1))$ makes $2\|M\|\eta \le c$. □

10.2 The stable branch

The stable-branch theorem gives uniform local exponential decay on every finite forward solution segment, without silently excluding solutions that are not globally defined.

Theorem 468 (Lyapunov's indirect method, stable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1, and let x_{eq} be an equilibrium. If its Jacobian A at the equilibrium is Hurwitz, then x_{eq} is locally exponentially stable (definition 292) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems*.*

Proof. Solve the identity-forced Lyapunov equation for a positive-definite P; after absorbing the $o(\|x - x_{\mathrm{eq}}\|)$ linearization remainder on a small ball, $V(x) = (x - x_{\mathrm{eq}})^{\mathsf{T}} P (x - x_{\mathrm{eq}})$ is an exponential Lyapunov function with exponent 2 there, so theorem 331 applies. □

10.3 Time-varying systems

For a time-varying system the linearization is itself time-varying, and exponential stability transfers between the two in both directions. The converse Lyapunov function of the nonlinear system serves for its linearization; conversely, the solution of the differential Lyapunov equation of the linearization serves for the nonlinear system.

Theorem 469 (Exponential stability passes to the linearization)**.** *Let x_{eq} be an equilibrium of $\dot{x} = f(t, x)$ on $\mathbb{R}^n$, with f continuously differentiable on $\mathbb{R} \times D$, $D = \{x : \|x - x_{\mathrm{eq}}\| < r\}$, and $f(t, \cdot)$ and $\partial f / \partial x(t, \cdot)$ Lipschitz on D, uniformly in t. If x_{eq} is locally exponentially stable (definition 292), then it is a locally exponentially stable equilibrium of the linearization*

$$\dot{x} = A(t)(x - x_{\mathrm{eq}}), \qquad A(t) = \frac{\partial f}{\partial x}(t, x_{\mathrm{eq}}).$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.15, "only if" direction. Khalil's bounded Jacobian is the Lipschitz bound on $f(t, \cdot)$.*

Proof. Let V be the converse Lyapunov function of the nonlinear system (theorem 334). Along the linearization,

$$\dot{V}_{\text{lin}} = \dot{V} + \frac{\partial V}{\partial x}(A(t)(x - x_{\text{eq}}) - f(t,x)) \leq -c_3\|x - x_{\text{eq}}\|^2 + c_4 L_1 \|x - x_{\text{eq}}\|^3,$$

by **??**. On a small ball this is at most $-\frac{c_3}{2}\|x - x_{\text{eq}}\|^2$, so V is an exponential Lyapunov function for the linearization (theorem 331). □

Theorem 470 (Lyapunov's indirect method for time-varying systems)**.** *Under the hypotheses of theorem 469, if x_{eq} is a locally exponentially stable equilibrium of the linearization $\dot{x} = A(t)(x - x_{\text{eq}})$, $A(t) = \frac{\partial f}{\partial x}(t, x_{\text{eq}})$, then it is locally exponentially stable for $\dot{x} = f(t,x)$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.13. Khalil's bounded Jacobian is the Lipschitz bound on $f(t,\cdot)$.*

Proof. In coordinates the linearization is $\dot{v} = A(t)v$ with A continuous and bounded, and exponentially stable (lemma 307). Theorem 396 with $Q = I$ gives a bounded $P(t)$ with $c_1\|x\|^2 \leq x^{\mathsf{T}}P(t)x \leq c_2\|x\|^2$ and derivative $-x^{\mathsf{T}}x$ along the linearization. Along f, $V = (x - x_{\text{eq}})^{\mathsf{T}}P(t)(x - x_{\text{eq}})$ has derivative

$$\dot{V} \leq -c_3\|x - x_{\text{eq}}\|^2 + 2\|P\|L_1\|x - x_{\text{eq}}\|^3,$$

by **??**, which is at most $-\frac{c_3}{2}\|x - x_{\text{eq}}\|^2$ on a small ball; conclude by theorem 331. □

Theorem 471 (Exponential stability and the linearization)**.** *Under the hypotheses of theorem 469, x_{eq} is locally exponentially stable for $\dot{x} = f(t,x)$ if and only if it is for the linearization.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.15.*

Proof. Theorem 469 and theorem 470. □

10.4 The unstable branch

The quadratic certificate is fed to Chetaev's theorem (theorem 352).

Theorem 472 (Lyapunov's indirect method, unstable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1 with $f(x_{\text{eq}}) = 0$. If the Jacobian at x_{eq} has a complex eigenvalue with positive real part, then x_{eq} is unstable (definition 284) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems *(Lyapunov's indirect method).*

Proof. Build the real quadratic certificate supplied by theorem 437, absorb the first-order nonlinear remainder on a small ball, and apply the exponential Chetaev criterion. □

Theorem 473 (Instability of a linear system)**.** *If $A \in \mathbb{R}^{n\times n}$ has a complex eigenvalue μ with $\operatorname{Re}\mu > 0$, then the origin is unstable (definition 284) for any system on $\mathbb{R}^n$ whose state equation is $\dot{x} = Ax$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7.*

Proof. The field $x \mapsto Ax$ is C^1, vanishes at 0 and is its own Jacobian, so this is theorem 472 at $x_{\text{eq}} = 0$. □

Chapter 11

Controllability and observability

11.1 Observability and controllability matrices

For a finite-dimensional linear system

$$\dot{x} = Ax + Bu, \qquad y = Cx,$$

with state matrix A, input matrix B, and output matrix C, this section collects the (finite-horizon) controllability and observability matrices and the bridge theorem connecting the textbook observability predicate to the kernel of the observability matrix.

11.1.1 Observability matrix

Definition 474 (Observability matrix)**.** The *observability matrix* of a pair (A, C) with $A \in \mathbb{F}^{n\times n}$ and $C \in \mathbb{F}^{p\times n}$ is the block-row matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ C\,A \\ C\,A^2 \\ \vdots \\ C\,A^{n-1} \end{bmatrix} \in \mathbb{F}^{(np)\times n}.$$

Rows are indexed by $\mathrm{Fin}\, n \times \mathrm{Fin}\, p$, so that A^k is available without casting $k : \mathrm{Fin}\, n$ through Fin.val.

Definition 475 (Observability)**.** A linear system (A, C) is *observable* when the only state $x \in \mathbb{F}^n$ for which

$$C\,A^k\,x = 0 \qquad \text{for every } k = 0, 1, \ldots, n-1$$

is the zero state. This phrasing does not mention $\mathcal{O}(A, C)$, so the bridge theorem 481 has real content.

Lemma 476 (Entries of the observability matrix)**.** *Block-row entry shape: at row (k, i) and column j, the observability matrix coincides with the (i, j) entry of $C\,A^k$:*

$$\mathcal{O}(A, C)_{(k,i),\,j} = (C\,A^k)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 474. □

Lemma 477 (Action of the observability matrix)**.** *For every state $x \in \mathbb{F}^n$ and every $(k, i) \in \operatorname{Fin} n \times \operatorname{Fin} p$,*

$$(\mathcal{O}(A, C) \cdot x)_{(k,i)} = (C\, A^k \cdot x)_i.$$

This is the definitional bridge that powers theorem 481.

Proof. Holds by definitional unfolding of $\mathcal{O}(A, C)$ (see definition 474). □

11.1.2 Controllability matrix

Definition 478 (Controllability matrix)**.** The *controllability matrix* of a pair (A, B) with $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$ is the block-column matrix

$$\mathcal{C}(A, B) = \begin{bmatrix} B & A\,B & A^2\,B & \cdots & A^{n-1}\,B \end{bmatrix} \in \mathbb{F}^{n\times(nm)}.$$

Columns are indexed by $\operatorname{Fin} n \times \operatorname{Fin} m$, so that A^k is available without casting $k : \operatorname{Fin} n$ through Fin.val.

Lemma 479 (Entries of the controllability matrix)**.** *Block-column entry shape: at row i and column (k, j), the controllability matrix coincides with the (i, j) entry of $A^k\,B$:*

$$\mathcal{C}(A, B)_{i,\,(k,j)} = (A^k\,B)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 478. □

Definition 480 (Controllability)**.** A linear system (A, B) is *controllable* when every target state $x \in \mathbb{F}^n$ is reachable from the origin in n steps: there exist input vectors $u_0, u_1, \ldots, u_{n-1} \in \mathbb{F}^m$ such that

$$x = \sum_{k=0}^{n-1} A^k\,B\,u_k.$$

This phrasing does not name the controllability matrix, so the bridge theorem 483 has real content.

11.1.3 Milestone: observability iff trivial kernel of $\mathcal{O}$

Theorem 481 (Observability via a trivial kernel)**.** *A finite-dimensional linear system (A, C) is observable in the sense of definition 475 if and only if the observability matrix $\mathcal{O}(A, C)$ has trivial kernel under matrix-vector multiplication:*

$$\operatorname{IsObservable}(A, C) \iff (\forall x,\ \mathcal{O}(A, C) \cdot x = 0 \Rightarrow x = 0).$$

Proof. Both directions follow from the bridge lemma 477, which identifies the (k, i)-coordinate of $\mathcal{O}(A, C) \cdot x$ with the i-coordinate of $(C\,A^k) \cdot x$. The forward direction reads the per-power kernel out of the assembled kernel coordinatewise; the reverse direction packs them back. □

11.1.4 Rank characterizations

Theorem 482 (Observability rank test)**.** *A finite-dimensional system (A, C) is observable if and only if the observability matrix $\mathcal{O}(A, C)$ has full column rank, i.e.*

$$\operatorname{rank} \mathcal{O}(A, C) = n,$$

where n is the state dimension.

Proof. Chain the kernel-form milestone theorem 481 with the matrix-level bridge $(\forall x,\ M \cdot x = 0 \Rightarrow x = 0) \iff \operatorname{rank} M = n$ from 'MatrixAlgebra.Rank', applied with $M = \mathcal{O}(A, C)$. □

Theorem 483 (Controllability rank test)**.** *A linear system (A, B) is controllable if and only if the controllability matrix has full row rank,*

$$\operatorname{rank} \mathcal{C}(A, B) = n,$$

where n is the state dimension.

Proof. Combine the reindexing identity $\mathcal{C}(A, B) \cdot u = \sum_k A^k B \cdot u_k$ (with $u_k(j) = u(k, j)$) with the matrix-level bridge $(\forall y,\ \exists x,\ M\,x = y) \iff \operatorname{rank} M = n$ from 'MatrixAlgebra.Rank', applied with $M = \mathcal{C}(A, B)$. □

11.1.5 Hautus observability

Definition 484 (Unobservable subspace)**.** The *unobservable subspace* of (A, C) is the A-invariant subspace

$$\mathcal{N}(A, C) \;=\; \{\, v \in \mathbb{C}^n \;:\; C\,A^k\,v = 0 \text{ for every } k = 0, 1, \dots, n-1 \,\}.$$

Equivalently, $\mathcal{N}(A, C) = \ker \mathcal{O}(A, C)$, but here it is phrased without naming $\mathcal{O}$ so the bridge theorem 485 reads as content.

Theorem 485 (Observability via the unobservable subspace)**.** *A finite-dimensional system (A, C) is observable in the sense of definition 475 if and only if its unobservable subspace is trivial:*

$$\mathcal{N}(A, C) = \{0\} \iff \mathrm{IsObservable}(A, C).$$

Proof. Both directions are membership unfoldings of definition 484 against definition 475. □

Lemma 486 (A-invariance of the unobservable subspace)**.** *The unobservable subspace is closed under the action of A: for every $v \in \mathcal{N}(A, C)$, also $A\,v \in \mathcal{N}(A, C)$.*

Proof. For $k = 0, \dots, n-2$ this is direct from the definition. For $k = n-1$ we land at $C\,A^n\,v$, which Cayley-Hamilton rewrites as a $\mathbb{C}$-linear combination of $C\,A^i\,v$ for $i < n$. Each of those is zero, so the combination is zero. □

Definition 487 (Hautus observability matrix)**.** The *Hautus observability matrix* of (A, C) at a complex number μ is the block-row matrix

$$H_{A,C}(\mu) \;=\; \begin{bmatrix} \mu I - A \\ C \end{bmatrix} \in \mathbb{C}^{(n+p)\times n}.$$

Lemma 488 (Kernel of the Hautus matrix)**.** *For every $\mu \in \mathbb{C}$ and $v \in \mathbb{C}^n$,*

$$H_{A,C}(\mu) \cdot v = 0 \iff (A\,v = \mu\,v \ \wedge\ C\,v = 0).$$

Proof. □

Theorem 489 (Unobservability gives a Hautus rank drop)**.** *If (A, C) is not observable, then there exists $\mu \in \mathbb{C}$ and a nonzero vector $v \in \mathbb{C}^n$ such that*

$$H_{A,C}(\mu) \cdot v = 0,$$

i.e. μ is an eigenvalue of A with an eigenvector in $\ker C$*.*

Proof. The unobservable subspace definition 484 is nontrivial by theorem 485. It is A-invariant by lemma 486, hence finite-dimensional and invariant; over $\mathbb{C}$ this guarantees a nonzero eigenvector inside it. That vector is automatically in $\ker C$ (membership at $k = 0$), so it lies in the kernel of $H_{A,C}(\mu)$ via lemma 488. □

Theorem 490 (A Hautus rank drop gives unobservability)**.** *If $\mu \in \mathbb{C}$ and $v \neq 0$ satisfy $H_{A,C}(\mu) \cdot v = 0$, then (A, C) is not observable.*

Proof. The witness gives $Av = \mu v$ and $Cv = 0$. By induction on k, $A^k v = \mu^k v$, so $C\,A^k\,v = \mu^k\,Cv = 0$ for every k. The vector v is therefore in the unobservable subspace and is nonzero, so observability fails. □

Theorem 491 (Hautus (PBH) test for observability)**.** *A finite-dimensional system (A, C) over $\mathbb{C}$ is observable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,C}(\mu)$ has trivial kernel:*

$$\mathrm{IsObservable}(A, C) \iff \forall \mu \in \mathbb{C},\ \ker H_{A,C}(\mu) = \{0\}.$$

Proof. Combine theorem 489 (failure direction) and theorem 490 (converse). □

11.1.6 Hautus controllability

Definition 492 (Hautus controllability matrix)**.** The *Hautus controllability matrix* of (A, B) at a complex number μ is the block-column matrix

$$H^{\mathrm{ctrl}}_{A,B}(\mu) \;=\; \begin{bmatrix} \mu I - A & B \end{bmatrix} \in \mathbb{C}^{n \times (n+m)}.$$

Theorem 493 (Hautus (PBH) test for controllability)**.** *A finite-dimensional system (A, B) over $\mathbb{C}$ is controllable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H^{\mathrm{ctrl}}_{A,B}(\mu)$ has full row rank:*

$$\mathrm{IsControllable}(A, B) \iff \forall \mu \in \mathbb{C},\ \operatorname{rank} H^{\mathrm{ctrl}}_{A,B}(\mu) = n.$$

Proof. Combine the duality bridge `isControllable_iff_isObservable_transpose` with the observability Hautus iff theorem 491, then convert the kernel-form RHS to the rank form via the matrix bridge `mulVec_kernel_trivial_iff_rank_eq_card_cols` from 'MatrixAlgebra.Rank', and finally identify the transposed Hautus matrices. □

11.2 Reachable subspace and Kalman decomposition

Definition 494 (Reachable subspace)**.** The reachable subspace of a pair (A, B) is the column span of its controllability matrix:

$$\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B).$$

Thus it is exactly the span of $B, AB, \dots, A^{n-1}B$.

Reference: Hespanha, *Linear Systems Theory*.

Lemma 495 (Membership in the reachable subspace)**.** *Let $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$, and let $\mathcal{C}(A, B) \in \mathbb{F}^{n\times(nm)}$ be their controllability matrix. For $x \in \mathbb{F}^n$,*

$$x \in \mathcal{R}(A, B) \quad \iff \quad \exists u : \operatorname{Fin}(n) \times \operatorname{Fin}(m) \to \mathbb{F}, \qquad \mathcal{C}(A, B)u = x.$$

Here $\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B)$ is the reachable subspace.

Proof. □

Theorem 496 (Controllability via the reachable subspace)**.** *A pair (A, B) is controllable if and only if its reachable subspace is the whole state space.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Lemma 497 (A-invariance of the reachable subspace)**.** *The reachable subspace is A-invariant: $A\mathcal{R}(A, B) \subseteq \mathcal{R}(A, B)$.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Definition 498 (Kalman decomposition)**.** A Kalman decomposition of (A, B, C) records four coordinate sectors adapted to the reachable and unobservable subspaces. Their direct sums recover those subspaces and the full state space. The chosen complements are noncanonical and are not individually asserted to be invariant under A.

References: Hespanha, *Linear Systems Theory*; Kailath, *Linear Systems*.

Theorem 499 (Existence of the Kalman decomposition)**.** *Every finite-dimensional complex state-space system has four coordinate sectors adapted to its reachable and unobservable subspaces. The complements are noncanonical vector-space complements and are not individually asserted to be invariant under A.*

References: Hespanha, Linear Systems Theory*; Kailath,* Linear Systems*.*

Proof. □

Definition 500 (Adapted coordinate equivalence)**.** Addition of the four Kalman coordinate sectors defines a linear equivalence $X_{c\bar{o}} \times X_{co} \times X_{\bar{c}\bar{o}} \times X_{\bar{c}o} \simeq \mathbb{C}^n$. Choosing sector bases therefore gives an adapted basis of the original state space.

Theorem 501 (Block-zero pattern in adapted coordinates)**.** *In a basis adapted to the four Kalman coordinate sectors, ordered as controllable-unobservable, controllable-observable, uncontrollable-unobservable, uncontrollable-observable, the system has the forced-zero pattern*

$$A' = \begin{bmatrix} * & * & * & * \\ 0 & * & 0 & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}, \qquad B' = \begin{bmatrix} * \\ \\ 0 \\ 0 \end{bmatrix}, \qquad C' = \begin{bmatrix} 0 & * & 0 & * \end{bmatrix}.$$

Every displayed zero is asserted entrywise; no vanishing claim is made for the starred blocks. The chosen complement sectors are not individually asserted to be invariant under A.

References: Hespanha, Linear Systems Theory*; Kalman, "Mathematical Description of Linear Dynamical Systems" (1963); Kailath,* Linear Systems*.*

Proof. Reachable-subspace invariance forces the lower-left state and input zeros, unobservable-subspace invariance forces the remaining state zeros, and the zeroth observability condition forces the output zeros. Coordinate-basis representation lemmas translate these component statements to individual matrix entries. □